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(54) INTEGRATED MODULAR SEATING SYSTEMS FOR ELECTRONIC GAMING SYSTEMS AND METHODS

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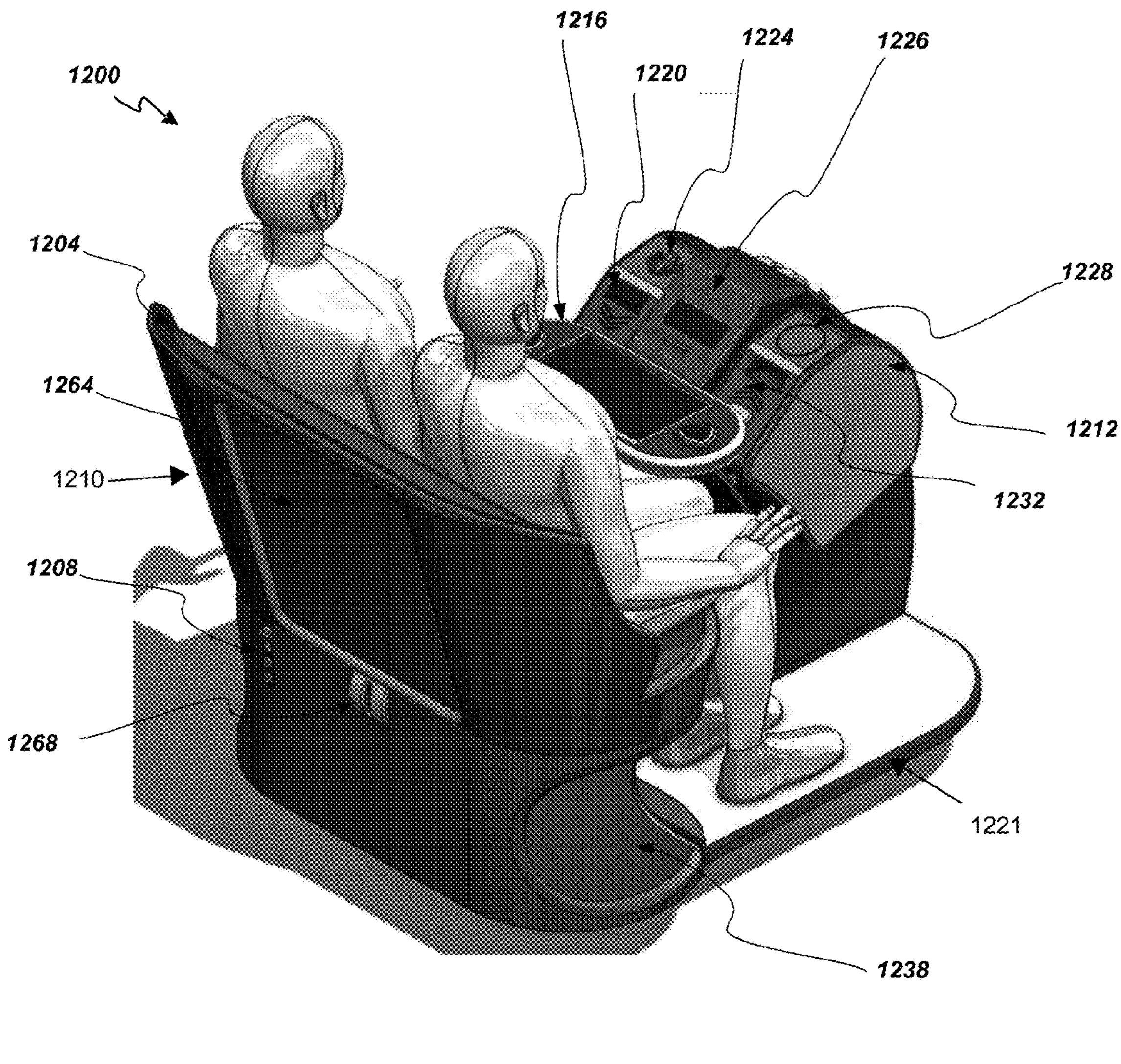
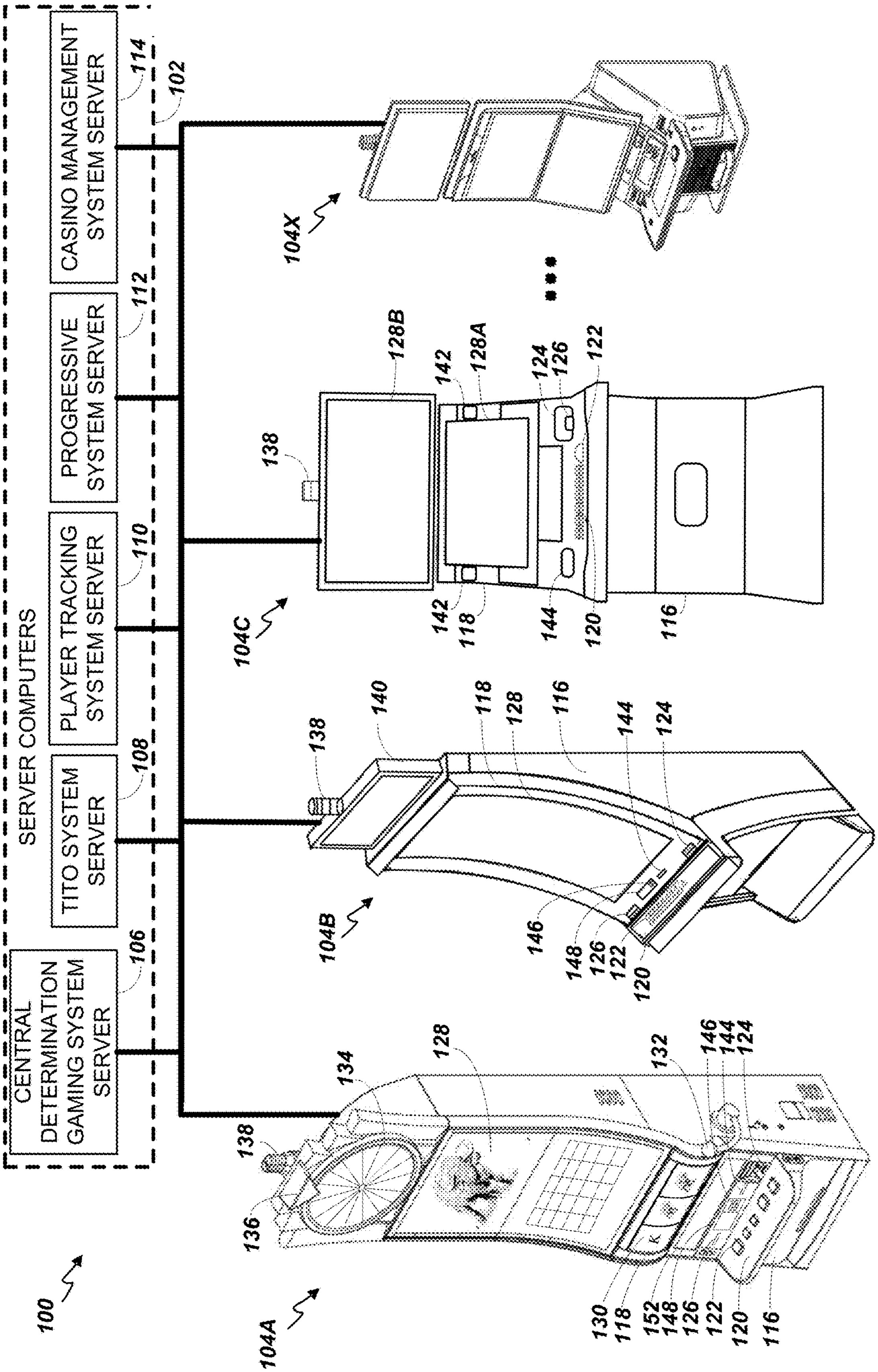
(52) **U.S. Cl.**
CPC *G07F 17/3216* (2013.01); *G07F 17/3209* (2013.01); *G07F 17/3211* (2013.01)

(57) **ABSTRACT**

Seating systems for electronic gaming on electronic gaming machines are provided. Some systems have a seat, a kiosk directly or indirectly coupled to the seat, and an operating console rotatably coupled to the kiosk and having a button deck, a first end piece detachably coupled to a first end of the operating console and having a first button, and a second end piece detachably coupled to a second end of the operating console, opposite the first end, and having a second button or a display. The first and second end pieces are configured to be detached from the operating console, the operating console is configured to receive a third end piece, the operating console is configured to be rotated between first and second orientations, and the button deck and the first and second end pieces are configured to be communicatively coupled to a game controller of the electronic gaming machine.

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SERVER COMPUTERS
CENTRAL DETERMINATION GAMING SYSTEM SERVER
106
TITO SYSTEM SERVER
108
PLAYER TRACKING SYSTEM SERVER
110
PROGRESSIVE SYSTEM SERVER
112
CASINO MANAGEMENT SYSTEM SERVER
114
102
104A
104B
104C
104X
116
118
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128B
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152

FIG. 1

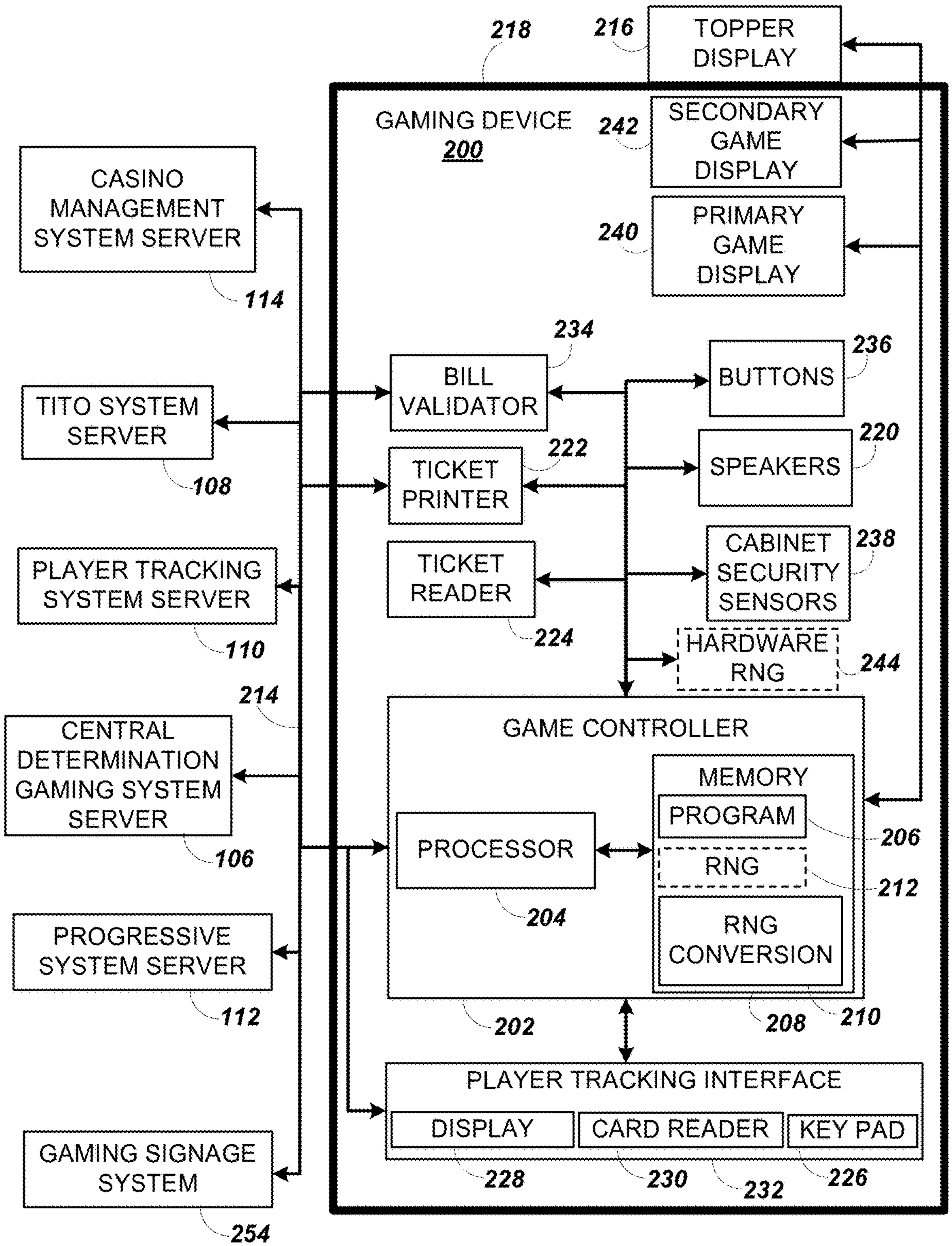
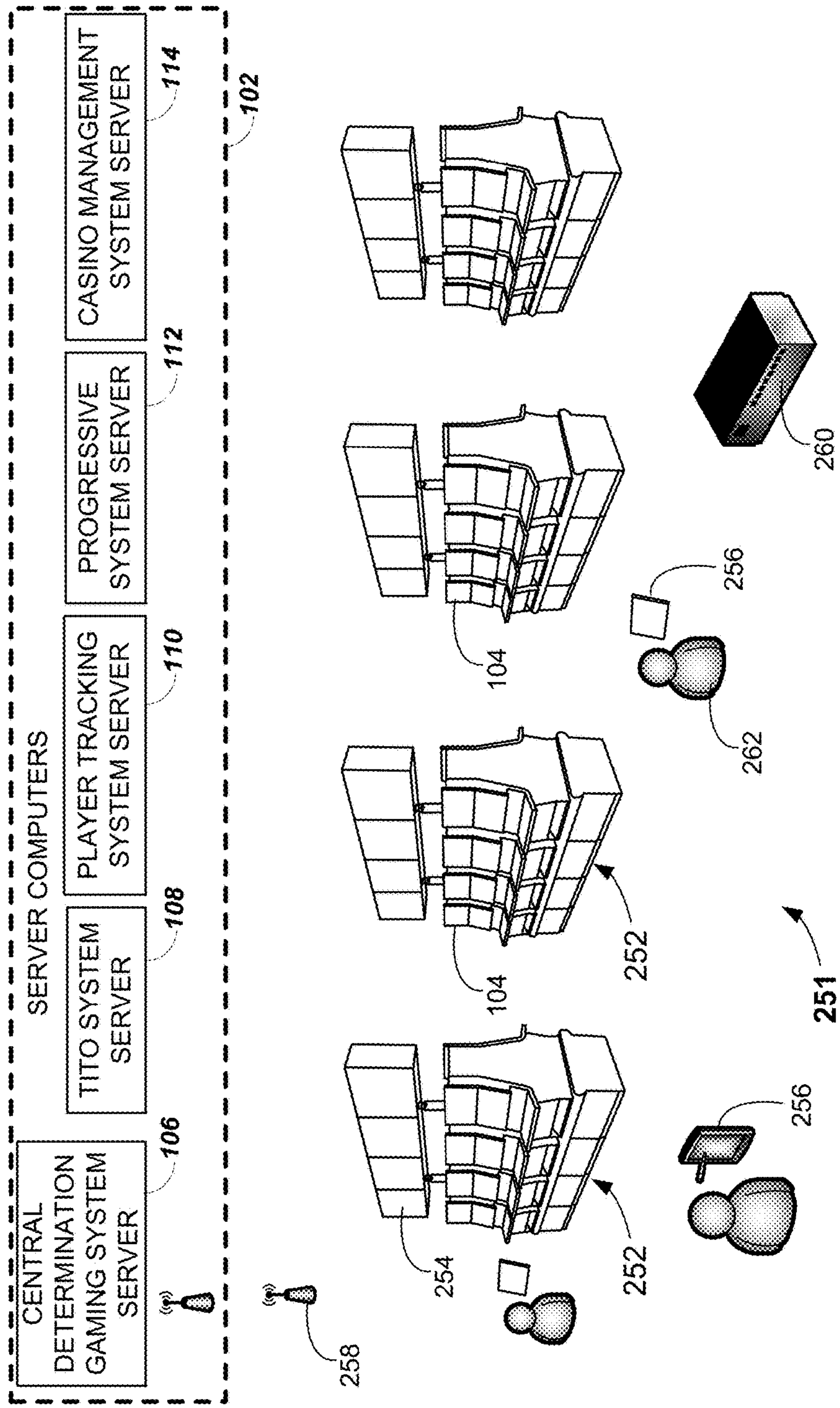

218
216
TOPPER DISPLAY
GAMING DEVICE
200
242
SECONDARY GAME DISPLAY
240
PRIMARY GAME DISPLAY
CASINO MANAGEMENT SYSTEM SERVER
114
TITO SYSTEM SERVER
108
PLAYER TRACKING SYSTEM SERVER
110
214
CENTRAL DETERMINATION GAMING SYSTEM SERVER
106
PROGRESSIVE SYSTEM SERVER
112
GAMING SIGNAGE SYSTEM
254
234
BILL VALIDATOR
236
BUTTONS
222
TICKET PRINTER
220
SPEAKERS
TICKET READER
224
238
CABINET SECURITY SENSORS
HARDWARE RNG
244
GAME CONTROLLER
MEMORY
PROGRAM
206
RNG
212
PROCESSOR
204
RNG CONVERSION
202
208
210
PLAYER TRACKING INTERFACE
DISPLAY
CARD READER
KEY PAD
228
230
232
226

FIG. 2A


SERVER COMPUTERS
CENTRAL DETERMINATION GAMING SYSTEM SERVER
106
TITO SYSTEM SERVER
108
PLAYER TRACKING SYSTEM SERVER
110
PROGRESSIVE SYSTEM SERVER
112
CASINO MANAGEMENT SYSTEM SERVER
114
102
258
254
252
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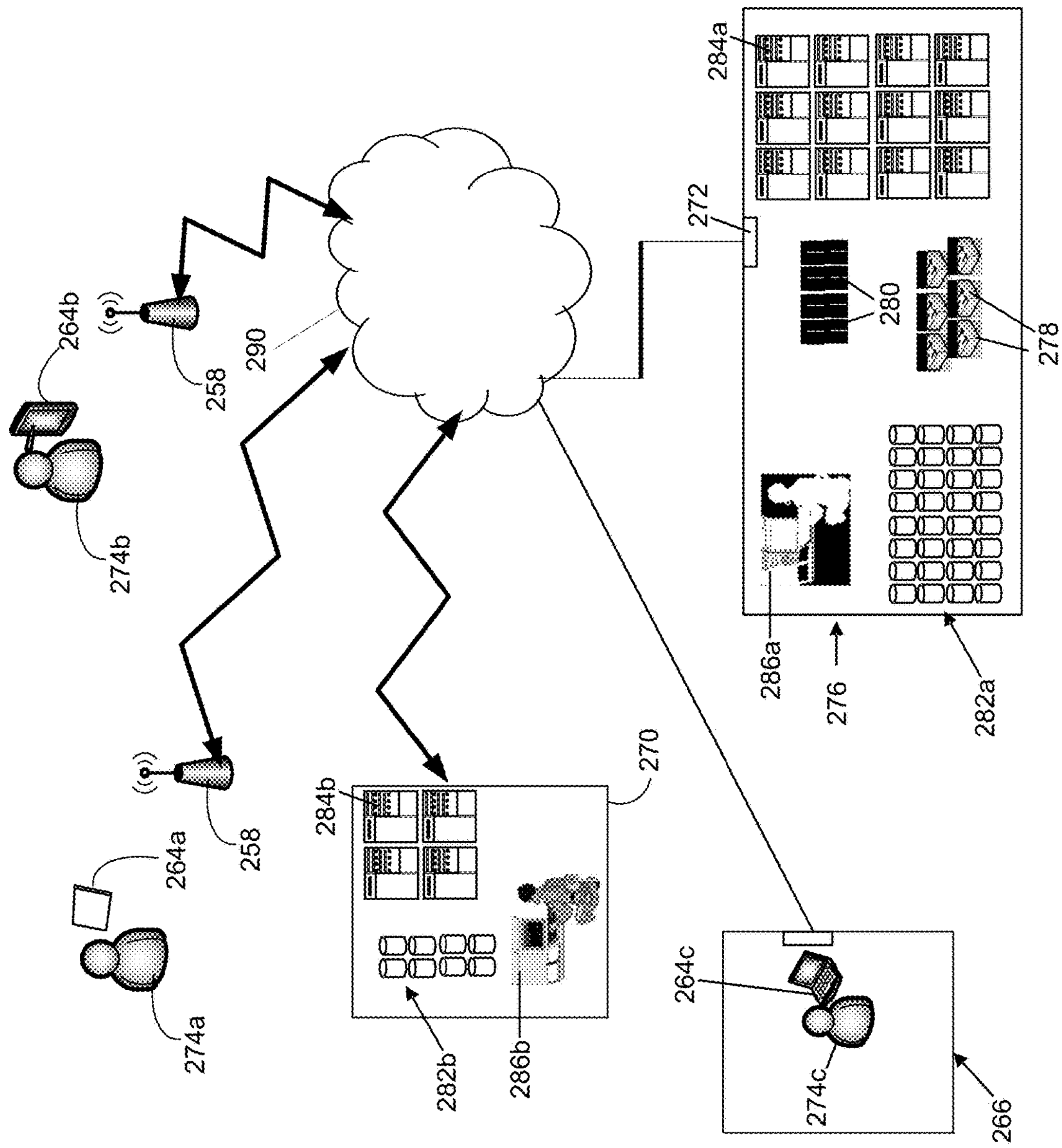
104
262
256
256
260

FIG. 2B



264b
258
290
274b
272
284a
280
278
286a
276
282a
258
264a
274a
270
284b
282b
286b
264c
274c
266

FIG. 2C

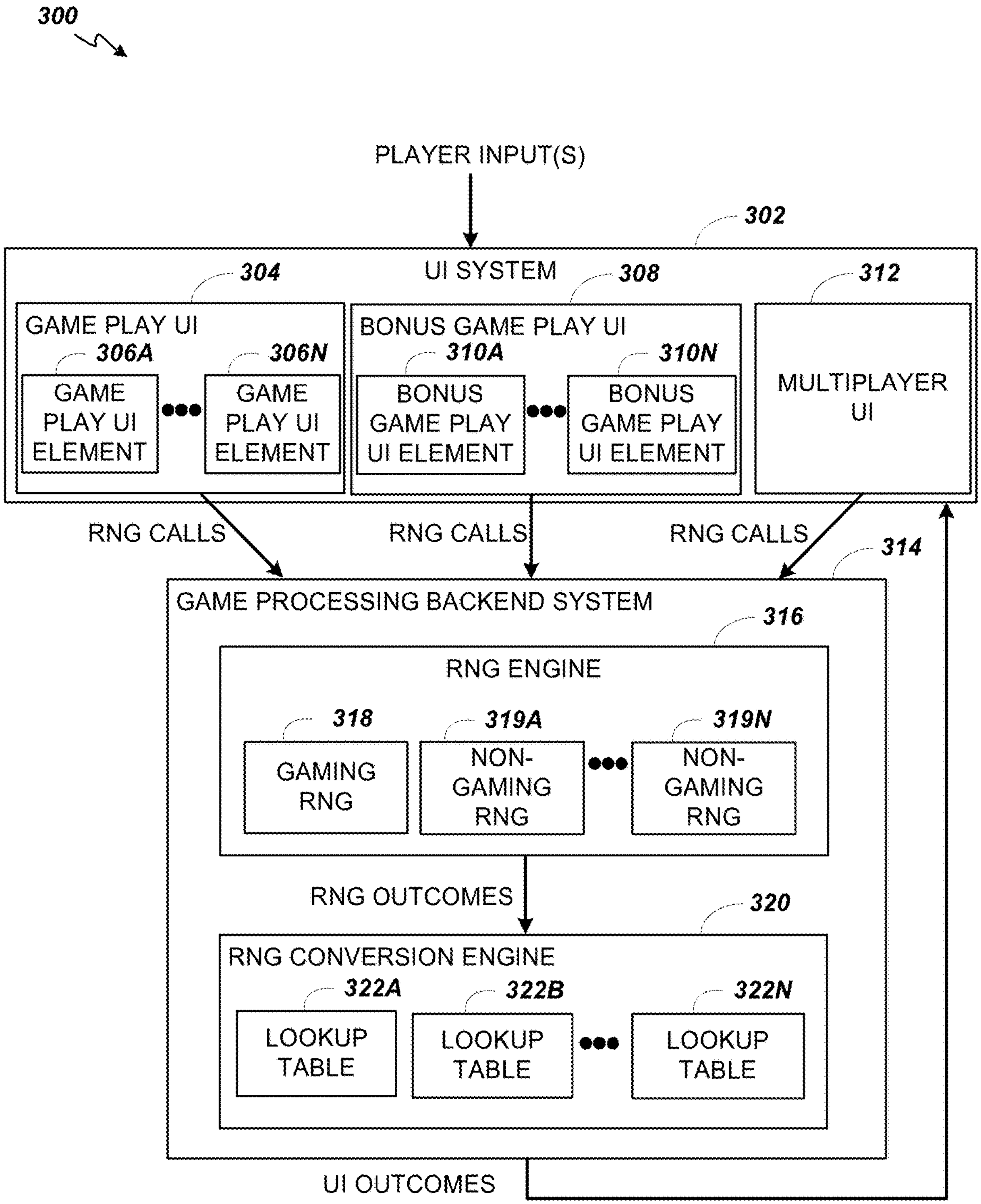

300
PLAYER INPUT(S)
302
UI SYSTEM
304
GAME PLAY UI
306A
GAME PLAY UI ELEMENT
306N
GAME PLAY UI ELEMENT
308
BONUS GAME PLAY UI
310A
BONUS GAME PLAY UI ELEMENT
310N
BONUS GAME PLAY UI ELEMENT
312
MULTIPLAYER UI
RNG CALLS
RNG CALLS
RNG CALLS
314
GAME PROCESSING BACKEND SYSTEM
316
RNG ENGINE
318
GAMING RNG
319A
NON-GAMING RNG
319N
NON-GAMING RNG
RNG OUTCOMES
320
RNG CONVERSION ENGINE
322A
LOOKUP TABLE
322B
LOOKUP TABLE
322N
LOOKUP TABLE
UI OUTCOMES

FIG. 3

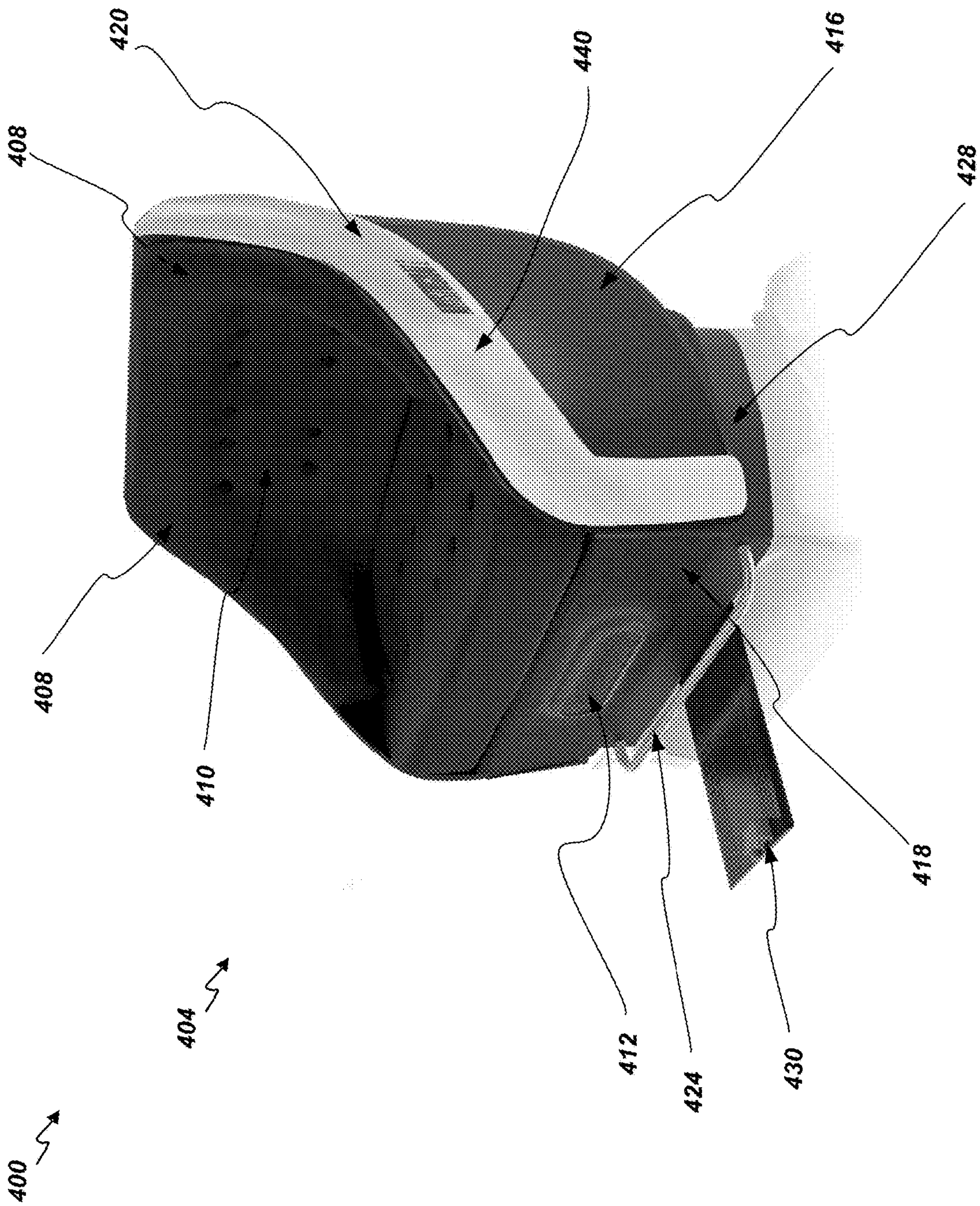


FIG. 4A

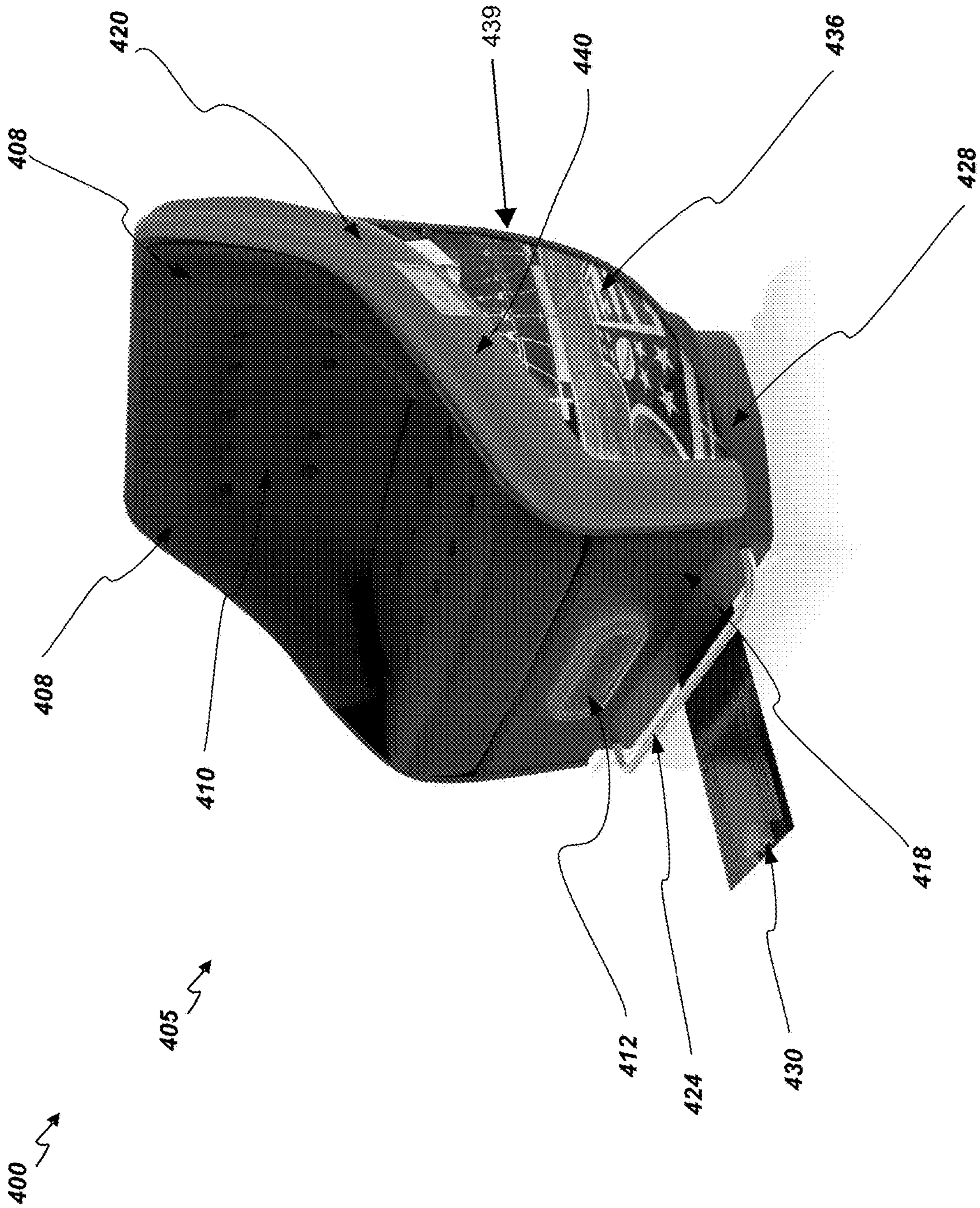


FIG. 4B

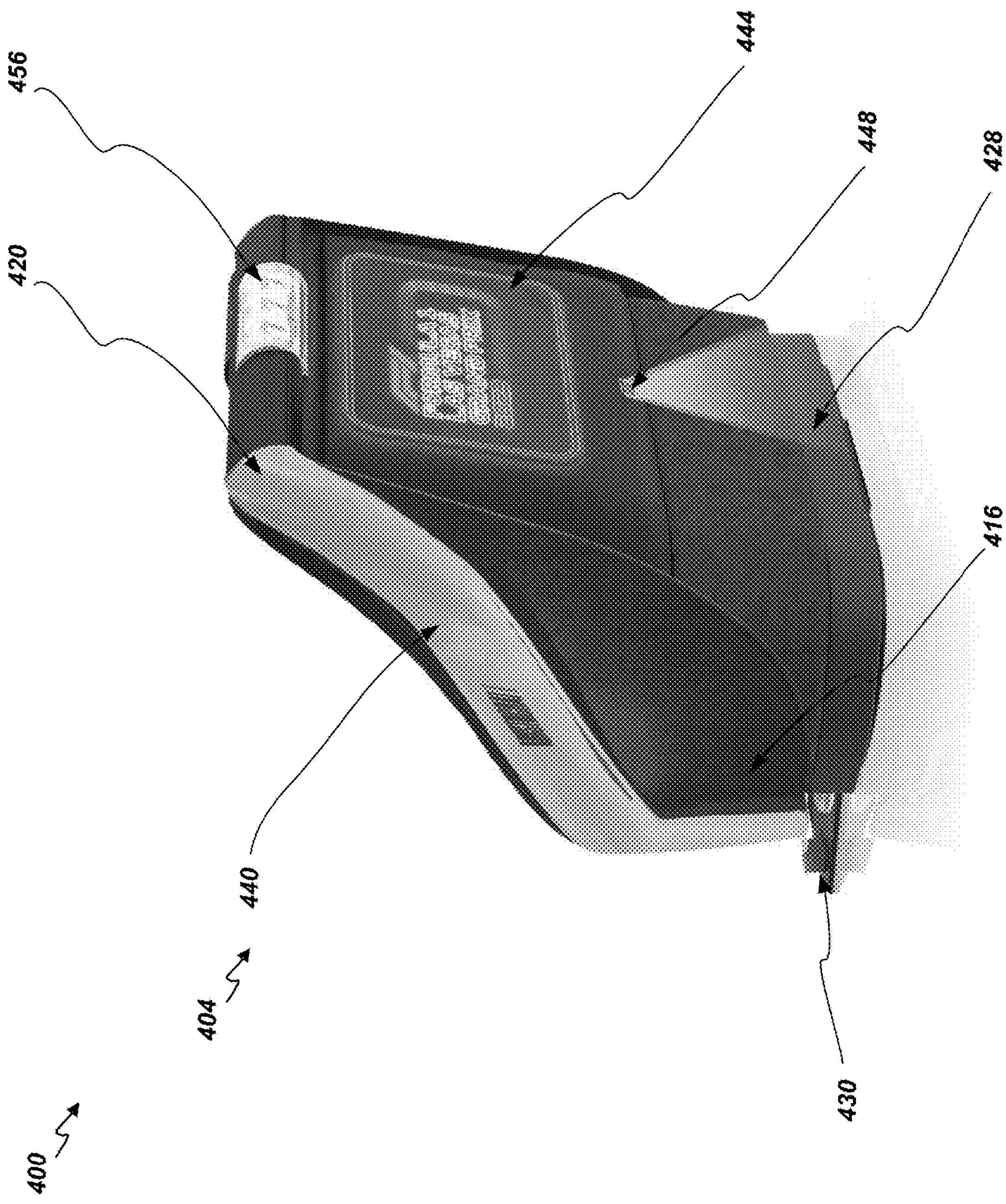


FIG. 4C

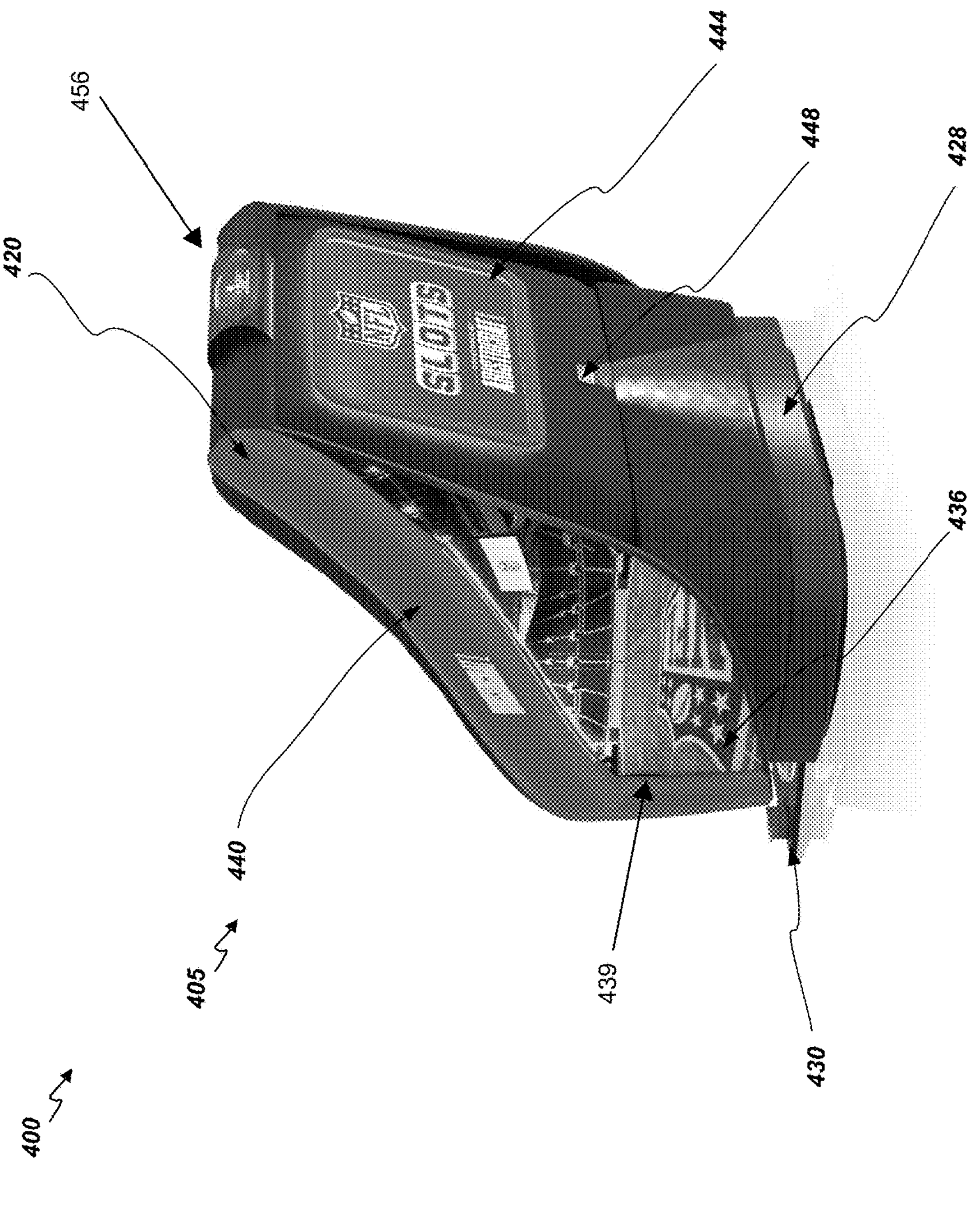


FIG. 4D

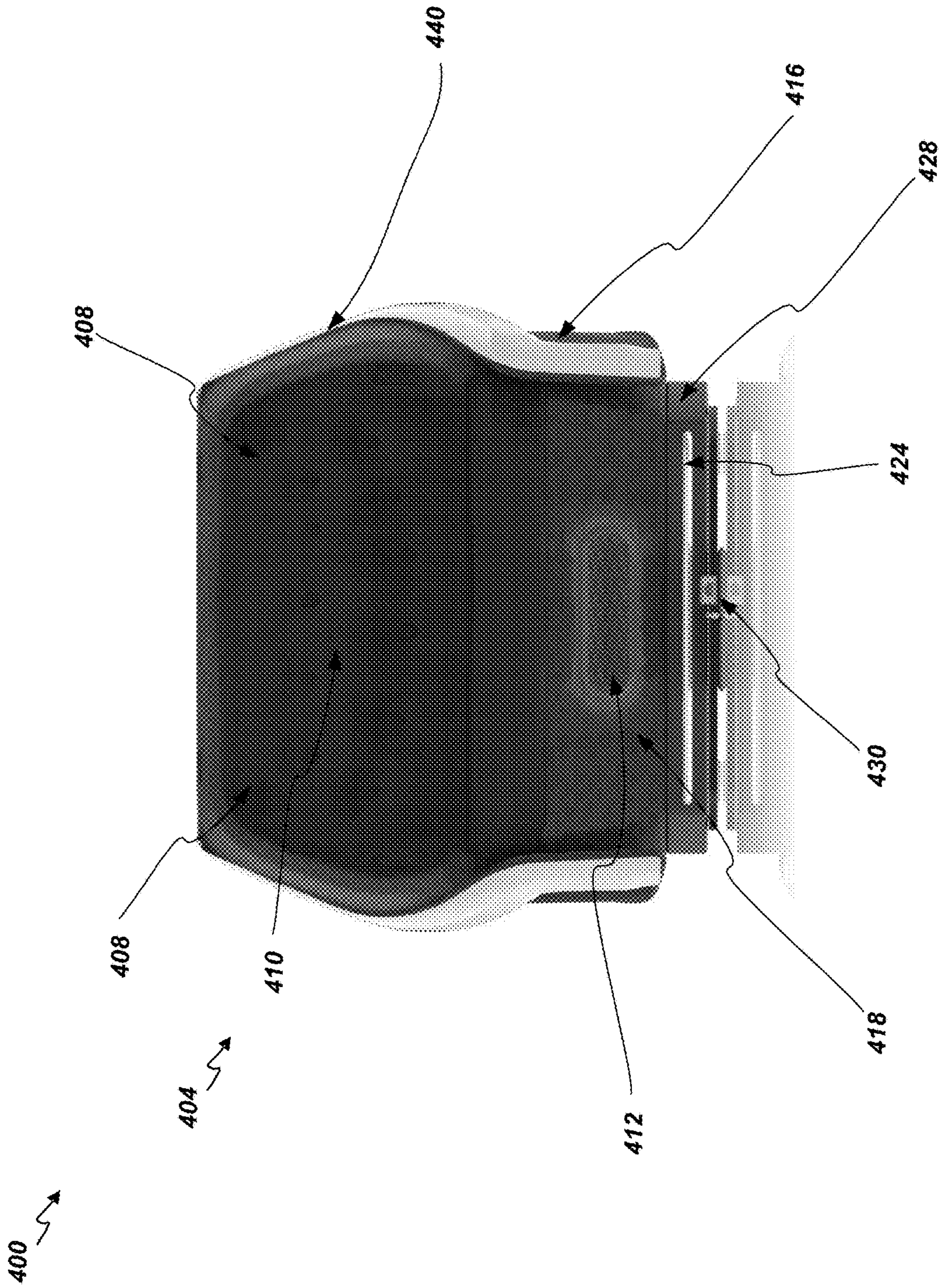


FIG. 4E

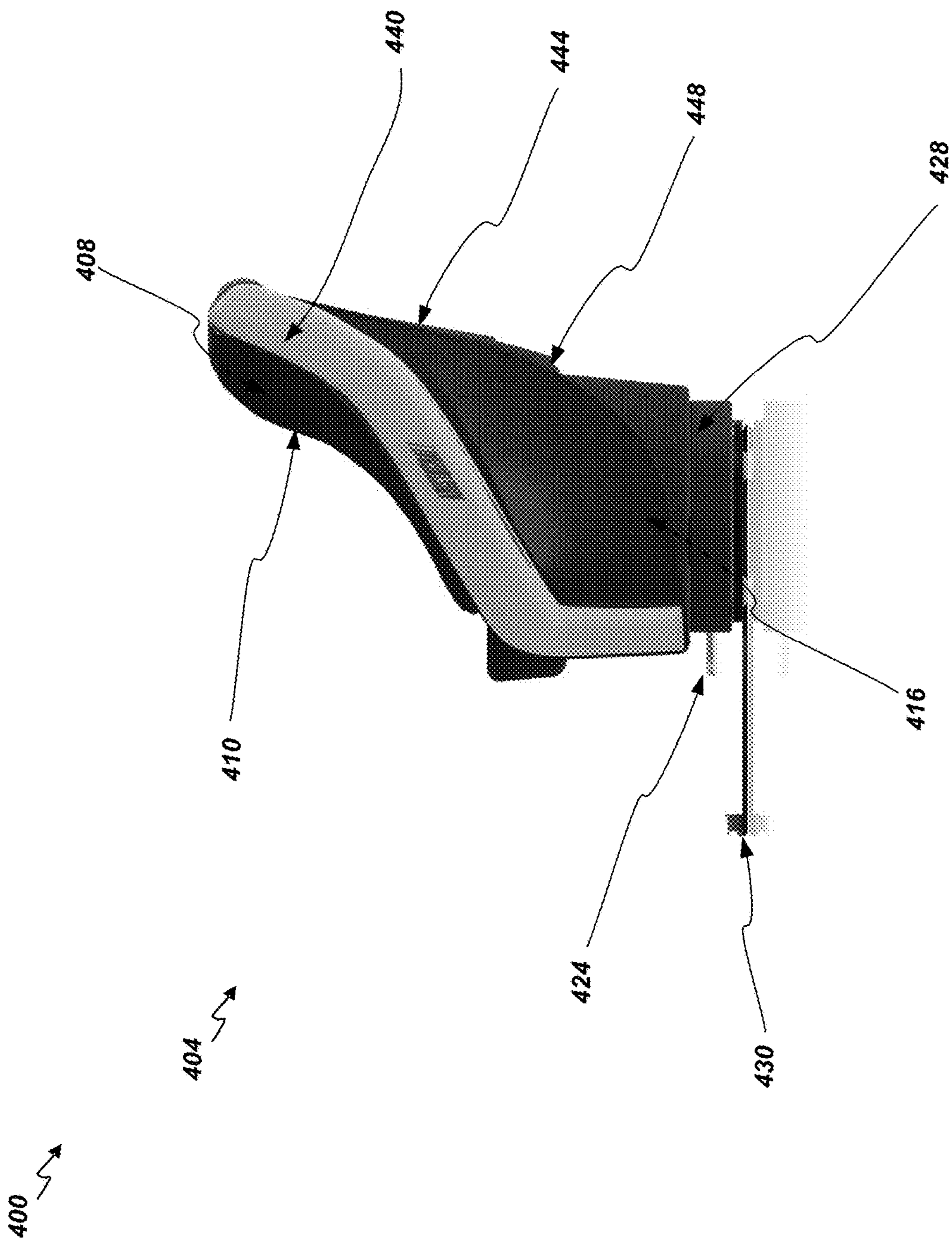


FIG. 4F

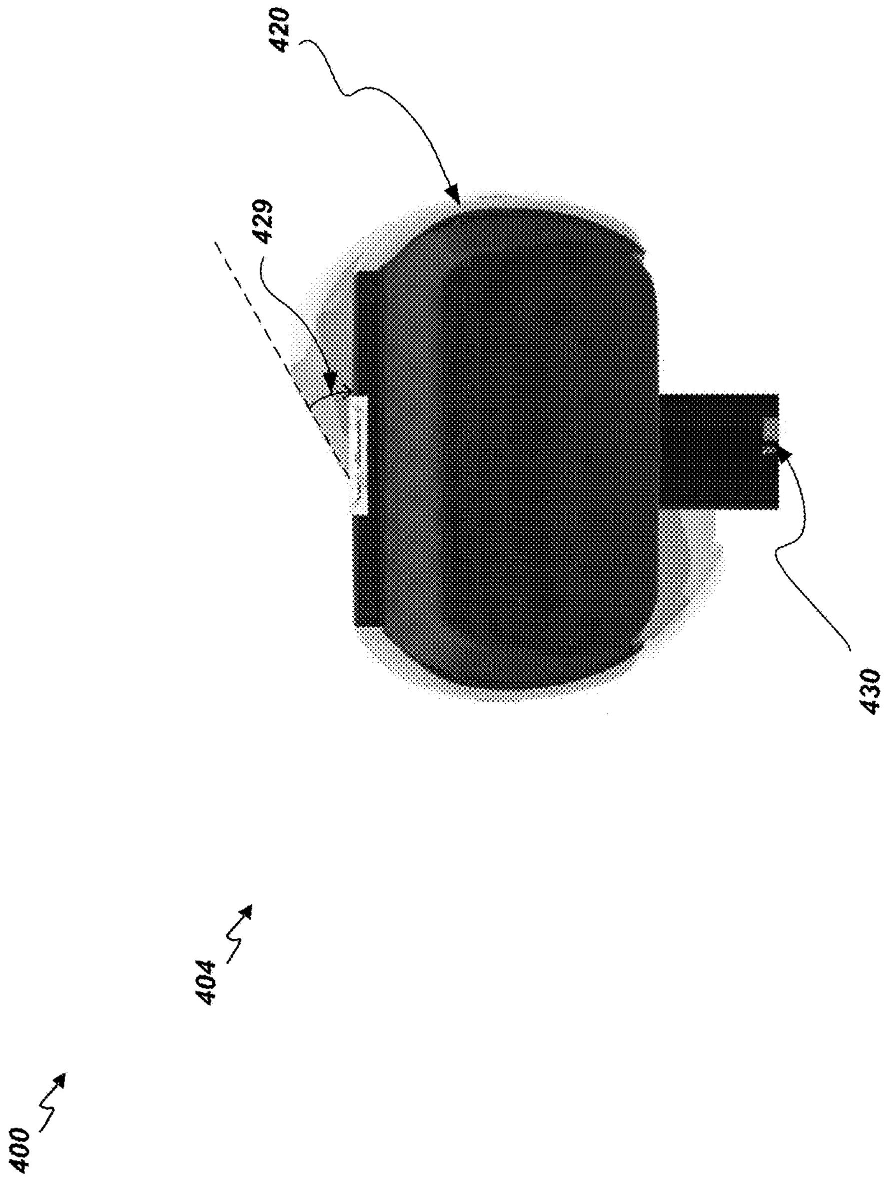


FIG. 4G

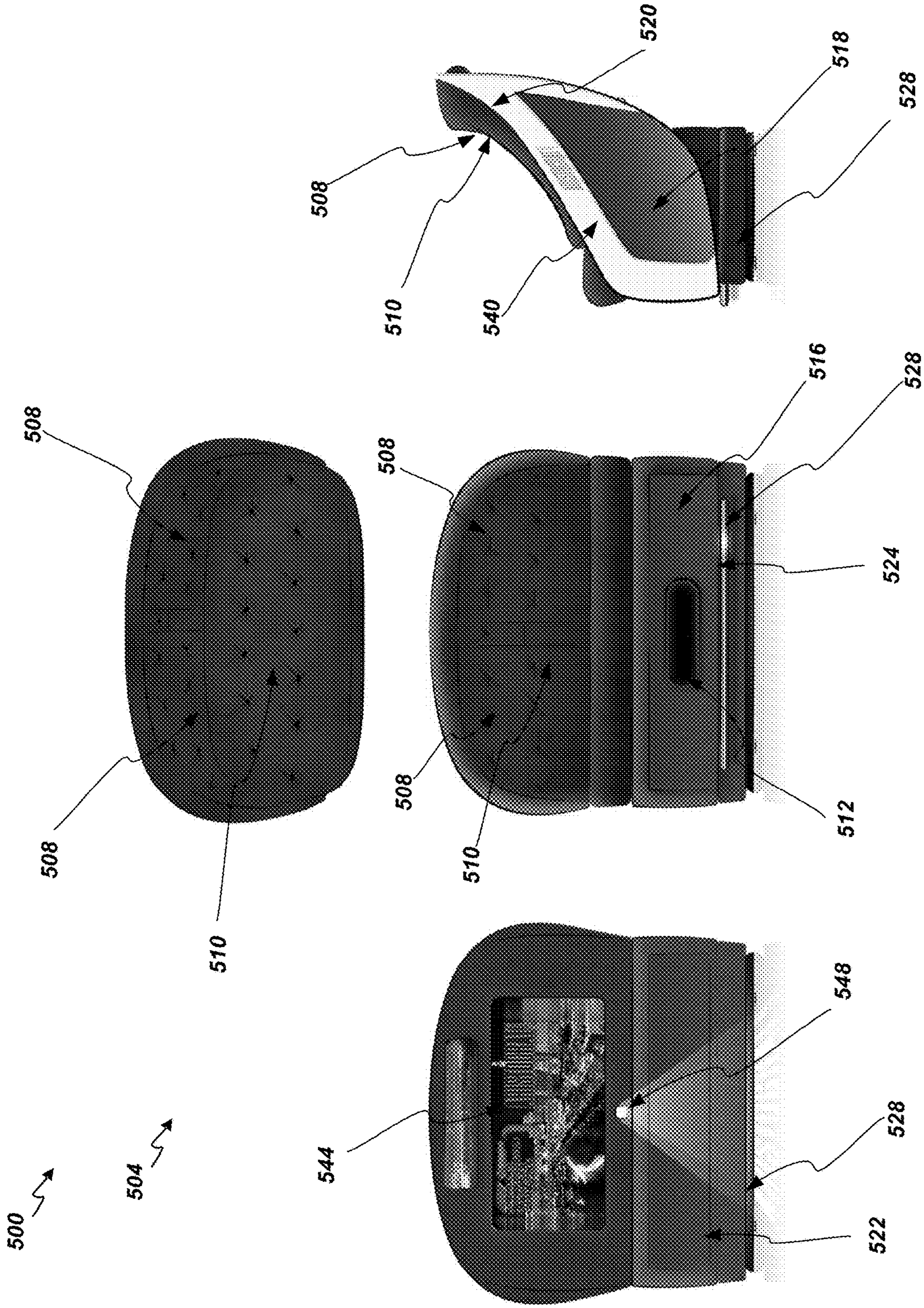


FIG. 5

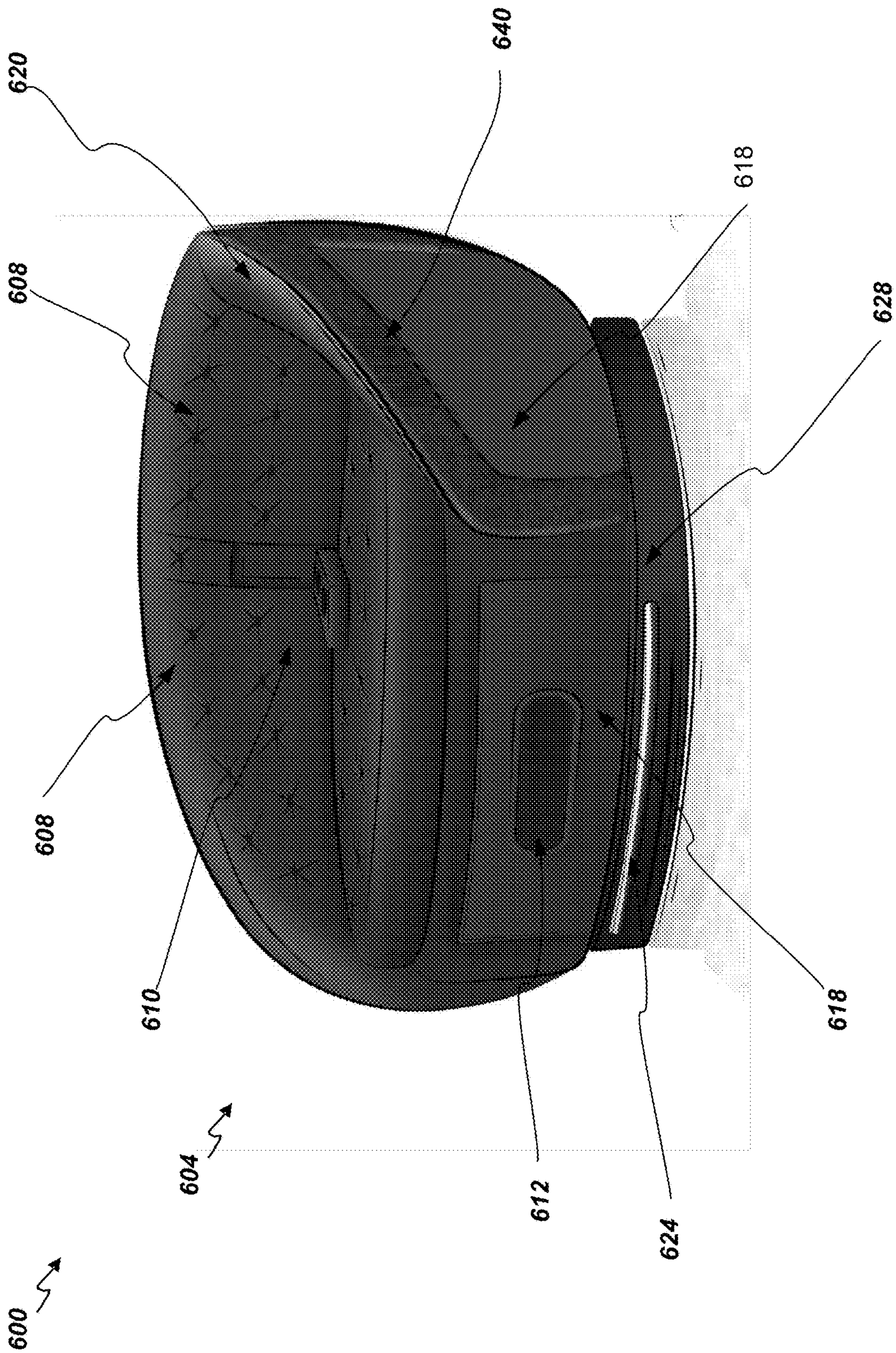


FIG. 6

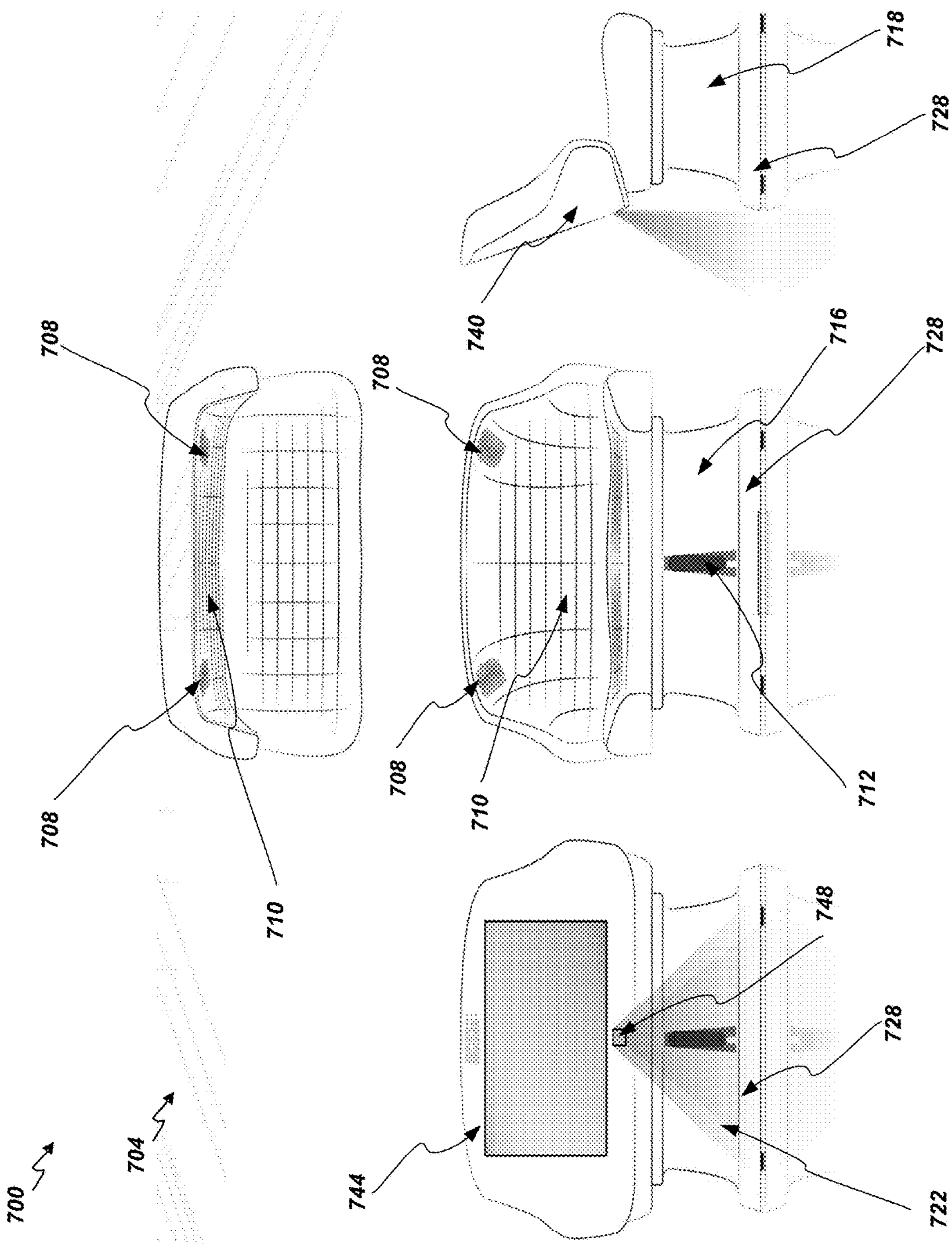

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FIG. 7

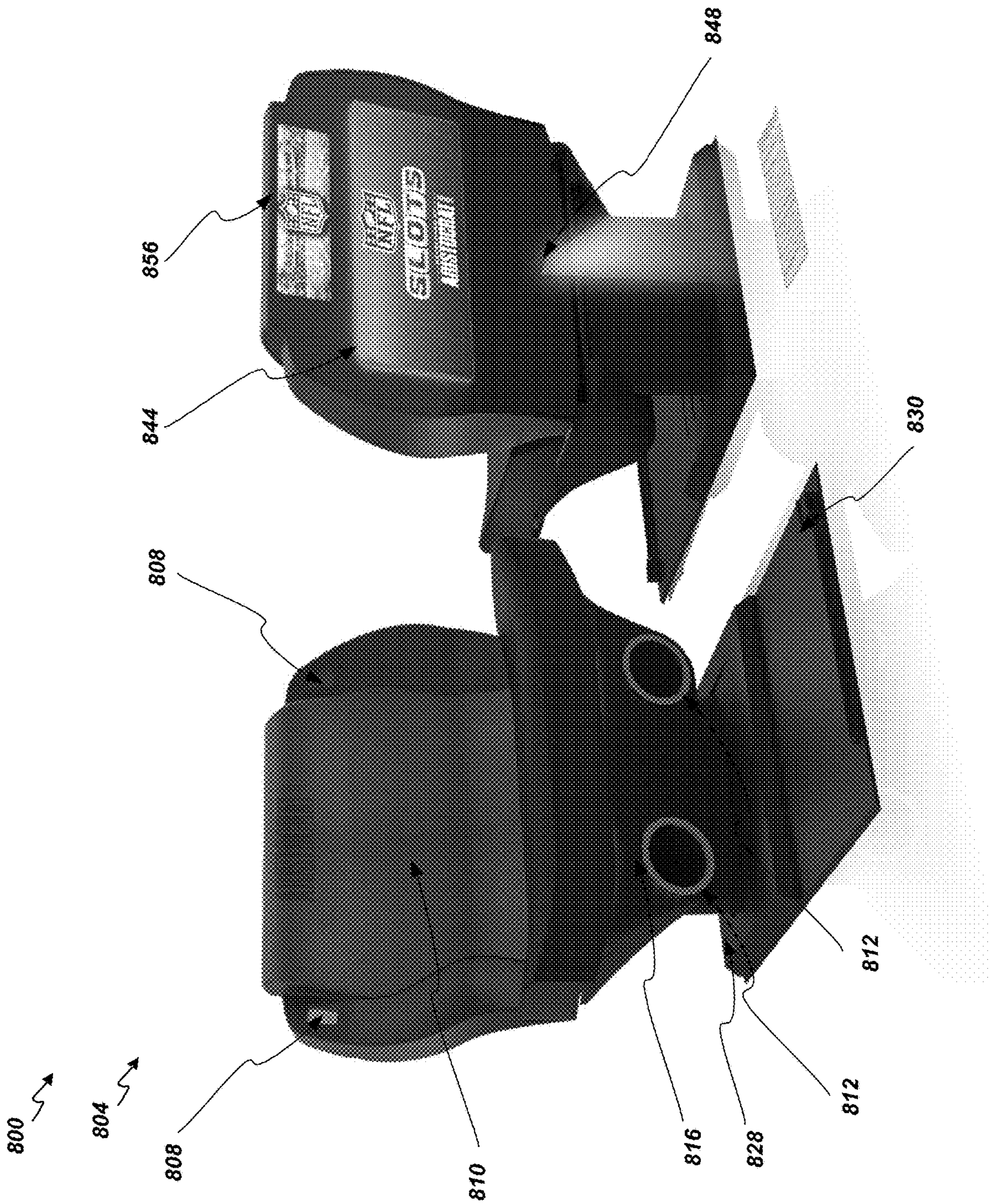

FIG. 8A

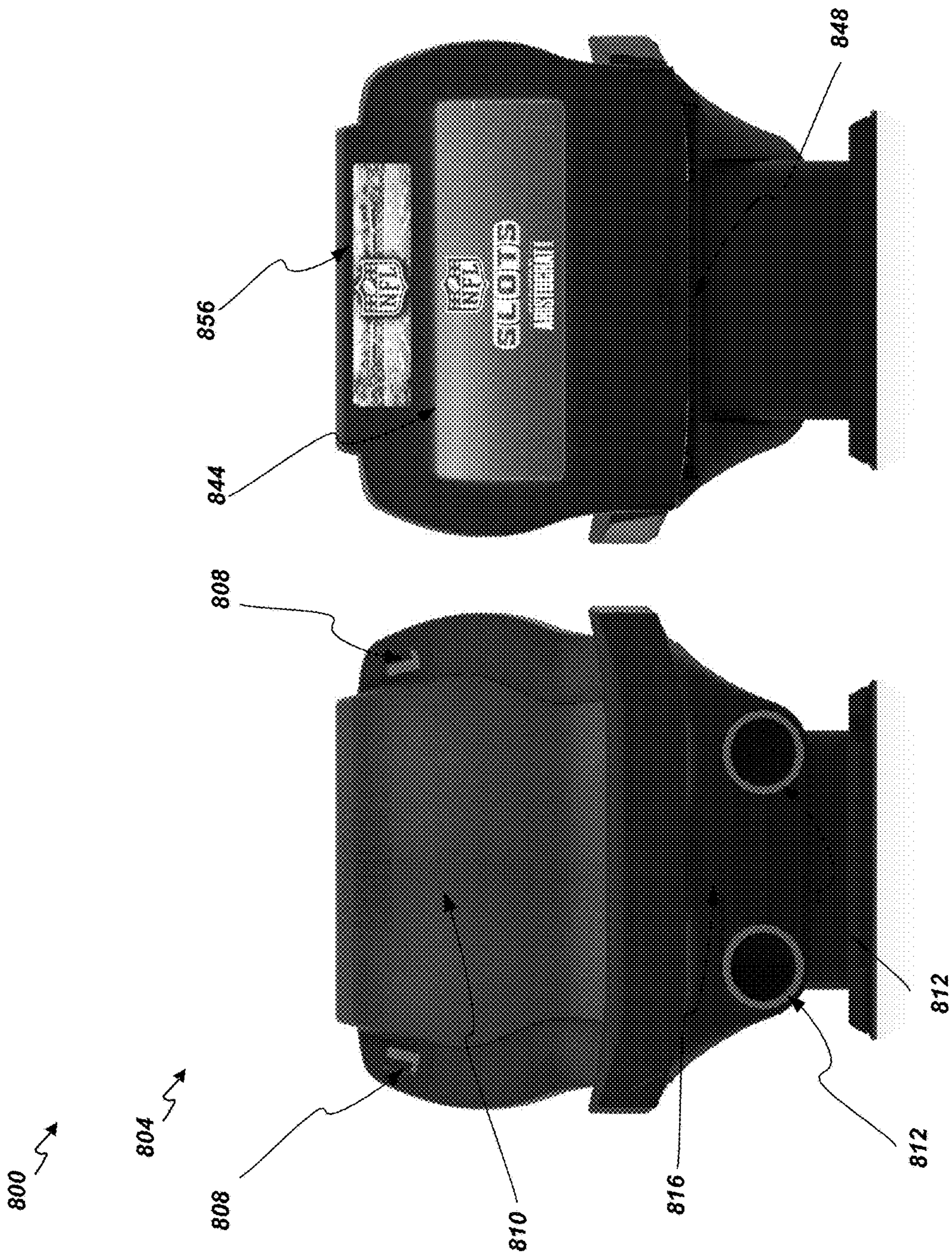

FIG. 8B

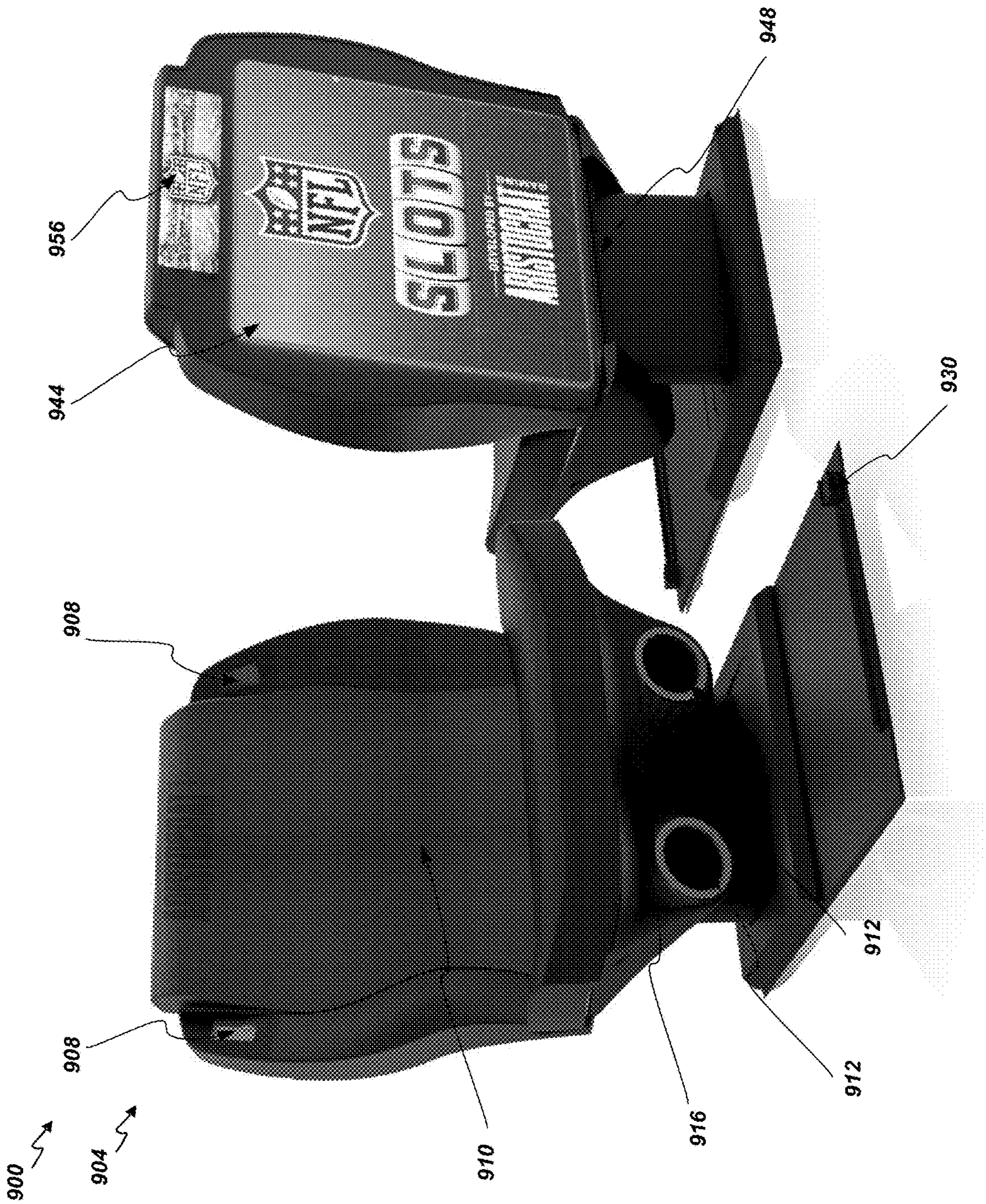

FIG. 9A

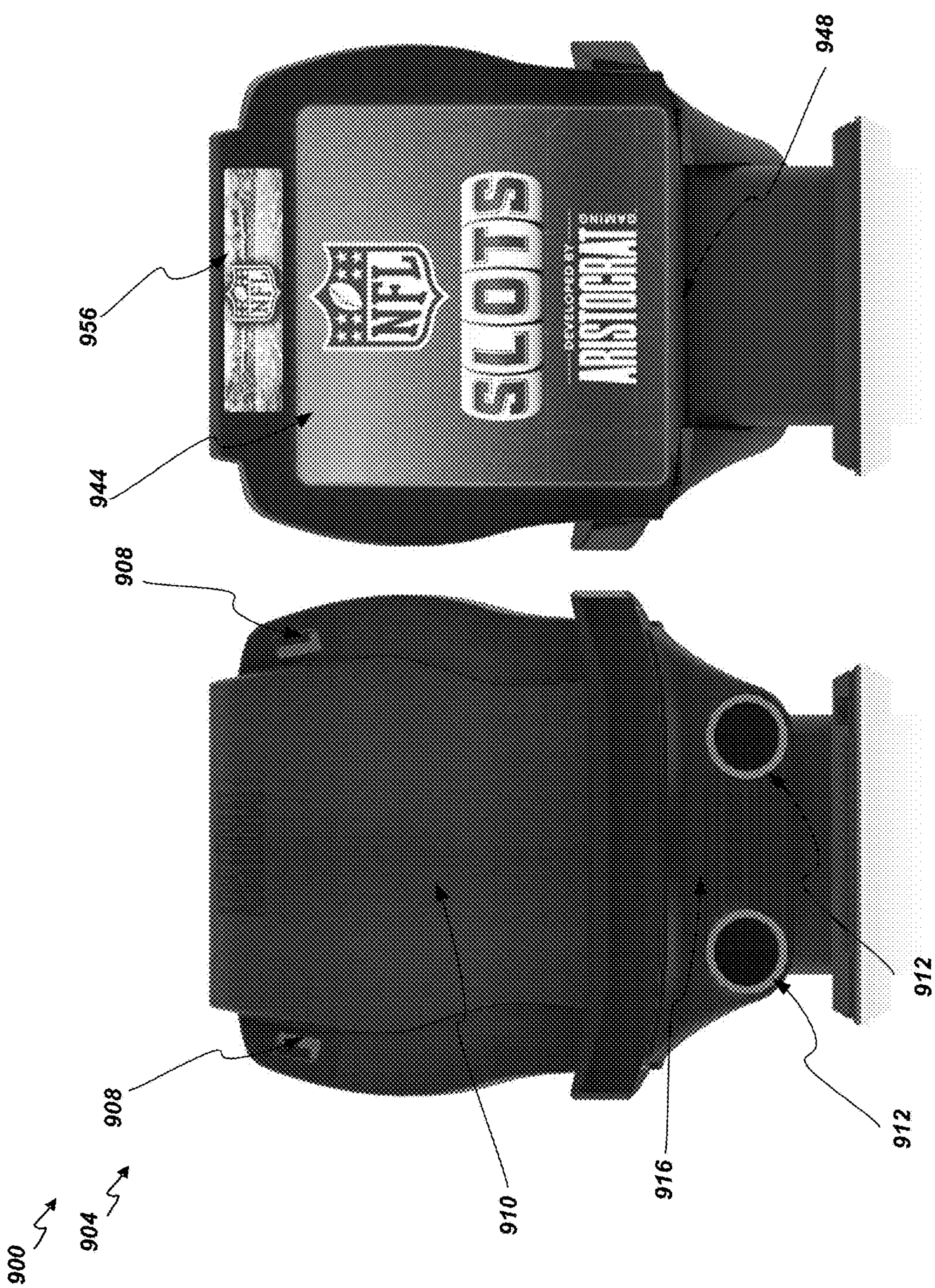

FIG. 9B

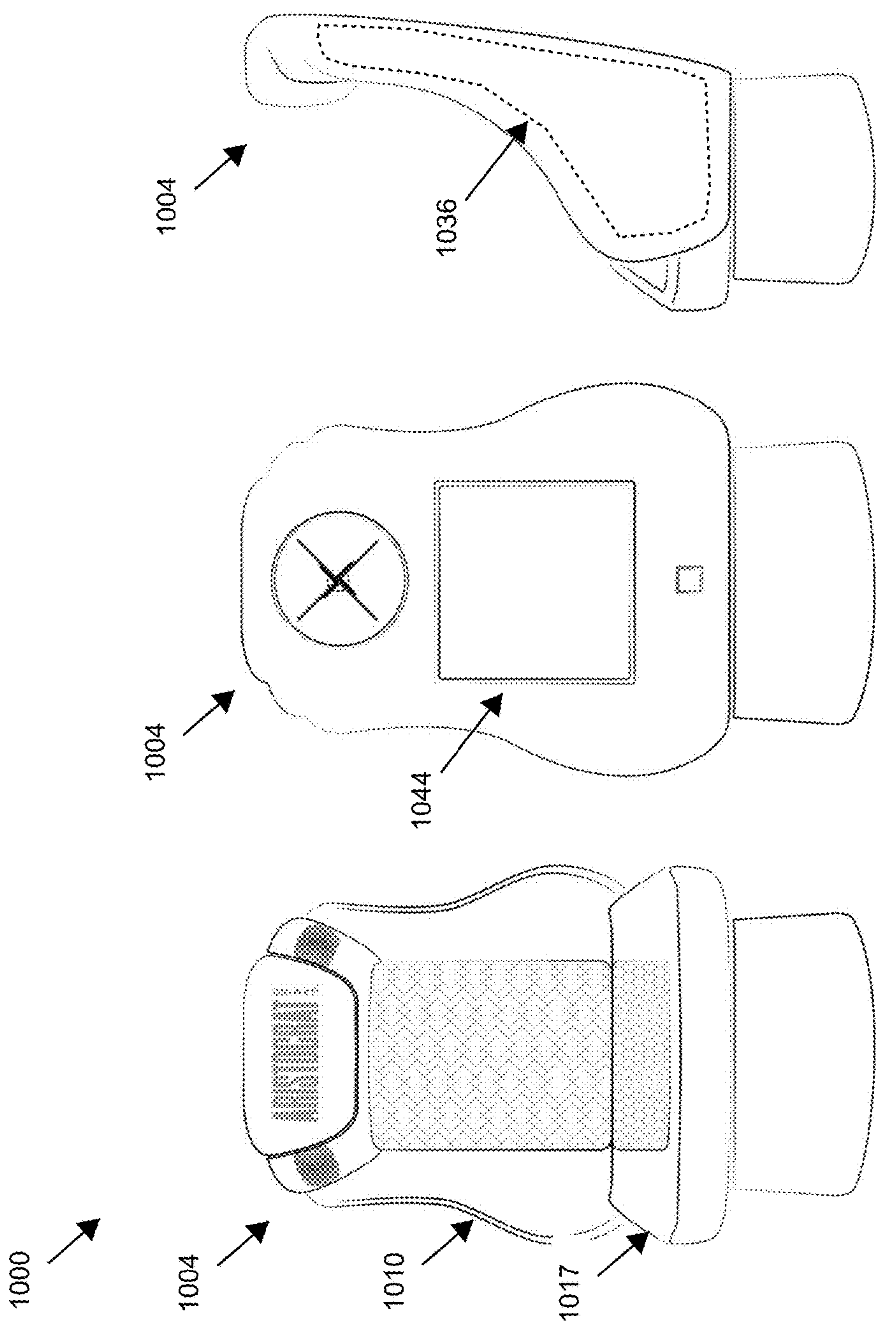

FIG. 10

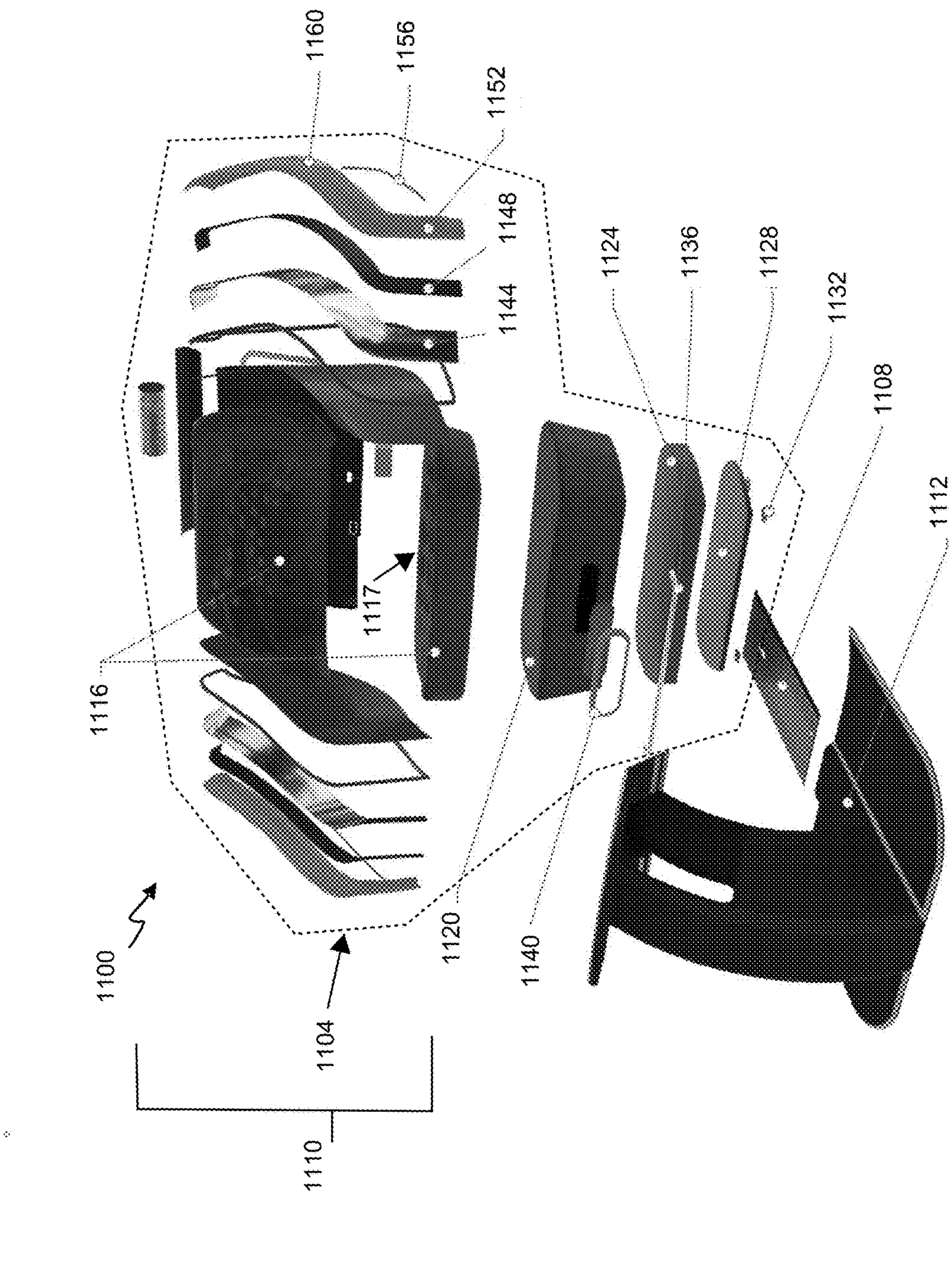
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FIG. 11A

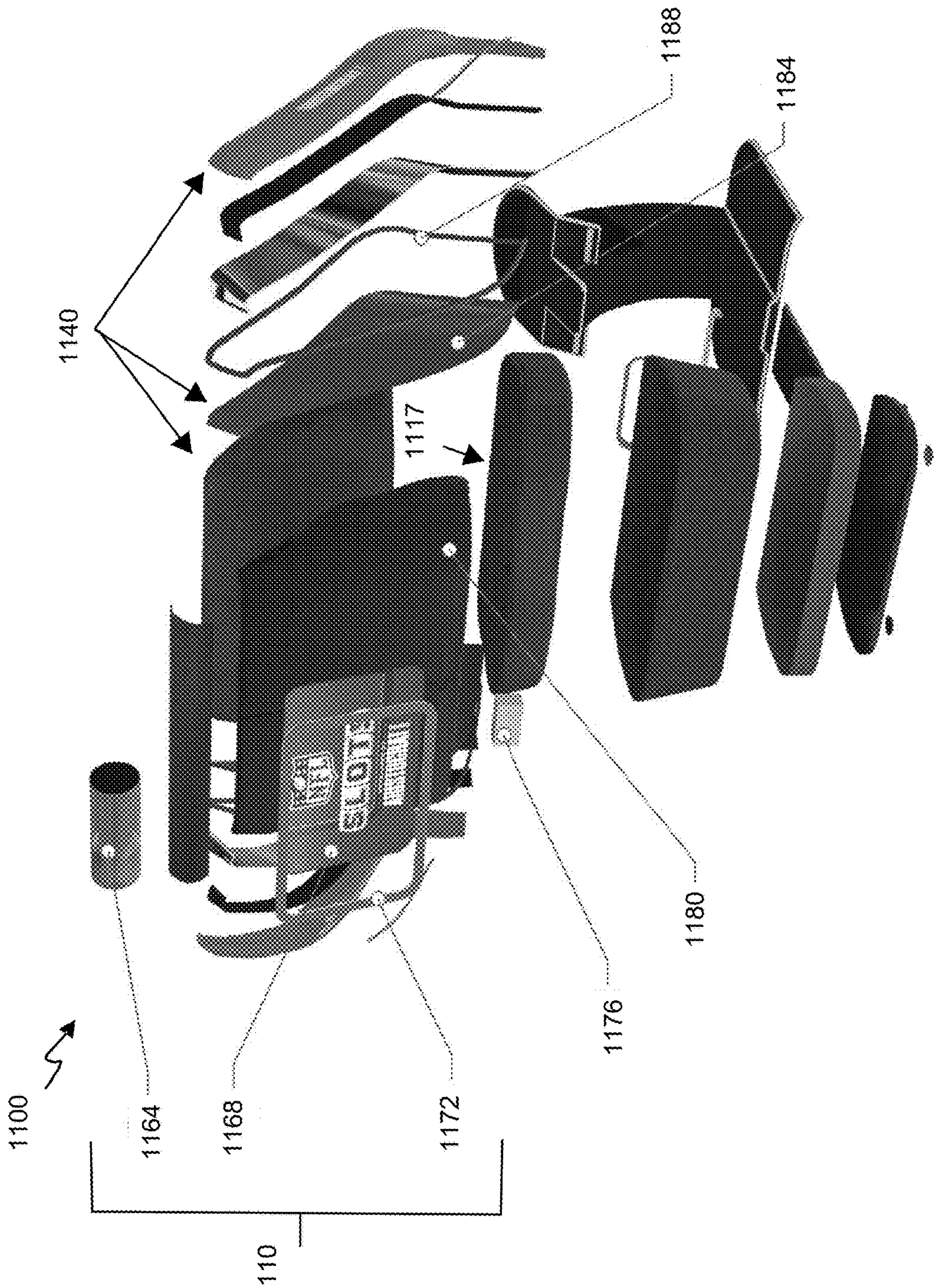

FIG. 11B

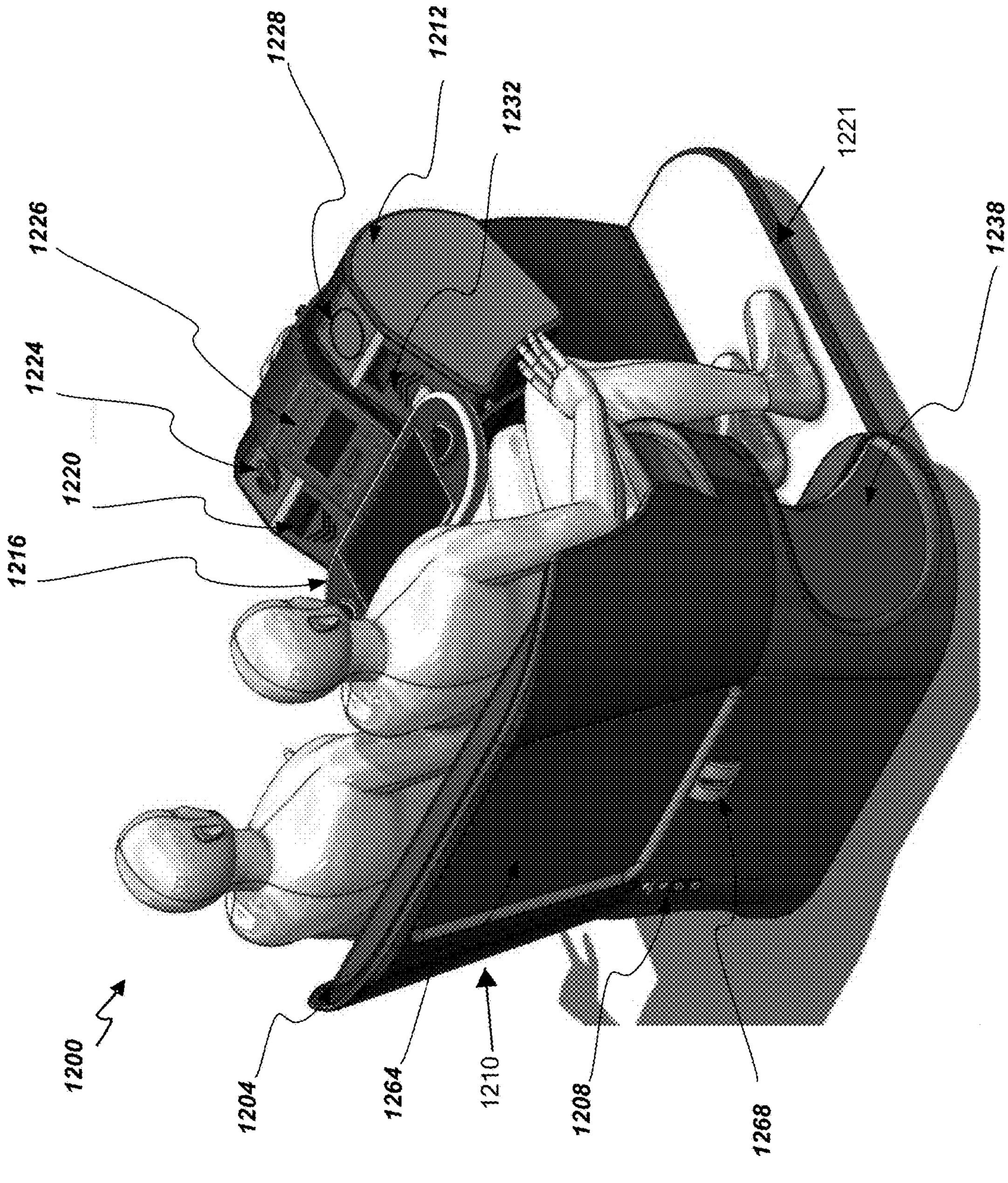

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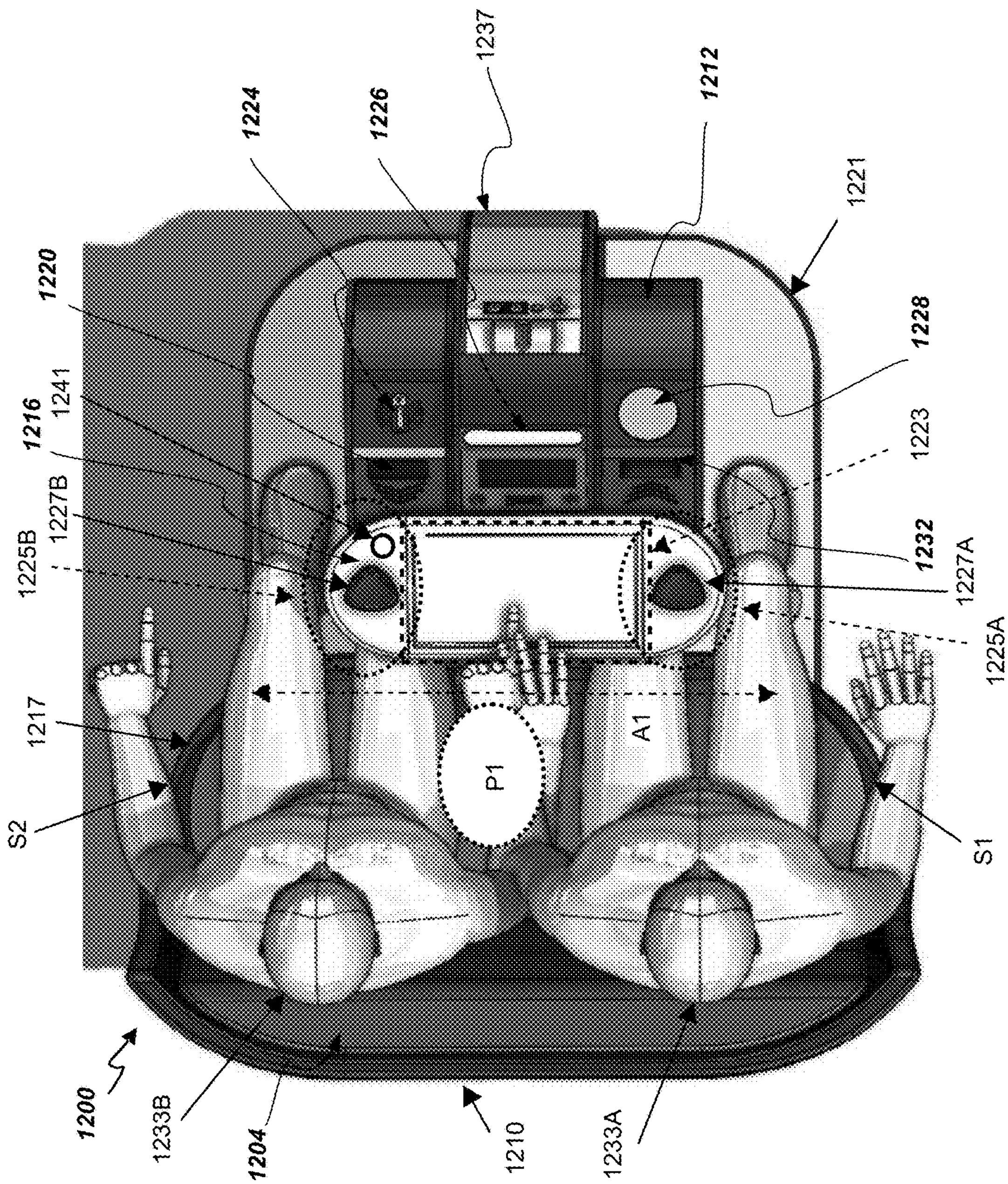
FIG. 12A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1216
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1227B
1225B
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S2
P1
A1
S1
1200
1233B
1204
1210
1233A

FIG. 12B

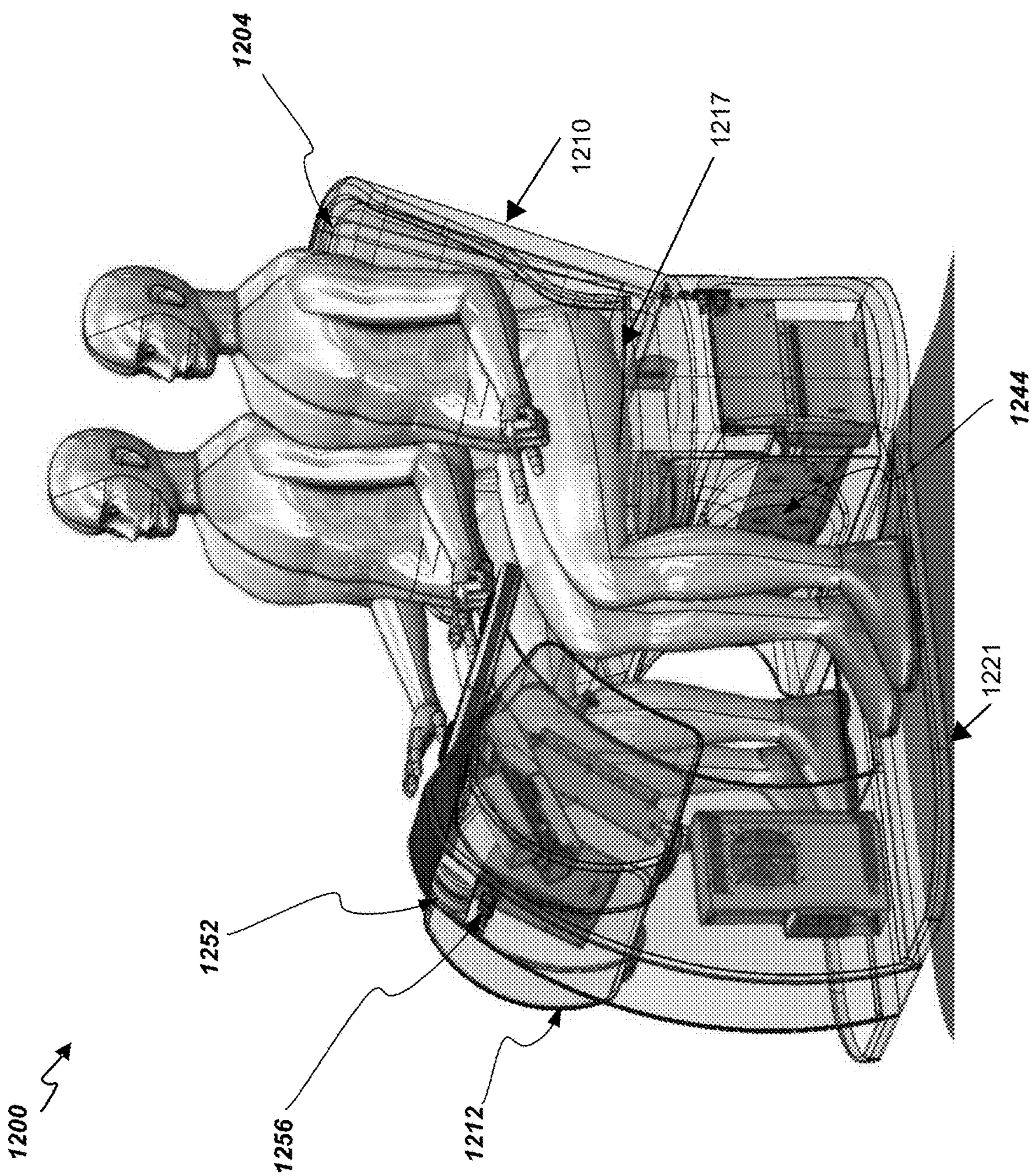

FIG. 12C

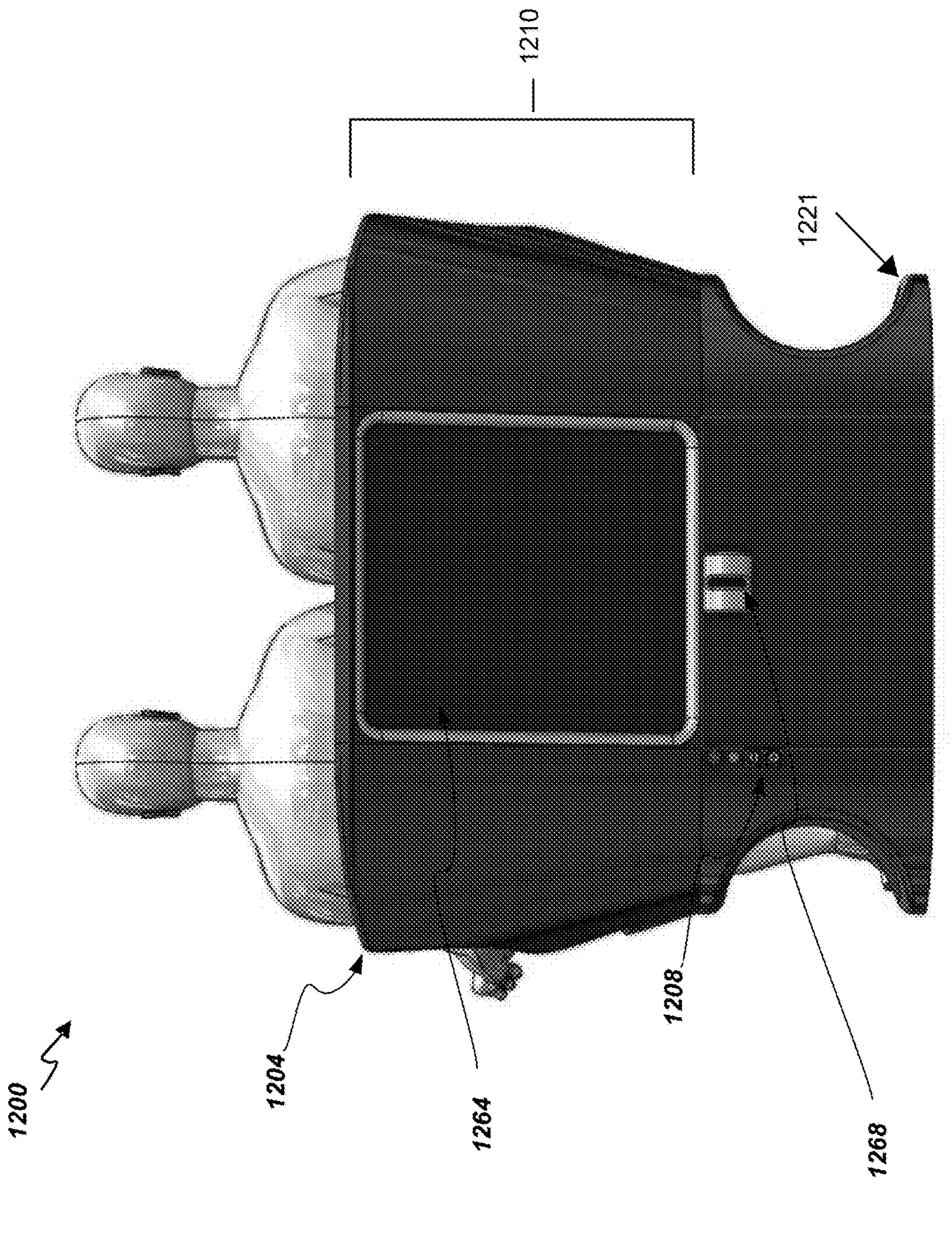

FIG. 12D

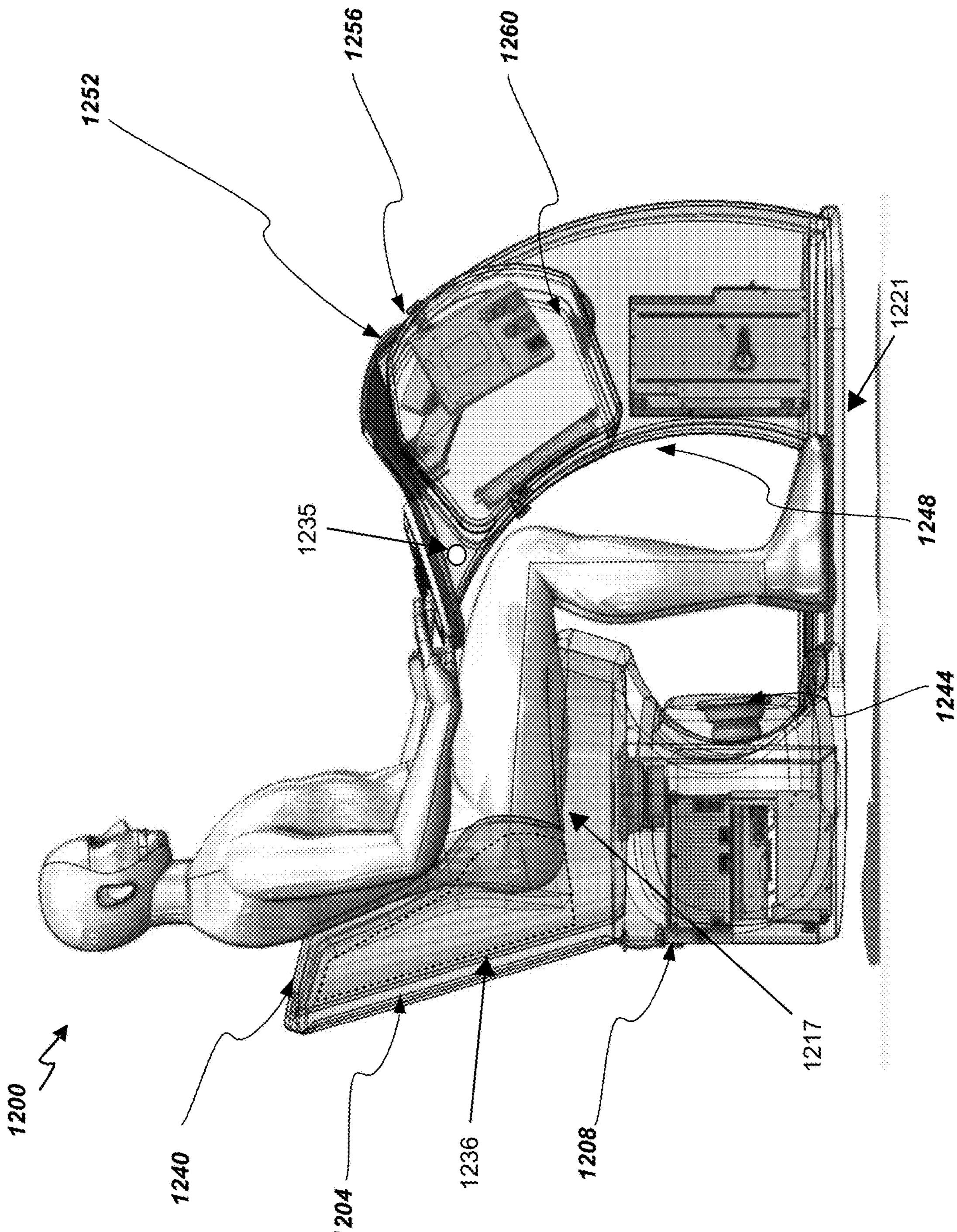
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FIG. 12E

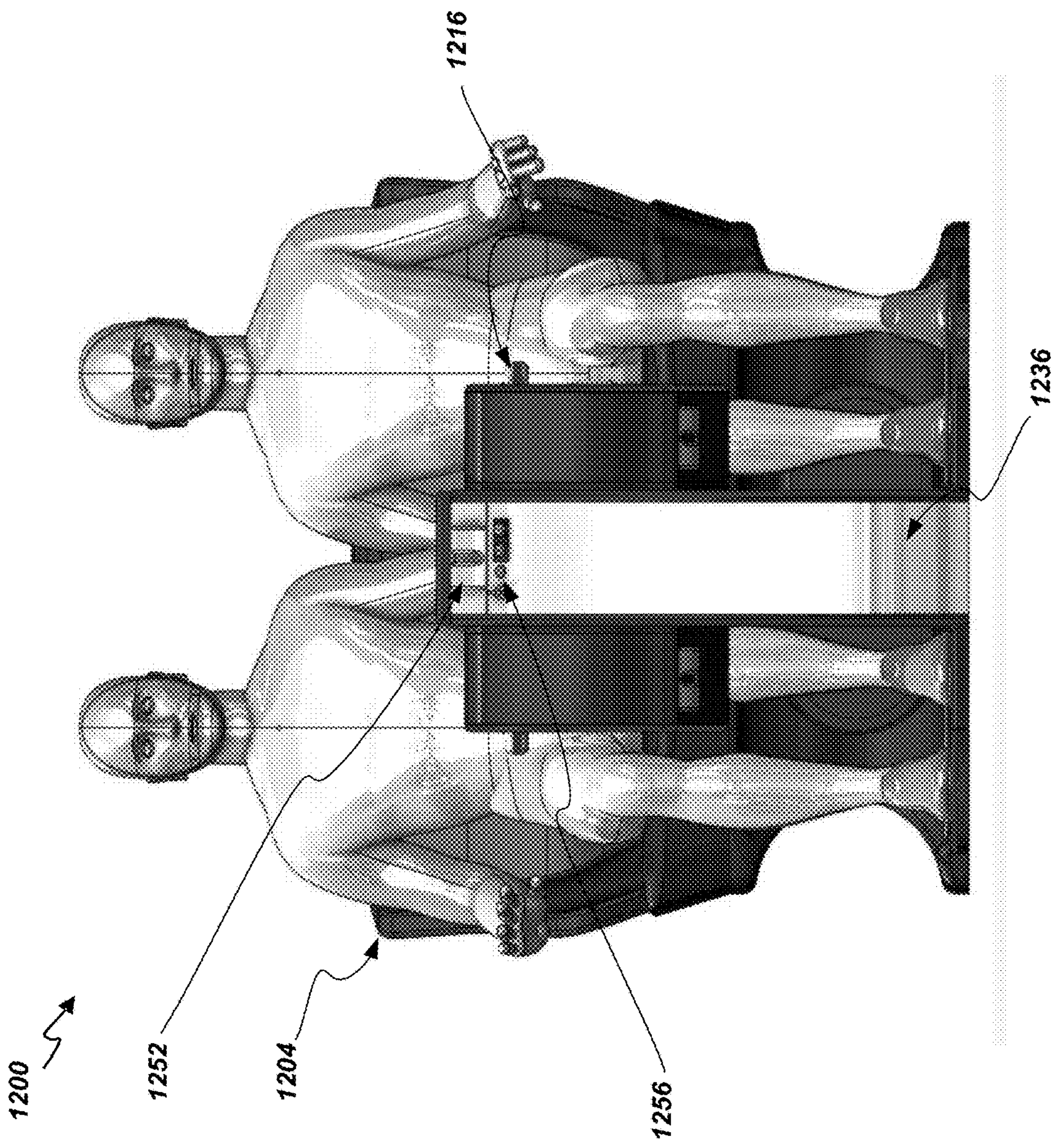


FIG. 12F

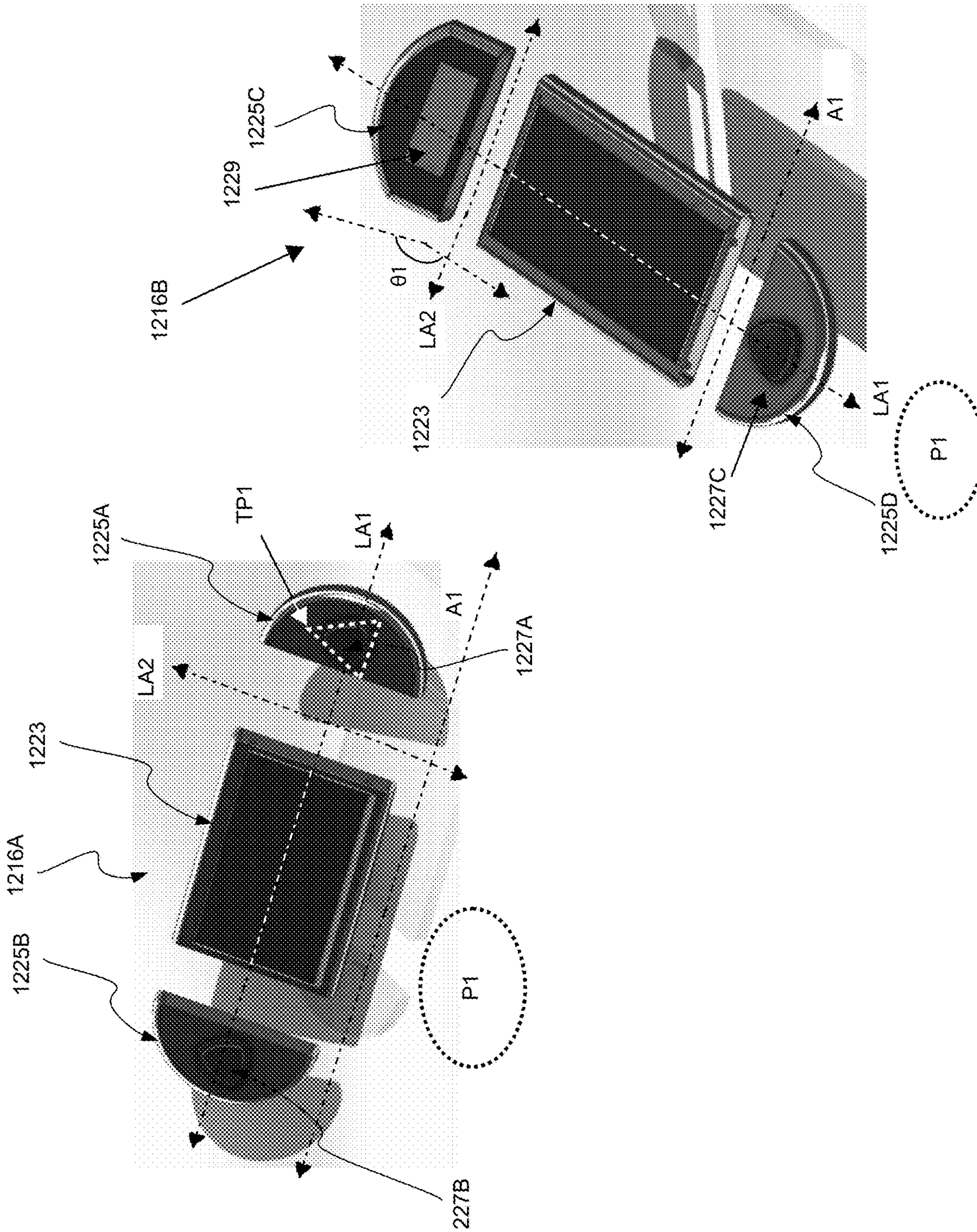
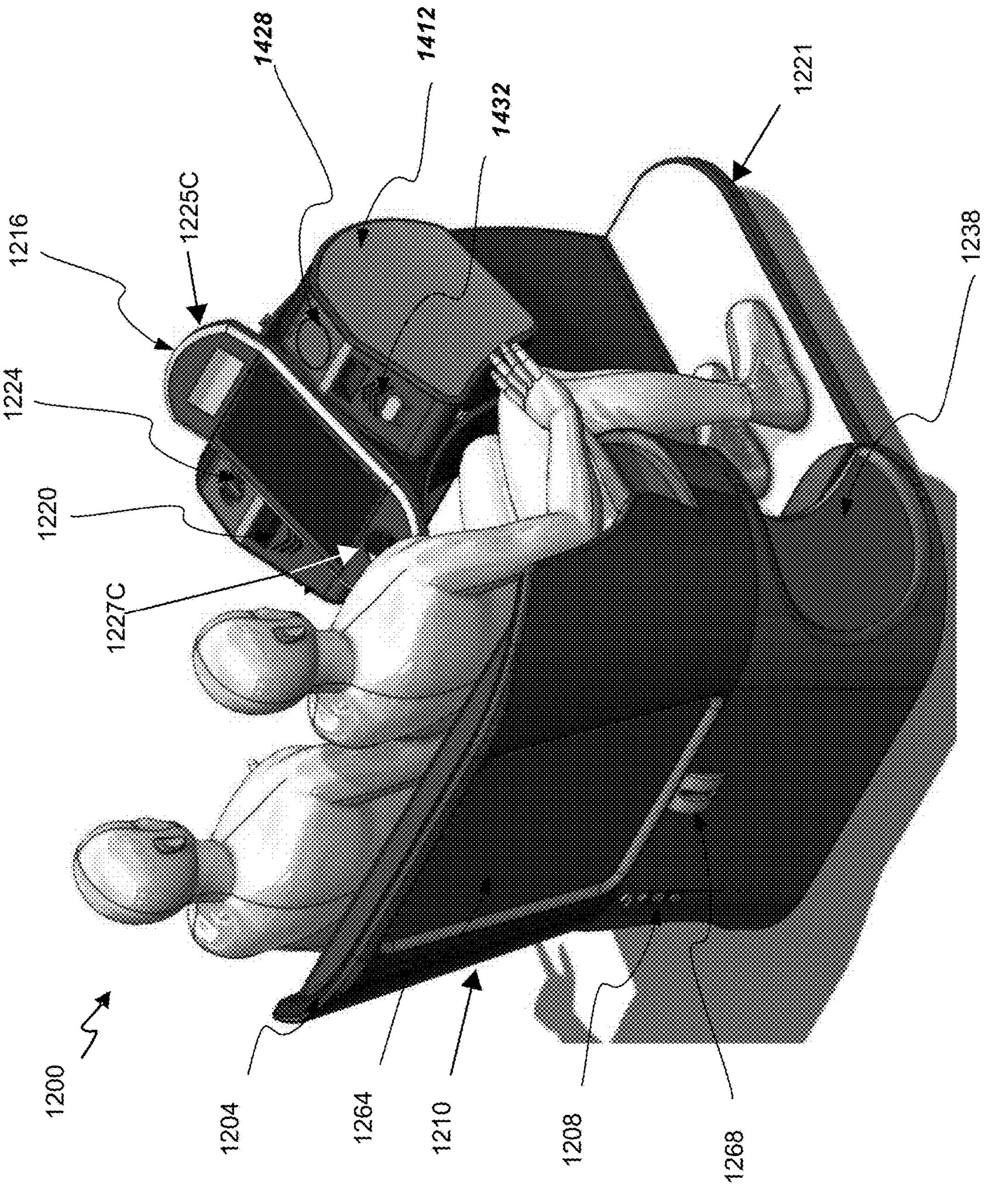

1216B
1225C
1229
θ1
LA2
1223
A1
LA1
P1
1227C
1225D
1225A
TP1
LA1
A1
1227A
LA2
1223
1216A
1225B
P1
1227B

FIG. 13



1200
1204
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1216
1225C
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1412
1432
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FIG. 14

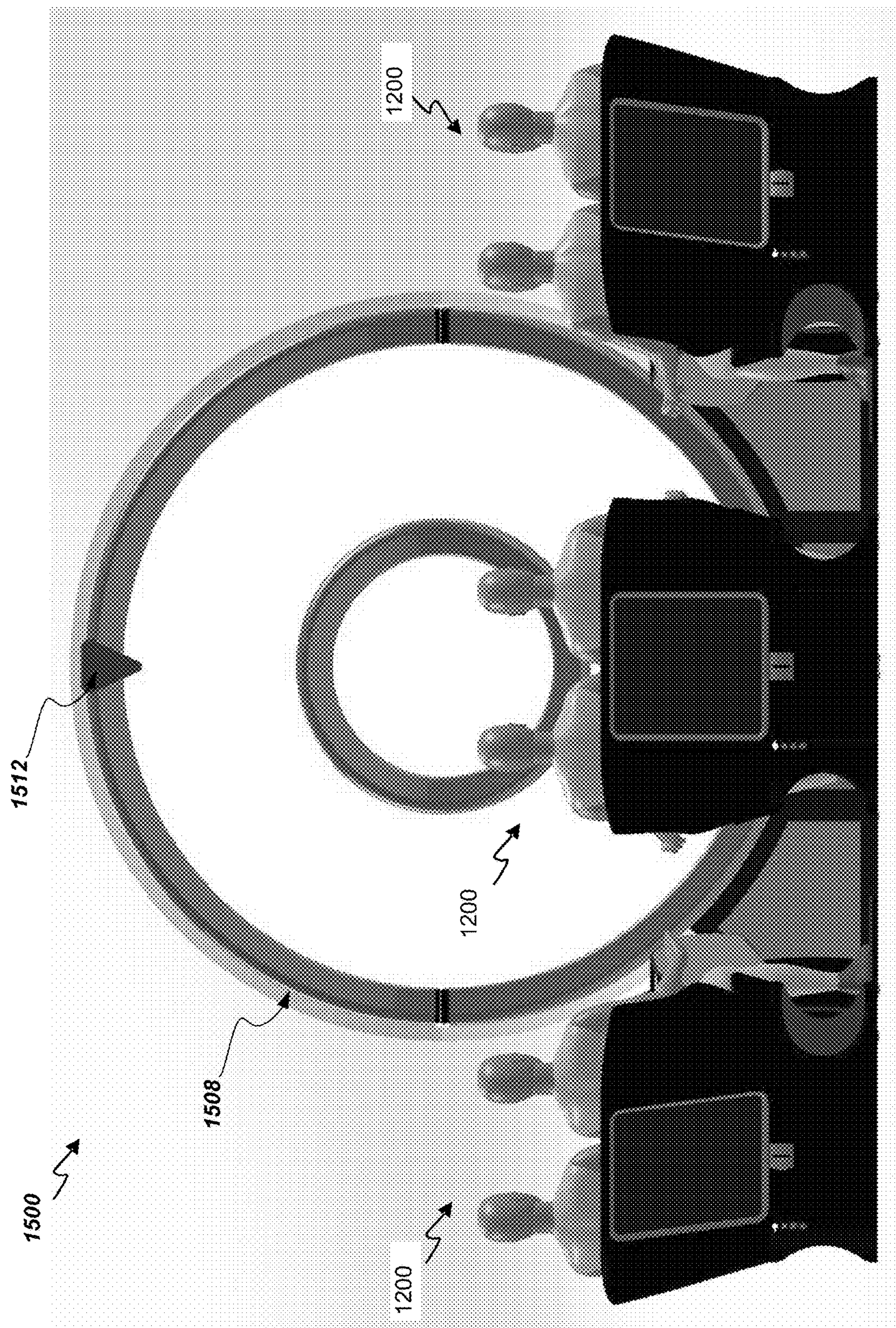

FIG. 15A

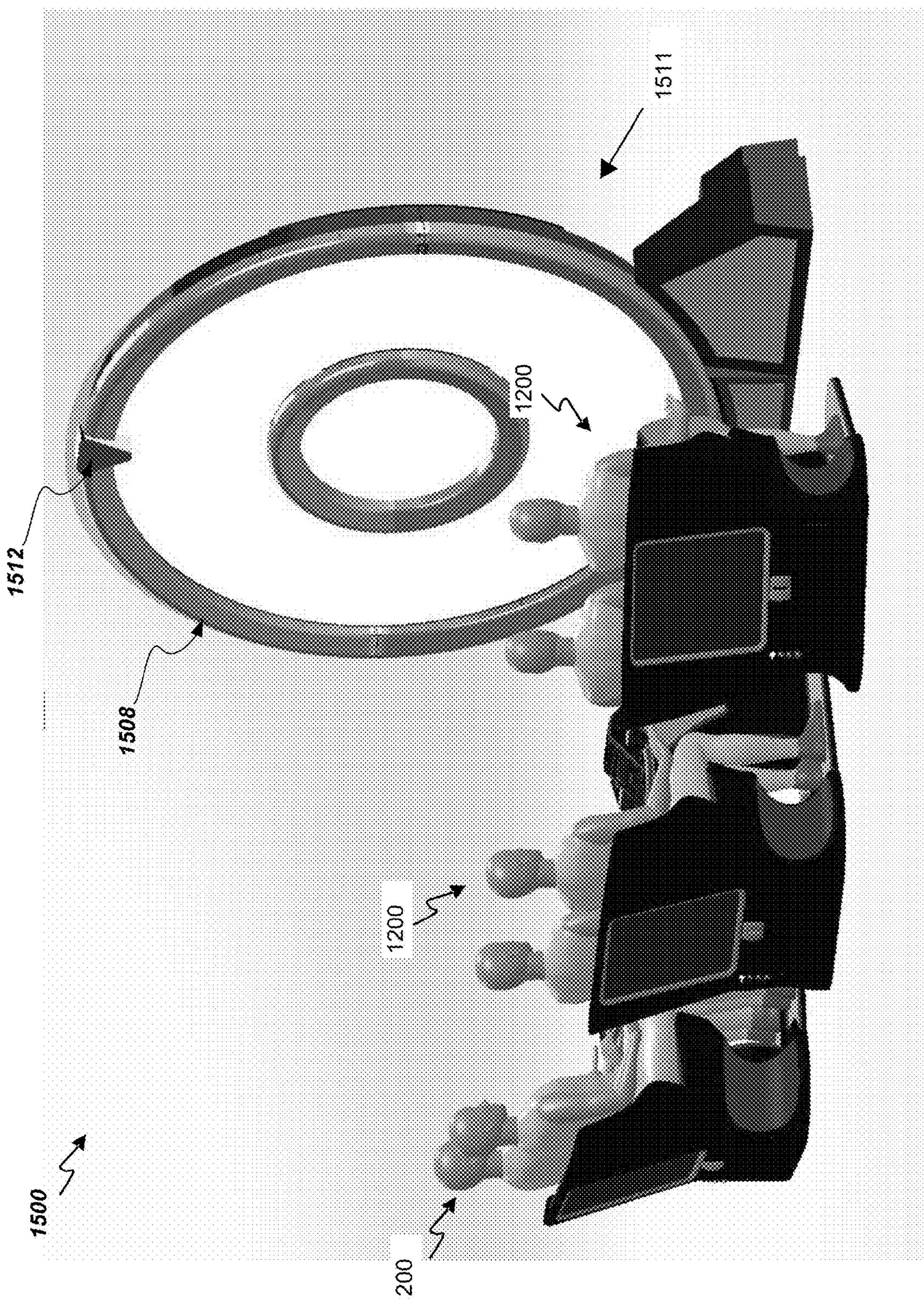

FIG. 15B

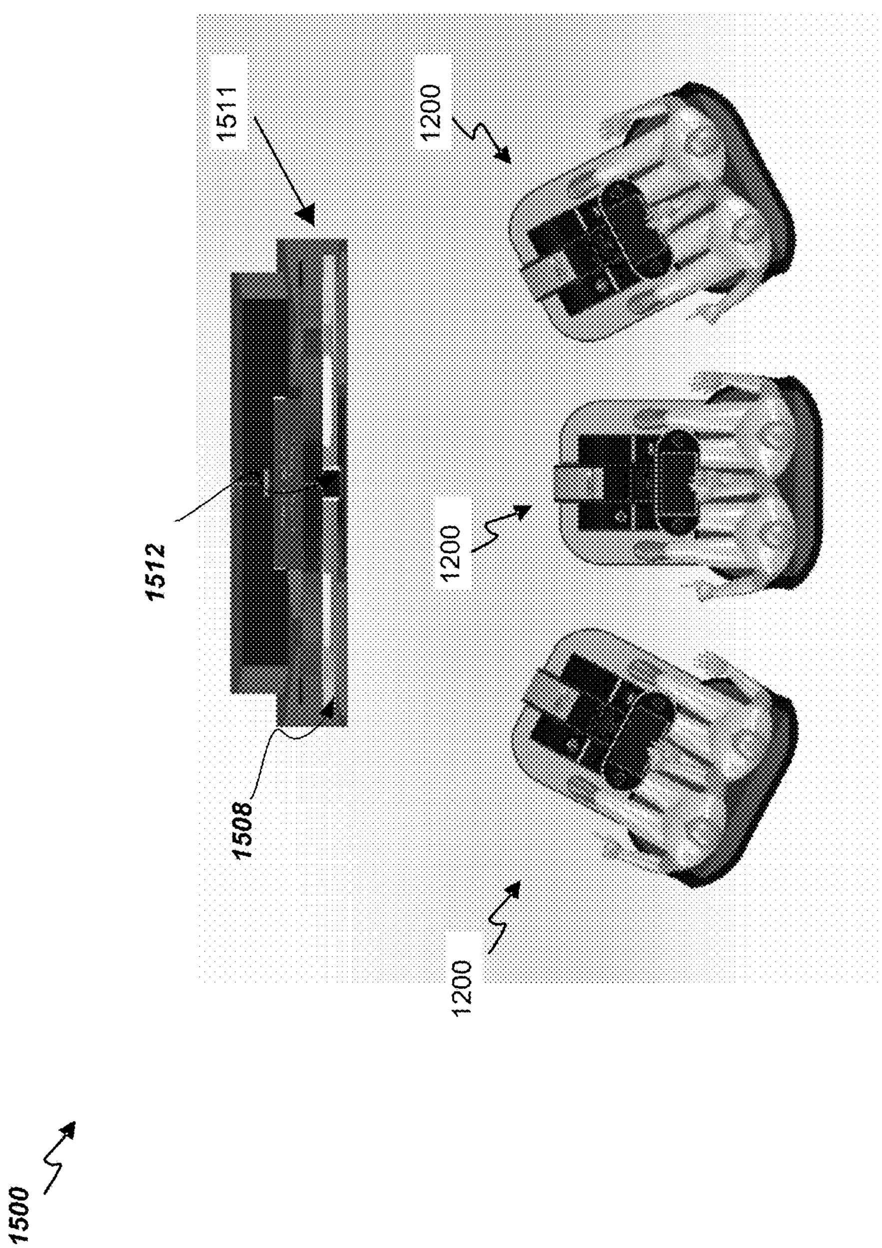

FIG. 15C

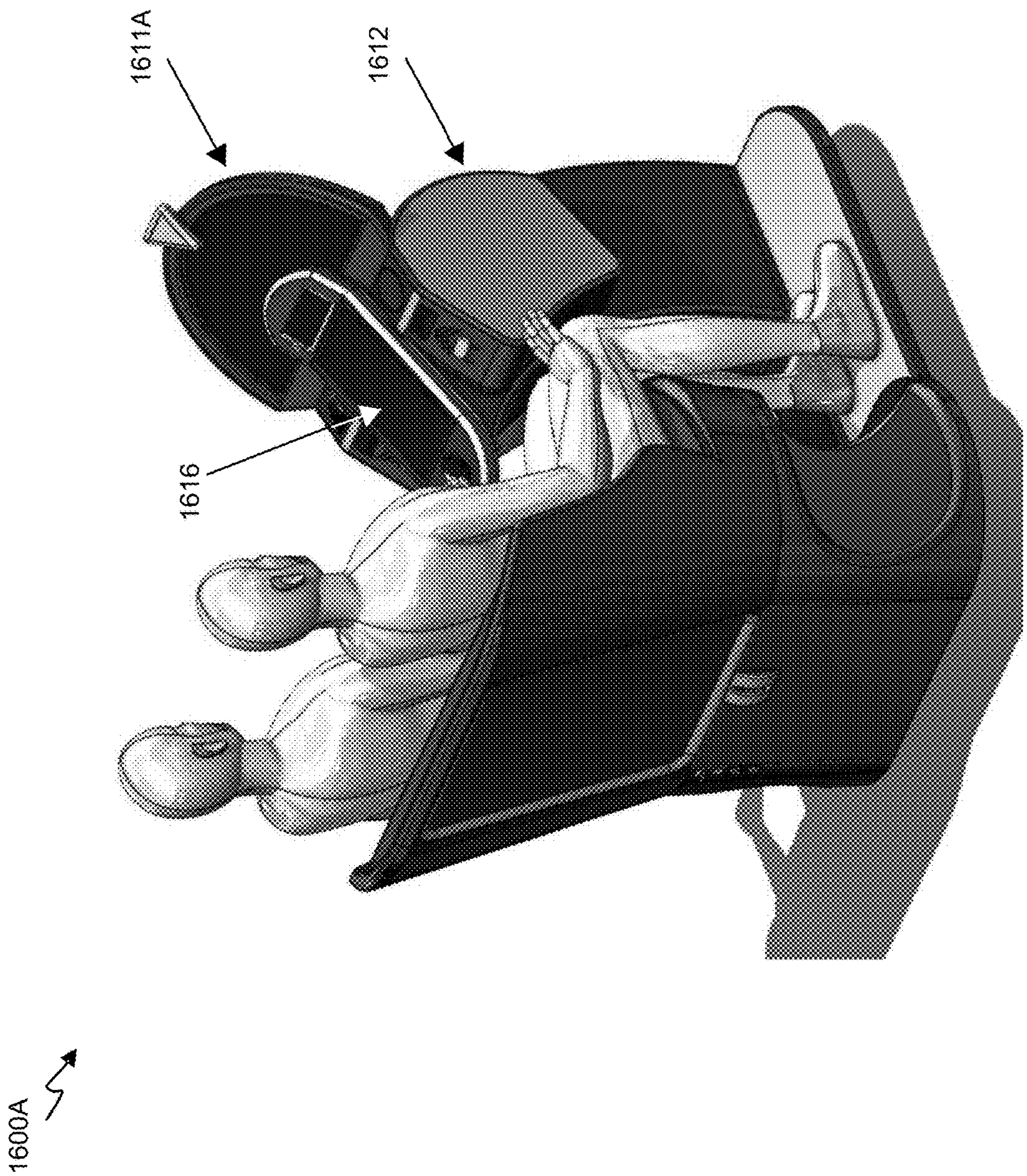

FIG. 16A

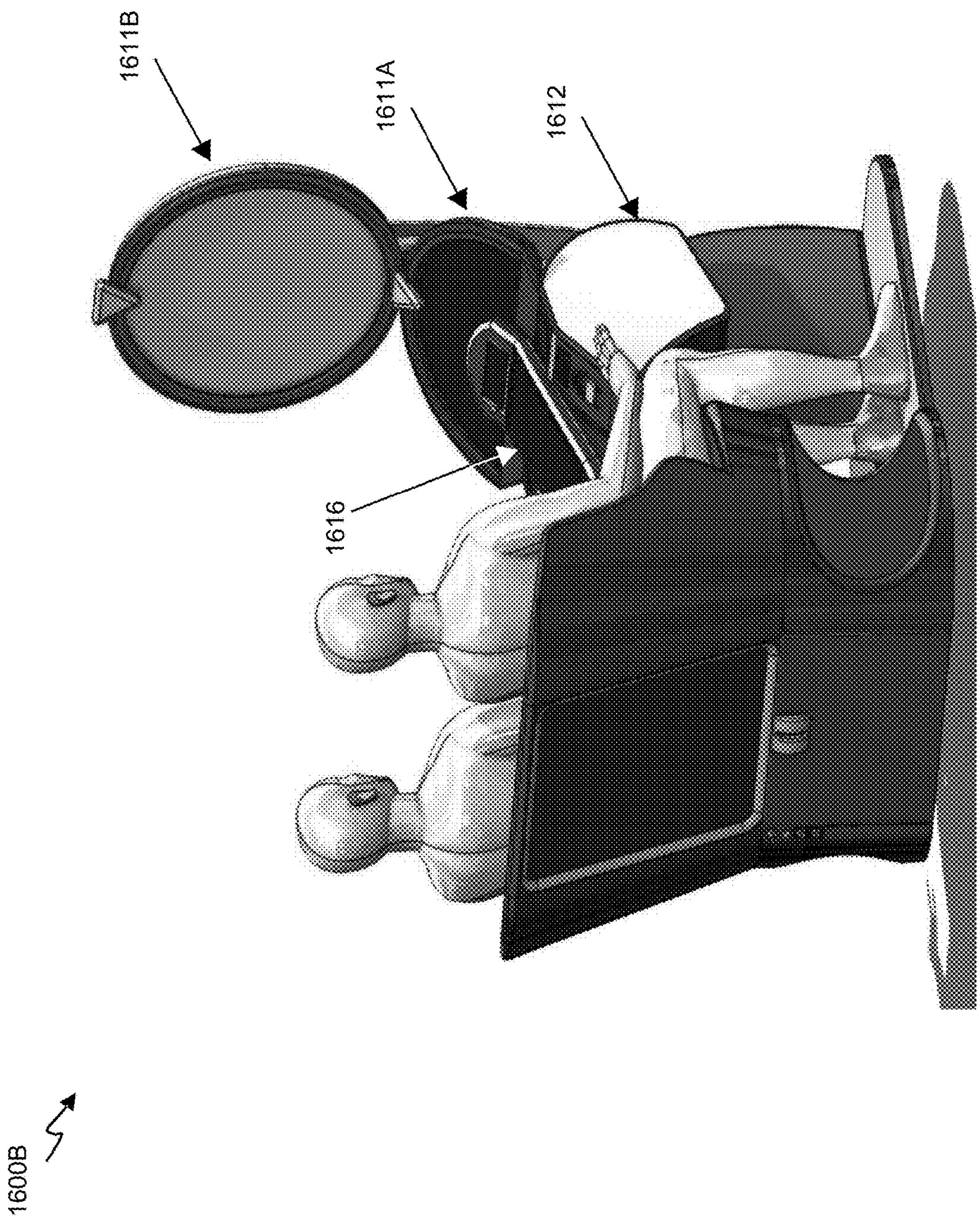

FIG. 16B

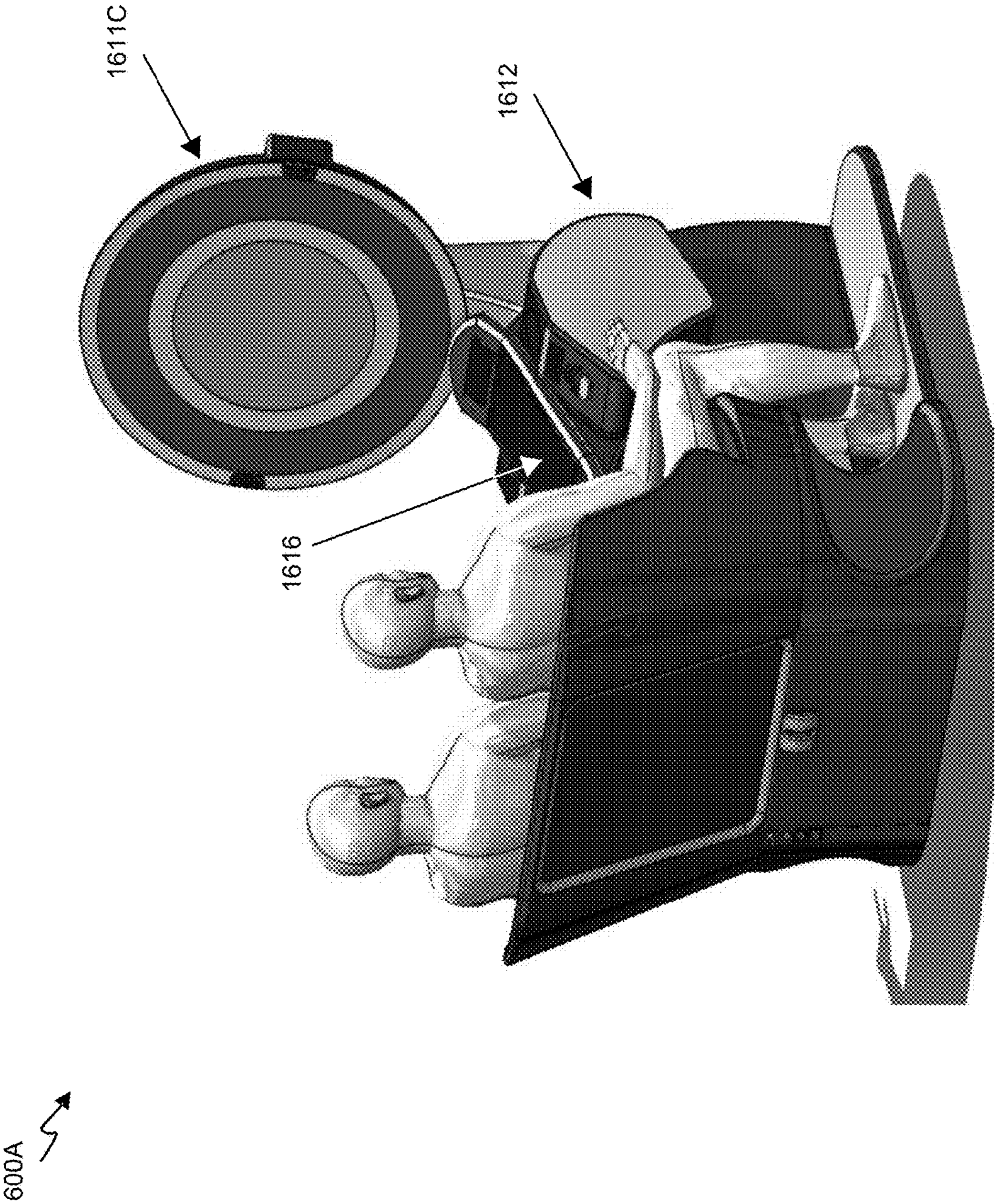

FIG. 16C

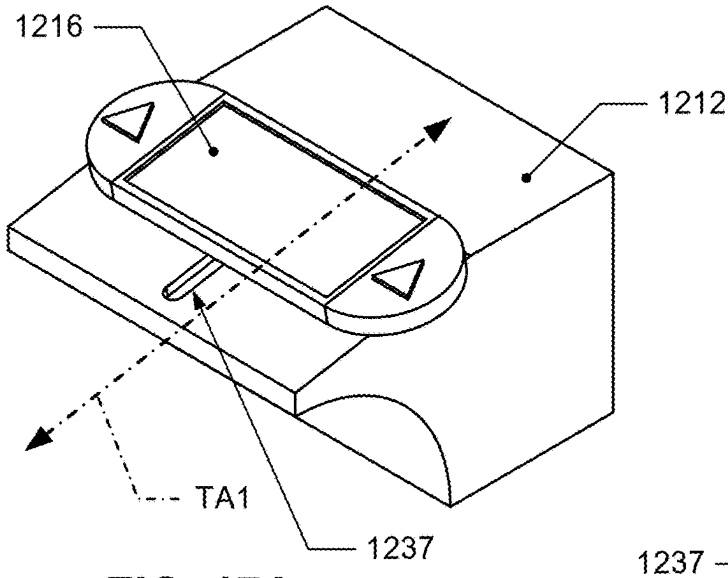

FIG. 17A

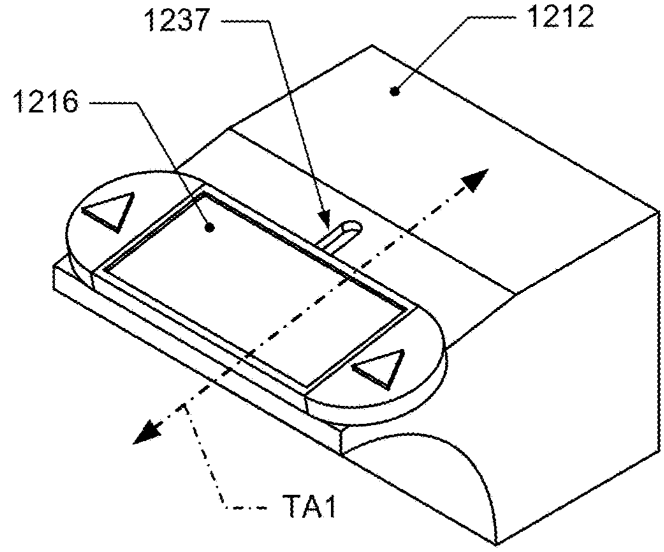

FIG. 17B

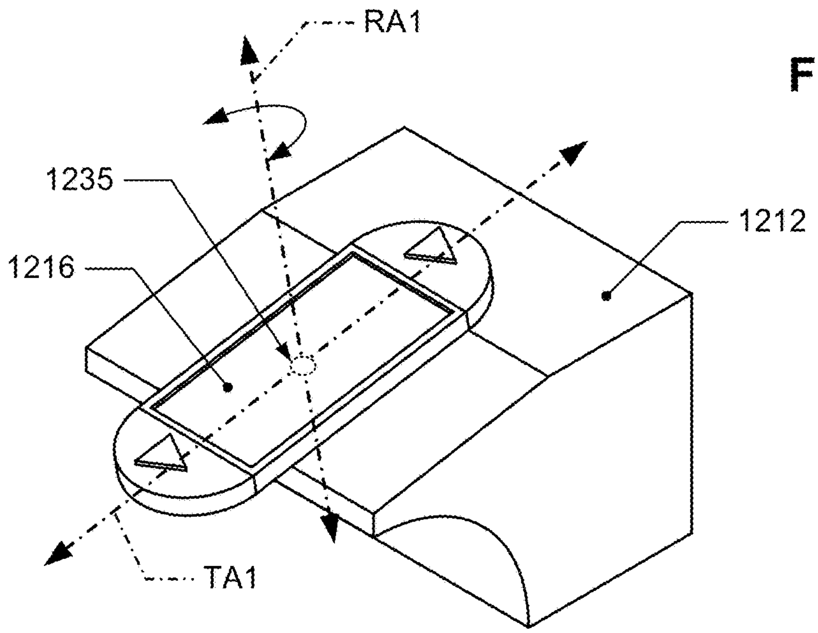

FIG. 17C

INTEGRATED MODULAR SEATING SYSTEMS FOR ELECTRONIC GAMING SYSTEMS AND METHODS

RELATED APPLICATIONS

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/557,272, filed Feb. 23, 2024, and titled "INTEGRATED MODULAR SEATING ASSEMBLY FOR ELECTRONIC GAMING SYSTEMS AND METHODS," which is hereby incorporated by reference in its entirety and for all purposes.

BACKGROUND

[0002] Electronic gaming machines ("EGMs") or gaming devices provide a variety of wagering games such as slot games, video poker games, video blackjack games, roulette games, video bingo games, keno games and other types of games that are frequently offered at casinos and other locations. Play on EGMs typically involves a player establishing a credit balance by inputting money, or another form of monetary credit, and placing a monetary wager (from the credit balance) on one or more outcomes of an instance (or single play) of a primary or base game. In some cases, a player may qualify for a special mode of the base game, a secondary game, or a bonus round of the base game by attaining a certain winning combination or triggering event in, or related to, the base game, or after the player is randomly awarded the special mode, secondary game, or bonus round. In the special mode, secondary game, or bonus round, the player is given an opportunity to win extra game credits, game tokens or other forms of payout. In the case of "game credits" that are awarded during play, the game credits are typically added to a credit meter total on the EGM and can be provided to the player upon completion of a gaming session or when the player wants to "cash out."

[0003] "Slot" type games are often displayed to the player in the form of various symbols arrayed in a row-by-column grid or matrix. Specific matching combinations of symbols along predetermined paths (or paylines) through the matrix indicate the outcome of the game. The display typically highlights winning combinations/outcomes for identification by the player. Matching combinations and their corresponding awards are usually shown in a "pay-table" which is available to the player for reference. Often, the player may vary his/her wager to include differing numbers of paylines and/or the amount bet on each line. By varying the wager, the player may sometimes alter the frequency or number of winning combinations, frequency, or number of secondary games, and/or the amount awarded.

[0004] Typical games use a random number generator (RNG) to randomly determine the outcome of each game. The game is designed to return a certain percentage of the amount wagered back to the player over the course of many plays or instances of the game, which is generally referred to as return to player (RTP). The RTP and randomness of the RNG ensure the fairness of the games and are highly regulated. Upon initiation of play, the RNG randomly determines a game outcome and symbols are then selected which correspond to that outcome. Notably, some games may include an element of skill on the part of the player and are therefore not entirely random.

[0005] Typical gaming systems employ user interfaces, monitors, and/or display devices installed on a cabinet to display gaming information and/or games to a player. Some of these gaming systems may include other apparatuses such as chairs for accommodating the player to interact or interface with the gaming systems. These other apparatuses are generally not designed, equipped, furnished, or personalized for the player, the games being offered, or the player comfort in mind, let alone matching any specific game characteristics.

[0006] As such, players may be unable to fully experience any or all impacts that gaming systems may be able to offer with these ordinary accommodation apparatuses or gaming systems. Further, when an electronic gaming system has related furnishings, e.g., chairs, installed or fixed on a casino floor, i.e., in the field, when changing any game offerings from a first game to a new game a gaming manufacturer may have to entirely remove the electronic gaming system including the furnishings from the floor to remove artwork or artworks related to the first game, replace the artwork for the new game, and return and re-install the electronic gaming system on the floor while complying with any security protocols for secure gaming.

[0007] The removal, replacement, and re-installation of seating for electronic gaming system involves processes that may be time consuming and be inflexible in varying electronic gaming system capabilities, and, in turn, cause a casino operator to incur cost from undergoing the processes and from the loss of game operation on the casino floor. Manual artwork replacement on seating may also present bottlenecks on removal, replacement, and re-installation when it requires the electronic gaming systems to be removed from the casino floor.

[0008] As such, conventional seating for gaming machines remains problematic. A modular seating assembly system for electronic gaming systems that offers a more complete game system immersion, greater flexibility in audio and display configurations, game play ergonomics and control, and less time consuming removal, replacement and re-installation when an electronic gaming system module is to be replaced, will be a welcome innovation for gaming establishments, e.g., casinos, that employ such gaming technology and the like.

SUMMARY

[0009] Details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. The following, non-limiting implementations are considered part of the disclosure; other implementations will be evident from the entirety of this disclosure and the accompanying drawings as well.

[0010] In some implementations, a seating system for electronic gaming on an electronic gaming machine may be provided. The seating system may have a seat with a seating surface and a backrest, a kiosk directly or indirectly coupled to the seat, and an operating console rotatably coupled to the kiosk and having a button deck, a first end piece detachably coupled to a first end of the operating console and having a first button, and a second end piece detachably coupled to a second end of the operating console, opposite the first end, and having a second button or a display. The first end piece and the second end piece may be configured to be detached from the operating console, the operating console may be

configured to receive a third end piece at the first end or the second end when the first end piece or the second end piece, respectively, has been detached, the operating console may be configured to be rotated between a first orientation and a second orientation 90 degrees apart from the first orientation, and the button deck, the first end piece, and the second end piece may be configured to be communicatively coupled to a game controller of the electronic gaming machine.

[0011] In some implementations, the button deck, the first end piece, the second end piece, or a combination thereof may be configured to receive input from a user, and provide information related to the input from the user to the game controller of the electronic gaming machine.

[0012] In some implementations, the operating console may be further configured to translate along a translation axis.

[0013] In some such implementations, the translation axis may be perpendicular to the axis of rotation.

[0014] In some such implementations, the operating console may be configured to translate along the translation axis without the use of tools.

[0015] In some such implementations, the translation axis may have a first end and a second end, the second end may be closer to the seat than the first end, and the operating console may be closer to the seat when positioned at the second end than when positioned at the first end.

[0016] In some implementations, the second end piece may be oriented at a non-parallel angle with respect to the button deck.

[0017] In some implementations, the first button may be a physical button.

[0018] In some implementations, the second end piece may have a display that is configured to display player related information.

[0019] In some implementations, the system may further have a display of the electronic gaming machine coupled to the kiosk and/or the seat, and the game controller that is configured to cause a game instance to be displayed on the display.

[0020] In some implementations, the electronic gaming machine may be without any physical coupling to the seat or the kiosk.

[0021] In some implementations, the seat may further have at least one display panel configured to display game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

[0022] In some such implementations, the system may further have a controller communicatively coupled with the game controller. The controller may be configured to display, based on an outcome of a game instance on the electronic gaming machine, game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

[0023] In some further such implementations, the game information, attractions, a portion of a game instance, images, videos, or a combination thereof displayed based on the outcome of the game instance may be related to a game theme of the game instance.

[0024] In some such implementations, the game information may include one or more wager amounts of the game instance.

[0025] In some such implementations, the at least one display panel may have one or more curved flexible display panel, the seat may further include at least one side panel adjacent to the seating surface and the backrest, and the curved flexible display panel may be positioned on the side panel.

[0026] In some further such implementations, the curved flexible display panel may have a flexible liquid crystal display (LCD), a flexible electrophoretic display (EPD), a flexible organic light emitting diode (OLED) display, a rollable TFT-driven OLED display, or an electronic paper display.

[0027] In some further such implementations, the side panel may include an armrest, and the curved flexible display panel may be positioned on the armrest.

[0028] In some implementations, a seating system for electronic gaming on an electronic gaming machine may be provided. The seating system may have a seat with a seating surface, a backrest, and at least one armrest adjacent to the seating surface and the backrest, at least one flexible curved display panel on the at least one armrest, a kiosk directly or indirectly coupled to the seat, and an operating console coupled to the kiosk and having a button deck configured to be communicatively coupled to a game controller of the electronic gaming machine. The at least one flexible curved display panel may be configured to display game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

[0029] In some implementations, the system may further have a controller communicatively coupled with the game controller. The controller may be configured to display, based on an outcome of a game instance on the electronic gaming machine, the game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

[0030] Additional aspects will be set forth in the detailed description which follows, and, in part, will be apparent from the disclosure, or may be learned by practice of the disclosed embodiments and/or the claimed subject matter.

[0031] The foregoing general description and the following detailed description are illustrative and explanatory and are intended to provide further explanation of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] FIG. **1** is an exemplary diagram showing several EGMs networked with various gaming-related servers.

[0033] FIG. **2**A is a block diagram showing various functional elements of an exemplary EGM.

[0034] FIG. **2**B depicts a casino gaming environment according to one example.

[0035] FIG. **2**C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure.

[0036] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture algorithm that implements a game processing pipeline for the play of a game in accordance with various implementations described herein.

[0037] FIGS. **4**A-**4**G illustrate different views of a first version of a seating system for electronic gaming.

[0038] FIG. **5** illustrates different views of a second version of a seating system for electronic gaming.

[0039] FIG. **6** illustrates a third version of a seating system for electronic gaming.

[0040] FIG. **7** illustrates different views of a fourth version of a seating system for electronic gaming.

[0041] FIGS. **8**A and **8**B illustrate different views of a fifth version of a seating system for electronic gaming.

[0042] FIGS. **9**A and **9**B illustrate different views of a sixth version of a seating system for electronic gaming.

[0043] FIG. **10** illustrates different views of a seventh version of a seating system for electronic gaming.

[0044] FIGS. **11**A and **11**B illustrate exploded views of another seating system for electronic gaming.

[0045] FIGS. **12**A-**12**F illustrate different views of yet another version of a seating system for electronic gaming.

[0046] FIG. **13** illustrates an operating console for of the seating system of FIG. **12**A in multiple configurations.

[0047] FIG. **14** illustrates the seating system of FIGS. **12**A-**12**F in another configuration.

[0048] FIGS. **15**A-**15**C illustrate different views of an electronic gaming system.

[0049] FIGS. **16**A-**16**C illustrate different views of an electronic gaming system.

[0050] FIGS. **17**A-**17**C illustrate off-angle views of the kiosk and the operating console in various configurations.

DETAILED DESCRIPTION

[0051] Examples in this disclosure provide seating systems for electronic gaming machines that may include an immersive and modular gaming seat, or bench which may be used synonymously with "seat". The seat may have a seating surface, seat back, a kiosk, and an operating console that is configured to receive user input, to rotate, and that has end pieces detachably coupled to a button deck of the operating console. The end pieces may also be configured to receive a user's input, which may include having a button, or to display information to a user. The operating console is communicatively connected, or coupled, to the EGM to communicate the user's input received at the operating console to the EGM. The seating system may be physically separate, or uncoupled, from the EGM in some instances, and may be physically coupled to the EGM in other instances. The kiosk may be in data communication with the seat and may include one or more projection devices to project gaming information or games from the kiosk. The bench may also include a swivel base that allows the bench to rotate with respect to a reference point. The seat may also include different modular panels that may include static graphics or artwork and may be modularly replaced with like panels that include flexible LCD or LED display panels. This may include flexible displays positioned on the side panels or armrests of the seat. Not only does the bench provide sounds and lights, but the bench may also be able to provide massaging service to a player while the player is seated. Moreover, the bench may provide the ability to connect to newly designed button deck style stand-alone kiosks with Cinema style LCD options, or in front of a large LED wheel display, as well as the ability to plug directly into standard gaming cabinets. The ability of the bench to connect to standard gaming cabinets is similar to how gaming chairs are currently connected in the field.

[0052] FIG. **1** illustrates several different models of EGMs which may be networked to various gaming related servers. Shown is a system **100** in a gaming environment including one or more server computers **102** (e.g., slot servers of a casino) that are in communication, via a communications network, with one or more gaming devices **104**A-**104**X (EGMs, slots, video poker, bingo machines, etc.) that can implement one or more aspects of the present disclosure. The gaming devices **104**A-**104**X may alternatively be portable and/or remote gaming devices such as, but not limited to, a smart phone, a tablet, a laptop, or a game console. Gaming devices **104**A-**104**X utilize specialized software and/or hardware to form non-generic, particular machines or apparatuses that comply with regulatory requirements regarding devices used for wagering or games of chance that provide monetary awards.

[0053] Communication between the gaming devices **104**A-**104**X and the server computers **102**, and among the gaming devices **104**A-**104**X, may be direct or indirect using one or more communication protocols. As an example, gaming devices **104**A-**104**X and the server computers **102** can communicate over one or more communication networks, such as over the Internet through a website maintained by a computer on a remote server or over an online data network including commercial online service providers, Internet service providers, private networks (e.g., local area networks and enterprise networks), and the like (e.g., wide area networks). The communication networks could allow gaming devices **104**A-**104**X to communicate with one another and/or the server computers **102** using a variety of communication-based technologies, such as radio frequency (RF) (e.g., wireless fidelity (WiFi®) and Bluetooth®), cable TV, satellite links and the like.

[0054] In some implementations, server computers **102** may not be necessary and/or preferred. For example, in one or more implementations, a stand-alone gaming device such as gaming device **104**A, gaming device **104**B or any of the other gaming devices **104**C-**104**X can implement one or more aspects of the present disclosure. However, it is typical to find multiple EGMs connected to networks implemented with one or more of the different server computers **102** described herein.

[0055] The server computers **102** may include a central determination gaming system server **106**, a ticket-in-ticket-out (TITO) system server **108**, a player tracking system server **110**, a progressive system server **112**, and/or a casino management system server **114**. Gaming devices **104**A-**104**X may include features to enable operation of any or all servers for use by the player and/or operator (e.g., the casino, resort, gaming establishment, tavern, pub, etc.). For example, game outcomes may be generated on a central determination gaming system server **106** and then transmitted over the network to any of a group of remote terminals or remote gaming devices **104**A-**104**X that utilize the game outcomes and display the results to the players.

[0056] Gaming device **104**A is often of a cabinet construction which may be aligned in rows or banks of similar devices for placement and operation on a casino floor. The gaming device **104**A often includes a main door which provides access to the interior of the cabinet. Gaming device **104**A typically includes a button area or button deck **120** accessible by a player that is configured with input switches or buttons **122**, an access channel for a bill validator **124**, and/or an access channel for a ticket-out printer **126**.

[0057] In FIG. **1**, gaming device **104**A is shown as a Relm XL™ model gaming device manufactured by Aristocrat® Technologies, Inc. As shown, gaming device **104**A is a reel machine having a gaming display area **118** comprising a number (typically 3 or 5) of mechanical reels **130** with various symbols displayed on them. The mechanical reels **130** are independently spun and stopped to show a set of symbols within the gaming display area **118** which may be used to determine an outcome to the game.

[0058] In many configurations, the gaming device **104**A may have a main display **128** (e.g., video display monitor) mounted to, or above, the gaming display area **118**. The main display **128** can be a high-resolution liquid crystal display (LCD), plasma, light emitting diode (LED), or organic light emitting diode (OLED) panel which may be flat or curved as shown, a cathode ray tube, or other conventional electronically controlled video monitor.

[0059] In some implementations, the bill validator **124** may also function as a "ticket-in" reader that allows the player to use a casino issued credit ticket to load credits onto the gaming device **104**A (e.g., in a cashless ticket ("TITO") system). In such cashless implementations, the gaming device **104**A may also include a "ticket-out" printer **126** for outputting a credit ticket when a "cash out" button is pressed. Cashless TITO systems are used to generate and track unique bar-codes or other indicators printed on tickets to allow players to avoid the use of bills and coins by loading credits using a ticket reader and cashing out credits using a ticket-out printer **126** on the gaming device **104**A. The gaming device **104**A can have hardware meters for purposes including ensuring regulatory compliance and monitoring the player credit balance. In addition, there can be additional meters that record the total amount of money wagered on the gaming device, total amount of money deposited, total amount of money withdrawn, total amount of winnings on gaming device **104**A.

[0060] In some implementations, a player tracking card reader **144**, a transceiver for wireless communication with a mobile device (e.g., a player's smartphone), a keypad **146**, and/or an illuminated display **148** for reading, receiving, entering, and/or displaying player tracking information is provided in gaming device **104**A. In such implementations, a game controller within the gaming device **104**A can communicate with the player tracking system server **110** to send and receive player tracking information.

[0061] Gaming device **104**A may also include a bonus topper wheel **134**. When bonus play is triggered (e.g., by a player achieving a particular outcome or set of outcomes in the primary game), bonus topper wheel **134** is operative to spin and stop with indicator arrow **136** indicating the outcome of the bonus game. Bonus topper wheel **134** is typically used to play a bonus game, but it could also be incorporated into play of the base or primary game.

[0062] A candle **138** may be mounted on the top of gaming device **104**A and may be activated by a player (e.g., using a switch or one of buttons **122**) to indicate to operations staff that gaming device **104**A has experienced a malfunction or the player requires service. The candle **138** is also often used to indicate a jackpot has been won and to alert staff that a hand payout of an award may be needed.

[0063] There may also be one or more information panels **152** which may be a back-lit, silkscreened glass panel with lettering to indicate general game information including, for example, a game denomination (e.g., $0.25 or $1), pay lines, pay tables, and/or various game related graphics. In some implementations, the information panel(s) **152** may be implemented as an additional video display.

[0064] Gaming devices **104**A have traditionally also included a handle **132** typically mounted to the side of main cabinet **116** which may be used to initiate game play.

[0065] Many or all the above-described components can be controlled by circuitry (e.g., a game controller) housed inside the main cabinet **116** of the gaming device **104**A, the details of which are shown in FIG. **2**A.

[0066] An alternative example gaming device **104**B illustrated in FIG. **1** is the Arc™ model gaming device manufactured by Aristocrat® Technologies, Inc. Note that where possible, reference numerals identifying similar features of the gaming device **104**A implementation are also identified in the gaming device **104**B implementation using the same reference numbers. Gaming device **104**B does not include physical reels and instead shows game play functions on main display **128**. An optional topper screen **140** may be used as a secondary game display for bonus play, to show game features or attraction activities while a game is not in play, or any other information or media desired by the game designer or operator. In some implementations, the optional topper screen **140** may also or alternatively be used to display progressive jackpot prizes available to a player during play of gaming device **104**B.

[0067] Example gaming device **104**B includes a main cabinet **116** including a main door which opens to provide access to the interior of the gaming device **104**B. The main or service door is typically used by service personnel to refill the ticket-out printer **126** and collect bills and tickets inserted into the bill validator **124**. The main or service door may also be accessed to reset the machine, verify and/or upgrade the software, and for general maintenance operations.

[0068] Another example gaming device **104**C shown is the Helix™ model gaming device manufactured by Aristocrat® Technologies, Inc. Gaming device **104**C includes a main display **128**A that is in a landscape orientation. Although not illustrated by the front view provided, the main display **128**A may have a curvature radius from top to bottom, or alternatively from side to side. In some implementations, the main display **128**A is a flat panel display. Main display **128**A is typically used for primary game play while secondary display **128**B is typically used for bonus game play, to show game features or attraction activities while the game is not in play, or any other information or media desired by the game designer or operator. In some implementations, example gaming device **104**C may also include speakers **142** to output various audio such as game sound, background music, etc.

[0069] Many different types of games, including mechanical slot games, video slot games, video poker, video black jack, video pachinko, keno, bingo, and lottery, may be provided with or implemented within the depicted gaming devices **104**A-**104**C and other similar gaming devices. Each gaming device may also be operable to provide many different games. Games may be differentiated according to themes, sounds, graphics, type of game (e.g., slot game vs. card game vs. game with aspects of skill), denomination, number of paylines, maximum jackpot, progressive or non-progressive, bonus games, and may be deployed for operation in Class **2** or Class **3**, etc.

[0070] FIG. **2**A is a block diagram depicting exemplary internal electronic components of a gaming device **200** connected to various external systems. All or parts of the gaming device **200** shown could be used to implement any one of the example gaming devices **104**A-X depicted in FIG. **1**. As shown in FIG. **2**A, gaming device **200** includes a topper display **216** or another form of a top box (e.g., a topper wheel, a topper screen, etc.) that sits above cabinet **218**. Cabinet **218** or topper display **216** may also house a number of other components which may be used to add features to a game being played on gaming device **200**, including speakers **220**, a ticket printer **222** which prints bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, a ticket reader **224** which reads bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, and a player tracking interface **232**. Player tracking interface **232** may include a keypad **226** for entering information, a player tracking display **228** for displaying information (e.g., an illuminated or video display), a card reader **230** for receiving data and/or communicating information to and from media or a device such as a smart phone enabling player tracking. FIG. **2** also depicts utilizing a ticket printer **222** to print tickets for a TITO system server **108**. Gaming device **200** may further include a bill validator **234**, player-input buttons **236** for player input, cabinet security sensors **238** to detect unauthorized opening of the cabinet **218**, a primary game display **240**, and a secondary game display **242**, each coupled to and operable under the control of game controller **202**.

[0071] The games available for playing on the gaming device **200** are controlled by a game controller **202** that includes one or more processors **204**. Processor **204** represents a general-purpose processor, a specialized processor intended to perform certain functional tasks, or a combination thereof. As an example, processor **204** can be a central processing unit (CPU) that has one or more multi-core processing units and memory mediums (e.g., cache memory) that function as buffers and/or temporary storage for data. Alternatively, processor **204** can be a specialized processor, such as an application specific integrated circuit (ASIC), graphics processing unit (GPU), field-programmable gate array (FPGA), digital signal processor (DSP), or another type of hardware accelerator. In another example, processor **204** is a system on chip (SoC) that combines and integrates one or more general-purpose processors and/or one or more specialized processors. Although FIG. **2**A illustrates that game controller **202** includes a single processor **204**, game controller **202** is not limited to this representation and instead can include multiple processors **204** (e.g., two or more processors).

[0072] FIG. **2**A illustrates that processor **204** is operatively coupled to memory **208**. Memory **208** is defined herein as including volatile and nonvolatile memory and other types of non-transitory data storage components. Volatile memory is memory that does not retain data values upon loss of power. Nonvolatile memory is memory that do retain data upon a loss of power. Examples of memory **208** include random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, universal serial bus (USB) flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, and/or other memory components, or a combination of any two or more of these memory components. In addition, examples of RAM include static random access memory (SRAM), dynamic random access memory (DRAM), magnetic random access memory (MRAM), and other such devices. Examples of ROM include a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device. Even though FIG. **2**A illustrates that game controller **202** includes a single memory **208**, game controller **202** could include multiple memories **208** for storing program instructions and/or data.

[0073] Memory **208** can store one or more game programs **206** that provide program instructions and/or data for carrying out various implementations (e.g., game mechanics) described herein. Stated another way, game program **206** represents an executable program stored in any portion or component of memory **208**. In one or more implementations, game program **206** is embodied in the form of source code that includes human-readable statements written in a programming language or machine code that contains numerical instructions recognizable by a suitable execution system, such as a processor **204** in a game controller or other system. Examples of executable programs include: (1) a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of memory **208** and run by processor **204**; (2) source code that may be expressed in proper format such as object code that is capable of being loaded into a random access portion of memory **208** and executed by processor **204**; and (3) source code that may be interpreted by another executable program to generate instructions in a random access portion of memory **208** to be executed by processor **204**.

[0074] Alternatively, game programs **206** can be set up to generate one or more game instances based on instructions and/or data that gaming device **200** exchanges with one or more remote gaming devices, such as a central determination gaming system server **106** (not shown in FIG. **2**A but shown in FIG. **1**). For purpose of this disclosure, the term "game instance" refers to a play or a round of a game that gaming device **200** presents (e.g., via a user interface (UI)) to a player. The game instance is communicated to gaming device **200** via the network **214** and then displayed on gaming device **200**. For example, gaming device **200** may execute game program **206** as video streaming software that allows the game to be displayed on gaming device **200**. When a game is stored on gaming device **200**, it may be loaded from memory **208** (e.g., from a read only memory (ROM)) or from the central determination gaming system server **106** to memory **208**.

[0075] Gaming devices, such as gaming device **200**, are highly regulated to ensure fairness and, in many cases, gaming device **200** is operable to award monetary awards (e.g., typically dispensed in the form of a redeemable voucher). Therefore, to satisfy security and regulatory requirements in a gaming environment, hardware and software architectures are implemented in gaming devices **200** that differ significantly from those of general-purpose computers. Adapting general purpose computers to function as gaming devices **200** is not simple or straightforward because of: (1) the regulatory requirements for gaming devices **200**, (2) the harsh environment in which gaming devices **200** operate, (3) security requirements, (4) fault tolerance requirements, and (5) the requirement for additional special purpose componentry enabling functionality of an EGM. These differences require substantial engineering effort with respect to game design implementation, game mechanics, hardware components, and software.

[0076] One regulatory requirement for games running on gaming device **200** generally involves complying with a certain level of randomness. Typically, gaming authorities mandate that gaming devices **200** satisfy a minimum level of randomness without specifying how a gaming device **200** should achieve this level of randomness. To comply, FIG. **2**A illustrates that gaming device **200** could include an RNG **212** that utilizes hardware and/or software to generate RNG outcomes that lack any pattern. The RNG operations are often specialized and non-generic to comply with regulatory and gaming requirements. For example, in a slot game, game program **206** can initiate multiple RNG calls to RNG **212** to generate RNG outcomes, where each RNG call and RNG outcome corresponds to an outcome for a reel. In another example, gaming device **200** can be a Class II gaming device where RNG **212** generates RNG outcomes for creating Bingo cards. In one or more implementations, RNG **212** could be one of a set of RNGs operating on gaming device **200**. More generally, an output of the RNG **212** can be the basis on which game outcomes are determined by the game controller **202**. Game developers could vary the degree of true randomness for each RNG (e.g., pseudorandom) and utilize specific RNGs depending on game requirements. The output of the RNG **212** can include a random number or pseudorandom number (either is generally referred to as a "random number").

[0077] In FIG. **2**A, RNG **212** and hardware RNG **244** are shown in dashed lines to illustrate that RNG **212**, hardware RNG **244**, or both can be included in gaming device **200**. In one implementation, instead of including RNG **212**, gaming device **200** could include a hardware RNG **244** that generates RNG outcomes. Analogous to RNG **212**, hardware RNG **244** performs specialized and non-generic operations to comply with regulatory and gaming requirements. For example, because of regulation requirements, hardware RNG **244** could be a random number generator that securely produces random numbers for cryptography use. The gaming device **200** then uses the secure random numbers to generate game outcomes for one or more game features. In another implementation, the gaming device **200** could include both hardware RNG **244** and RNG **212**. RNG **212** may utilize the RNG outcomes from hardware RNG **244** as one of many sources of entropy for generating secure random numbers for the game features.

[0078] Another regulatory requirement for running games on gaming device **200** includes ensuring a certain level of RTP. Similar to the randomness requirement discussed above, numerous gaming jurisdictions also mandate that gaming device **200** provides a minimum level of RTP (e.g., RTP of at least 75%). A game can use one or more lookup tables (also called weighted tables) as part of a technical solution that satisfies regulatory requirements for randomness and RTP. In particular, a lookup table can integrate game features (e.g., trigger events for special modes or bonus games; newly introduced game elements such as extra reels, new symbols, or new cards; stop positions for dynamic game elements such as spinning reels, spinning wheels, or shifting reels; or card selections from a deck) with random numbers generated by one or more RNGs, so as to achieve a given level of volatility for a target level of RTP. (In general, volatility refers to the frequency or probability of an event such as a special mode, payout, etc. For example, for a target level of RTP, a higher-volatility game may have a lower payout most of the time with an occasional bonus having a very high payout, while a lower-volatility game has a steadier payout with more frequent bonuses of smaller amounts.) Configuring a lookup table can involve engineering decisions with respect to how RNG outcomes are mapped to game outcomes for a given game feature, while still satisfying regulatory requirements for RTP. Configuring a lookup table can also involve engineering decisions about whether different game features are combined in a given entry of the lookup table or split between different entries (for the respective game features), while still satisfying regulatory requirements for RTP and allowing for varying levels of game volatility.

[0079] FIG. **2**A illustrates that gaming device **200** includes an RNG conversion engine **210** that translates the RNG outcome from RNG **212** to a game outcome presented to a player. To meet a designated RTP, a game developer can set up the RNG conversion engine **210** to utilize one or more lookup tables to translate the RNG outcome to a symbol element, stop position on a reel strip layout, and/or randomly chosen aspect of a game feature. As an example, the lookup tables can regulate a prize payout amount for each RNG outcome and how often the gaming device **200** pays out the prize payout amounts. The RNG conversion engine **210** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. The mapping between the RNG outcome to the game outcome controls the frequency in hitting certain prize payout amounts.

[0080] FIG. **2**A also depicts that gaming device **200** is connected over network **214** to player tracking system server **110**. Player tracking system server **110** may be, for example, an OASIS® system manufactured by Aristocrat® Technologies, Inc. Player tracking system server **110** is used to track play (e.g. amount wagered, games played, time of play and/or other quantitative or qualitative measures) for individual players so that an operator may reward players in a loyalty program. The player may use the player tracking interface **232** to access his/her account information, activate free play, and/or request various information. Player tracking or loyalty programs seek to reward players for their play and help build brand loyalty to the gaming establishment. The rewards typically correspond to the player's level of patronage (e.g., to the player's playing frequency and/or total amount of game plays at a given casino). Player tracking rewards may be complimentary and/or discounted meals, lodging, entertainment and/or additional play. Player tracking information may be combined with other information that is now readily obtainable by a casino management system.

[0081] When a player wishes to play the gaming device **200**, he/she can insert cash or a ticket voucher through a coin acceptor (not shown) or bill validator **234** to establish a credit balance on the gaming device. The player uses the credit balance to place wagers on instances of the game and to receive credit awards based on the outcome of winning instances. The credit balance is decreased by the amount of each wager and increased upon a win. The player can add additional credits to the balance at any time. The player may also optionally insert a loyalty club card into the card reader **230**. During the game, the player views with one or more UIs, the game outcome on one or more of the primary game displays **240** and secondary game display **242**. Other game and prize information may also be displayed.

[0082] For each game instance, a player may make selections, which may affect the play of the game. For example, the player may vary the total amount wagered by selecting the amount bet per line and the number of lines played. In many games, the player is asked to initiate or select options during the course of game play (such as spinning a wheel to begin a bonus round or selecting various items during a feature game). The player may make these selections using the player-input buttons **236**, the primary game display **240** which may be a touch screen, or using some other device which enables a player to input information into the gaming device **200**.

[0083] During certain game events, the gaming device **200** may display visual and auditory effects that can be perceived by the player. These effects add to the excitement of a game, which makes a player more likely to enjoy the playing experience. Auditory effects include various sounds that are projected by the speakers **220**. Visual effects include flashing lights, strobing lights or other patterns displayed from lights on the gaming device **200** or from lights behind the information panel **152** (FIG. **1**).

[0084] When the player is done, he/she cashes out the credit balance (typically by pressing a cash out button to receive a ticket from the ticket printer **222**). The ticket may be "cashed-in" for money or inserted into another machine to establish a credit balance for play.

[0085] Additionally, or alternatively, gaming devices **104**A-**104**X and **200** can include or be coupled to one or more wireless transmitters, receivers, and/or transceivers (not shown in FIGS. **1** and **2**A) that communicate (e.g., Bluetooth® or other near-field communication technology) with one or more mobile devices to perform a variety of wireless operations in a casino environment. Examples of wireless operations in a casino environment include detecting the presence of mobile devices, performing credit, points, comps, or other marketing or hard currency transfers, establishing wagering sessions, and/or providing a personalized casino-based experience using a mobile application. In one implementation, to perform these wireless operations, a wireless transmitter or transceiver initiates a secure wireless connection between a gaming device **104**A-**104**X and **200** and a mobile device. After establishing a secure wireless connection between the gaming device **104**A-**104**X and **200** and the mobile device, the wireless transmitter or transceiver does not send and/or receive application data to and/or from the mobile device. Rather, the mobile device communicates with gaming devices **104**A-**104**X and **200** using another wireless connection (e.g., WiFi® or cellular network). In another implementation, a wireless transceiver establishes a secure connection to directly communicate with the mobile device. The mobile device and gaming device **104**A-**104**X and **200** sends and receives data utilizing the wireless transceiver instead of utilizing an external network. For example, the mobile device would perform digital wallet transactions by directly communicating with the wireless transceiver. In one or more implementations, a wireless transmitter could broadcast data received by one or more mobile devices without establishing a pairing connection with the mobile devices.

[0086] Although FIGS. **1** and **2**A illustrate specific implementations of a gaming device (e.g., gaming devices **104**A-**104**X and **200**), the disclosure is not limited to those implementations shown in FIGS. **1** and **2**. For example, not all gaming devices suitable for implementing implementations of the present disclosure necessarily include top wheels, top boxes, information panels, cashless ticket systems, and/or player tracking systems. Further, some suitable gaming devices have only a single game display that includes only a mechanical set of reels and/or a video display, while others are designed for bar counters or tabletops and have displays that face upwards. Gaming devices **104**A-**104**X and **200** may also include other processors that are not separately shown. Using FIG. **2**A as an example, gaming device **200** could include display controllers (not shown in FIG. **2**A) configured to receive video input signals or instructions to display images on game displays **240** and **242**. Alternatively, such display controllers may be integrated into game controller **202**. The use and discussion of FIGS. **1** and **2** are examples to facilitate case of description and explanation.

[0087] FIG. **2**B depicts a casino gaming environment according to one example. In this example, casino **251** includes banks **252** of EGMs **104**. In this example, each bank **252** of EGMs **104** includes a corresponding gaming signage system **254** (also shown in FIG. **2**A). According to this implementation, casino **251** also includes mobile gaming devices **256**, which are also configured to present wagering games in this example. The mobile gaming devices **256** may, for example, include tablet devices, cellular phones, smart phones and/or other handheld devices. In this example, mobile gaming devices **256** are configured for communication with one or more other devices in the casino **251**, including but not limited to one or more of the server computers **102**, via wireless access points **258**.

[0088] According to some examples, mobile gaming devices **256** may be configured for stand-alone determination of game outcomes. However, in some alternative implementations the mobile gaming devices **256** may be configured to receive game outcomes from another device, such as the central determination gaming system server **106**, one of the EGMs **104**, etc.

[0089] Some mobile gaming devices **256** may be configured to accept monetary credits from a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, via a patron casino account, etc. However, some mobile gaming devices **256** may not be configured to accept monetary credits via a credit or debit card. Some mobile gaming devices **256** may include a ticket reader and/or a ticket printer whereas some mobile gaming devices **256** may not, depending on the particular implementation.

[0090] In some implementations, casino **251** may include one or more kiosks **260** that are configured to facilitate monetary transactions involving the mobile gaming devices **256**, which may include cash out and/or cash in transactions. The kiosks **260** may be configured for wired and/or wireless communication with the mobile gaming devices **256**. The kiosks **260** may be configured to accept monetary credits from casino patrons **262** and/or to dispense monetary credits to casino patrons **262** via cash, a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, etc. According to some examples, the kiosks **260** may be configured to accept monetary credits from a casino patron and to provide a corresponding amount of monetary credits to a mobile gaming device **256** for wagering purposes, e.g., via a wireless link such as a near-field communications link.

[0091] In some such examples, when a casino patron **262** is ready to cash out, the casino patron **262** may select a cash out option provided by a mobile gaming device **256**, which may include a real button or a virtual button (e.g., a button provided via a graphical user interface) in some instances. In some such examples, the mobile gaming device **256** may send a "cash out" signal to a kiosk **260** via a wireless link in response to receiving a "cash out" indication from a casino patron. The kiosk **260** may provide monetary credits to the casino patron **262** corresponding to the "cash out" signal, which may be in the form of cash, a credit ticket, a credit transmitted to a financial account corresponding to the casino patron, etc.

[0092] In some implementations, a cash-in process and/or a cash-out process may be facilitated by the TITO system server **108**. For example, the TITO system server **108** may control, or at least authorize, ticket-in and ticket-out transactions that involve a mobile gaming device **256** and/or a kiosk **260**.

[0093] Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information. For example, some mobile gaming devices **256** may be configured for wireless communication with the player tracking system server **110**. Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information via wireless communication with a patron's player loyalty card, a patron's smartphone, etc.

[0094] According to some implementations, a mobile gaming device **256** may be configured to provide safeguards that prevent the mobile gaming device **256** from being used by an unauthorized person. For example, some mobile gaming devices **256** may include one or more biometric sensors and may be configured to receive input via the biometric sensor(s) to verify the identity of an authorized patron. Some mobile gaming devices **256** may be configured to function only within a predetermined or configurable area, such as a casino gaming area.

[0095] FIG. **2**C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure. As with other figures presented in this disclosure, the numbers, types, and arrangements of gaming devices shown in FIG. **2**C are merely shown by way of example. In this example, various gaming devices, including but not limited to end user devices (EUDs) **264***a*, **264***b* and **264***c* are capable of communication via one or more networks **417**. The networks **417** may, for example, include one or more cellular telephone networks, the Internet, etc. In this example, the EUDs **264***a* and **264***b* are mobile devices: according to this example the EUD **264***a* is a tablet device and the EUD **264***b* is a smart phone. In this implementation, the EUD **264***c* is a laptop computer that is located within a residence **266** at the time depicted in FIG. **2**C. Accordingly, in this example the hardware of EUDs is not specifically configured for online gaming, although each EUD is configured with software for online gaming. For example, each EUD may be configured with a web browser. Other implementations may include other types of EUD, some of which may be specifically configured for online gaming.

[0096] In this example, gaming data center **276** includes various devices that are configured to provide online wagering games via the networks **417**. The gaming data center **276** is capable of communication with the networks **417** via the gateway **272**. In this example, switches **278** and routers **280** are configured to provide network connectivity for devices of the gaming data center **276**, including storage devices **282***a*, servers **284***a* and one or more workstations **570***a*. The servers **284***a* may, for example, be configured to provide access to a library of games for online game play. In some examples, code for executing at least some of the games may initially be stored on one or more of the storage devices **282***a*. The code may be subsequently loaded onto server **284***a* after selection by a player via an EUD and communication of that selection from the EUD via the networks **417**. The server **284***a* onto which code for the selected game has been loaded may provide the game according to selections made by a player and indicated via the player's EUD. In other examples, code for executing at least some of the games may initially be stored on one or more of the servers **284***a*. Although only one gaming data center **276** is shown in FIG. **2**C, some implementations may include multiple gaming data centers **276**.

[0097] In this example, a financial institution data center **270** is also configured for communication via the networks **417**. Here, the financial institution data center **270** includes servers **284***b*, storage devices **282***b*, and one or more workstations **286***b*. According to this example, the financial institution data center **270** is configured to maintain financial accounts, such as checking accounts, savings accounts, loan accounts, etc. In some implementations one or more of the authorized users **274***a*-**274***c* may maintain at least one financial account with the financial institution that is serviced via the financial institution data center **270**.

[0098] According to some implementations, the gaming data center **276** may be configured to provide online wagering games in which money may be won or lost. According to some such implementations, one or more of the servers **284***a* may be configured to monitor player credit balances, which may be expressed in game credits, in currency units, or in any other appropriate manner. In some implementations, the server(s) **284***a* may be configured to obtain financial credits from and/or provide financial credits to one or more financial institutions, according to a player's "cash in" selections, wagering game results and a player's "cash out" instructions. According to some such implementations, the server(s) **284***a* may be configured to electronically credit or debit the account of a player that is maintained by a financial institution, e.g., an account that is maintained via the financial institution data center **270**. The server(s) **284***a* may, in some examples, be configured to maintain an audit record of such transactions.

[0099] In some alternative implementations, the gaming data center **276** may be configured to provide online wagering games for which credits may not be exchanged for cash or the equivalent. In some such examples, players may purchase game credits for online game play, but may not "cash out" for monetary credit after a gaming session. Moreover, although the financial institution data center **270** and the gaming data center **276** include their own servers and storage devices in this example, in some examples the financial institution data center **270** and/or the gaming data center **276** may use offsite "cloud-based" servers and/or storage devices. In some alternative examples, the financial institution data center **270** and/or the gaming data center **276** may rely entirely on cloud-based servers.

[0100] One or more types of devices in the gaming data center **276** (or elsewhere) may be capable of executing middleware, e.g., for data management and/or device communication. Authentication information, player tracking information, etc., including but not limited to information obtained by EUDs **264** and/or other information regarding authorized users of EUDs **264** (including but not limited to the authorized users **274***a*-**274***c*), may be stored on storage devices **282** and/or servers **284**. Other game-related information and/or software, such as information and/or software relating to leaderboards, players currently playing a game, game themes, game-related promotions, game competitions, etc., also may be stored on storage devices **282** and/or servers **284**. In some implementations, some such game-related software may be available as "apps" and may be downloadable (e.g., from the gaming data center **276**) by authorized users.

[0101] In some examples, authorized users and/or entities (such as representatives of gaming regulatory authorities) may obtain gaming-related information via the gaming data center **276**. One or more other devices (such EUDs **264** or devices of the gaming data center **276**) may act as intermediaries for such data feeds. Such devices may, for example, be capable of applying data filtering algorithms, executing data summary and/or analysis software, etc. In some implementations, data filtering, summary and/or analysis software may be available as "apps" and downloadable by authorized users.

[0102] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture **300** that implements a game processing pipeline for the play of a game in accordance with various implementations described herein. As shown in FIG. **3**, the gaming processing pipeline starts with having a UI system **302** receive one or more player inputs for the game instance. Based on the player input(s), the UI system **302** generates and sends one or more RNG calls to a game processing backend system **314**. Game processing backend system **314** then processes the RNG calls with RNG engine **316** to generate one or more RNG outcomes. The RNG outcomes are then sent to the RNG conversion engine **320** to generate one or more game outcomes for the UI system **302** to display to a player. The game processing architecture **300** can implement the game processing pipeline using a gaming device, such as gaming devices **104**A-**104**X and **200** shown in FIGS. **1** and **2**, respectively. Alternatively, portions of the gaming processing architecture **300** can implement the game processing pipeline using a gaming device and one or more remote gaming devices, such as central determination gaming system server **106** shown in FIG. **1**.

[0103] UI system **302** includes one or more UIs that a player can interact with. The UI system **302** could include one or more game play UIs **304**, one or more bonus game play UIs **308**, and one or more multiplayer UIs **312**, where each UI type includes one or more mechanical UIs and/or graphical UIs (GUIs). In other words, game play UI **304**, bonus game play UI **308**, and the multiplayer UI **312** may utilize a variety of UI elements, such as mechanical UI elements (e.g., physical "spin" button or mechanical reels) and/or GUI elements (e.g., virtual reels shown on a video display or a virtual button deck) to receive player inputs and/or present game play to a player. Using FIG. **3** as an example, the different UI elements are shown as game play UI elements **306**A-**306**N and bonus game play UI elements **310**A-**310**N.

[0104] The game play UI **304** represents a UI that a player typically interfaces with for a base game. During a game instance of a base game, the game play UI elements **306**A-**306**N (e.g., GUI elements depicting one or more virtual reels) are shown and/or made available to a user. In a subsequent game instance, the UI system **302** could transition out of the base game to one or more bonus games. The bonus game play UI **308** represents a UI that utilizes bonus game play UI elements **310**A-**310**N for a player to interact with and/or view during a bonus game. In one or more implementations, at least some of the game play UI element **306**A-**306**N are similar to the bonus game play UI elements **310**A-**310**N. In other implementations, the game play UI element **306**A-**306**N can differ from the bonus game play UI elements **310**A-**310**N.

[0105] FIG. **3** also illustrates that UI system **302** could include a multiplayer UI **312** purposed for game play that differs or is separate from the typical base game. For example, multiplayer UI **312** could be set up to receive player inputs and/or presents game play information relating to a tournament mode. When a gaming device transitions from a primary game mode that presents the base game to a tournament mode, a single gaming device is linked and synchronized to other gaming devices to generate a tournament outcome. For example, multiple RNG engines **316** corresponding to each gaming device could be collectively linked to determine a tournament outcome. To enhance a player's gaming experience, tournament mode can modify and synchronize sound, music, reel spin speed, and/or other operations of the gaming devices according to the tournament game play. After tournament game play ends, operators can switch back the gaming device from tournament mode to a primary game mode to present the base game. Although FIG. **3** does not explicitly depict that multiplayer UI **312** includes UI elements, multiplayer UI **312** could also include one or more multiplayer UI elements.

[0106] Based on the player inputs, the UI system **302** could generate RNG calls to a game processing backend system **314**. As an example, the UI system **302** could use one or more application programming interfaces (APIs) to generate the RNG calls. To process the RNG calls, the RNG engine **316** could utilize gaming RNG **318** and/or non-gaming RNGs **319**A-**319**N. Gaming RNG **318** could correspond to RNG **212** or hardware RNG **244** shown in FIG. **2**A. As previously discussed with reference to FIG. **2**A, gaming RNG **318** often performs specialized and non-generic operations that comply with regulatory and/or game requirements. For example, because of regulation requirements, gaming RNG **318** could correspond to RNG **212** by being a cryptographic RNG or pseudorandom number generator (PRNG) (e.g., Fortuna PRNG) that securely produces random numbers for one or more game features. To securely generate random numbers, gaming RNG **318** could collect random data from various sources of entropy, such as from an operating system (OS) and/or a hardware RNG (e.g., hardware RNG **244** shown in FIG. **2**A). Alternatively, non-gaming RNGs **319**A-**319**N may not be cryptographically secure and/or be computationally less expensive. Non-gaming RNGs **319**A-**319**N can, thus, be used to generate outcomes for non-gaming purposes. As an example, non-gaming RNGs **319**A-**319**N can generate random numbers for generating random messages that appear on the gaming device.

[0107] The RNG conversion engine **320** processes each RNG outcome from RNG engine **316** and converts the RNG outcome to a UI outcome that is feedback to the UI system **302**. With reference to FIG. **2**A, RNG conversion engine **320** corresponds to RNG conversion engine **210** used for game play. As previously described, RNG conversion engine **320** translates the RNG outcome from the RNG **212** to a game outcome presented to a player. RNG conversion engine **320** utilizes one or more lookup tables **322**A-**322**N to regulate a prize payout amount for each RNG outcome and how often the gaming device pays out the derived prize payout amounts. In one example, the RNG conversion engine **320** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. In this example, the mapping between the RNG outcome and the game outcome controls the frequency in hitting certain prize payout amounts. Different lookup tables could be utilized depending on the different game modes, for example, a base game versus a bonus game.

[0108] After generating the UI outcome, the game processing backend system **314** sends the UI outcome to UI system **302**. Examples of UI outcomes are symbols to display on a video reel or reel stops for a mechanical reel. In one example, if the UI outcome is for a base game, the UI system **302** updates one or more game play UI elements **306**A-**306**N, such as symbols, for the game play UI **304**. In another example, if the UI outcome is for a bonus game, the UI system could update one or more bonus game play UI elements **310**A-**310**N (e.g., symbols) for the bonus game play UI **308**. In response to updating the appropriate UI, the player may subsequently provide additional player inputs to initiate a subsequent game instance that progresses through the game processing pipeline.

[0109] Provided herein are new and novel seating systems and electronic gaming systems. It is desirable to provide unique, exciting, and immersive experiences for electronic gaming systems for individual players, multiple players, potential players, and onlookers. Various improvements to electronic gaming experiences may be made with seating systems that provide additional lighting, sound, video, and other sensory enhancements. This may include providing a seat, which may also be referred to herein as a bench or gaming bench, on which one or more persons may sit to play or watch game instances. The seat may be a part of a seating system that has multiple features like one or more displays, a kiosk, and features for receiving input from a user such as a button, a button deck with physical buttons, or a virtual button deck. In some instances, the seating system may be physically separate from the electronic gaming machine (EGM) and its one or more displays. In other instances, the seating system may have one or more displays of the EGM, such that the seating system is physically coupled to the one or more displays of the EGM.

[0110] In some implementations, the seat of a seating system may have a curved flexible display panel positioned on a side panel and/or armrest of the seating system. The curved flexible display panel is configured to display various materials, such as images, videos, game information, attractions, a portion of a game instance, or a combination thereof. In some instances, the seating system may have one or more side panels with curved surfaces and positioned thereon is the curved flexible display panel. The seating system may also have an armrest, and in some implementations, the curved flexible display panel is positioned on the armrest. Some implementations may have curved flexible displays on both the side panel and the armrest of the seat.

[0111] The seating systems provided herein may also have an operating console that has detachable and reconfigurable end pieces, and that is configured to rotate and, in some cases, translate along an axis. As provided herein, the operating console is configured to receive input from a player and, in some implementations, display information to the player on the seat. The operating console is also configured to communicate with a game controller of the EGM and thereby transmit the player input to the game controller and receive information from the game controller. In some implementations, the operating console is configured to move between orientations and/or locations to provide different playing configurations, such as a one-player configuration and a two-player configuration. In a one-player example configuration, one of the operating console's end pieces may have a button for receiving a player's input, and the operating console is positioned such that this button is positioned in a suitable location for single player gameplay, such as in the center of the seat. In a two-player example configuration, each of the operating console's end pieces may have a button, respectively, for receiving a player's input, and the operating console is positioned such that each player can reach their own respective button. For instance, the operating console may be positioned such that one button on a first end piece is closer to a first side of the seat, and the other button on the second end piece is closer to a second side of the seat. This configuration of the operating console provides both players the opportunity to press their own button and play a game on the EGM.

[0112] As provided above, the operating console may be reconfigured with different end pieces that may have different functionality or configurations. For example, each end piece may have a button for receiving a player's input. One of these end pieces may be removed and replaced with another end piece that may have a different button, the same button, or a different feature or function. Such reconfigurability provides numerous benefits. For instance, if a button or end piece becomes non-operable, then a new end piece with a new button can be positioned on the operating console without replacing or repairing the entire operating console or seating system. In another example, one end piece may be removed and replaced with an end piece having a different button that may have a different size, function, shape, orientation, or a combination thereof. For instance, one end piece may have a circular button, and it may be removed and replaced with a button of a different shape, such as a triangle or tri-lobe button.

[0113] In another instance, end pieces may have buttons at different orientations to account for rotation of the operating console. This may advantageously position the buttons at the same orientation with respect to the seat and player, while the operating console is at different orientations. For instance, a first end piece may have a button in a first orientation with respect to the end piece and operating console. A second end piece may have the same shaped button in a second, different orientation, with respect to the end piece and operating console, such as 90 degrees different than the first orientation. Although these two end pieces have buttons at different orientations with respect to the operating console, these two end pieces are configured to position the button in the same orientation with respect to the seat when the operating console has been rotated between positions. In this example, the operating console may have the first end piece and be in a first position with respect to the seat, which orients the button in a play orientation with respect to the seat. When the operating console is rotated to a second position, the first end piece may be replaced with the second end piece which has the button positioned in the second, different orientation with respect to the operating console, but at the same play orientation with respect to the seat. This advantageously positions the button in the same orientation with respect to a player despite rotation of the operating console. This example is further discussed and illustrated below.

[0114] In some implementations, an end piece of the operating console may have a feature other than a button, such as a display for providing information to a player on the seat. Similar to the orientation of the buttons described above, some end pieces may have the displays at different orientations with respect to the operating console. For example, a first end piece may have a display at a first orientation with respect to the first end piece and the operating console, and a second end piece may have the display at a second orientation with respect to the second end piece and the operating console. These two end pieces are configured to account for rotation of the operating console by having their displays positioned at different orientations with respect to the operating console. When the operating console is rotated between different positions, the displays on the first and second end pieces, respectively, may be in the same orientation with respect to the seat.

[0115] Various implementations of seating systems will now be discussed. Provided herein are numerous examples of seating systems, and the discussion will begin with an example seating system **1200** and some of its features shown in FIGS. **12**A-**12**F, **13**, **14**, and **17**A-C. Additional seating systems are illustrated in FIGS. **4**A-**11**B as well as FIGS. **15**A-**16**C, and discussion of these systems are farther below. As used herein, both the phrases a "seating system" and an "electronic gaming machine," are used herein and for purposes of this disclosure, both these phrases may be used interchangeably.

[0116] Turning now to FIGS. **12**A-**12**F, these Figures illustrate different views of the example seating system **1200**. The last two numbers of each reference numeral used in FIGS. **12**A-**12**F refer to the same features with reference numerals having the same last two numbers in other figures. The seating system **1200** here has a seat **1204**, which may also be referred to herein as a "bench" or "gaming bench." The seat **1204** has a seating surface **1217** (identified in FIGS. **12**B, **12**C, and **12**E) on which one or more players may sit. The seat **1204** also has a backrest **1210** (identified in FIGS. **12**A-**12**D) which may be configured to support the back of a player sitting on the seat **1204**.

[0117] The seating system **1200** also has a kiosk **1212** that is coupled to the seat **1204**. This coupling may be direct or indirect. In FIGS. **12**A-**12**F, the kiosk **1212** is depicted as indirectly coupled to the seat **1204** via a base **1221** of the seating system **1200**. In these illustrations, the kiosk **1212** is directly coupled to the base **1221** and the seat **1204** is directly coupled to the base **1221**. In some implementations, the kiosk **1212** has various features. For example, the kiosk **1212** may have a ticket printer **1220** (similar to the ticket printer **222** of FIG. **2**), a wireless charging device **1224**, a player tracking device **1226** (similar to the player tracking interface **232** of FIG. **2**), a cup holder **1228**, a bill validator **1232** (similar to the bill validator **124** of FIG. **1**), a door access **1237**, or a combination thereof. FIG. **12**A also shows that the seat **1204** includes a bottom panel **1238** that may include a light panel with an etched pattern.

[0118] The seating systems are configured to be communicatively coupled with an electronic gaming machine, such as any EGM provided herein. These EGMs, which may also be considered electronic gaming systems, may have one or more displays and one or more game controllers, like described herein. The EGMs also have features configured to receive input from a player and provide this input to the EGM. These input receiving features may be located on the operating console of the seating system, as provided herein, and may include physical and/or virtual buttons.

[0119] For the seating system **1200** illustrated in FIGS. **12**A-**12**F, it has an operating console **1216** that is configured to receive input from a user or player, and that is configured to be reconfigurable and rotatable. As illustrated in FIG. **12**B, the operating console **1216** may have a button deck **1223** which is encompassed by a dashed rectangle and this button deck **1223** may have a rectangular shape. The button deck **1223** may have a display panel configured to provide a virtual button deck, may have one or more physical buttons, or both. The button deck **1223** is configured to receive player input and in some cases, display information related to a game on the EGM. The operating console **1216** also has a first end piece **1225**A detachably coupled at one end of the operating console **1216** and a second end piece **1225**B detachably coupled at an opposite end of the operating console **1216**, both of which are encircled with dashed ellipses.

[0120] Each end piece **1225**A and **1225**B may be configured to receive player input, provide information, or both. For example, in FIG. **12**B, each end piece **1225**A and **1225**B has a button **1227**A and **1227**B, respectively, configured to receive input from a player. The button may be a physical button that is configured to move (e.g., a pressable button or a "smash" button), as well as a virtual button. In some implementations, the button may be a physical button that also has a display provided therein which may provide various information to a player, such as game related information like a wager amount or prompts (e.g., "GO!", "START", "PLAY", "WIN!"). In some implementations, like described below, an end piece may have a display configured to display information, such as a player tracking information.

[0121] The operating console **1216** is configured such that its end pieces **1225**A and **1225**B are detachably coupled to the button deck **1223**. These end pieces may be detached (i.e., removed or uncoupled) from the operating console, and other end pieces may be detachably coupled to the operating console. The physical coupling between the operating console **1216** and the end pieces may include various connection features, such as clips, bolts, screws, or the like. This may include an end piece sliding onto the operating console **1216** and held in place with screws. There may also be electrical and/or data coupling between the operating console and the end pieces, such as by USB-A, USB-C, HDMI, Ethernet (RJ45), DisplayPort, VGA, DVI, Mini DisplayPort, serial connectors like RS-232 (DB-9), or the like. In some instances the operating console **1216** will have sensors configured to detect an end piece being removed. One example is a reed sensor with a magnet and a reed switch that together are configured to create a magnetic circuit, and when one component is moved relative to the other, the circuit is broken which may indicate an end piece has been removed.

[0122] In some implementations, a system provided herein may have one or more USB ports and/or Micro USB ports configured to provide charging of, and/or communication with, a device, such as a player's portable electronic device and/or a gaming device accessory, such as a remote button like a remote spin button. The communication between the one or more USB ports or Micro USB ports and the device may be through a wired or wireless communications interface. In some instances, the system may have a wireless communications interface configured to receive and transmit information between a controller of the system and the device. In some instances, the USB ports or Micro USB ports may provide wireless communications via a wireless antenna connected thereto. The one or more USB ports or Micro USB ports may be positioned on various aspects of the system, such as on an end piece, on the operating console **1216**, on the button deck **1223**, on the kiosk **1212**, on the seat **1204**, on an armrest, on the bottom panel **1238**, or a combination thereof. As illustrated in FIG. **12**B, a USB port **1241** is provided on the second end piece **1225**B.

[0123] The operating console **1216** is also configured to be rotated into different orientations and usable in any of the orientations. For instance, the operating console is configured to be rotated between a first orientation and a second orientation 90 degrees apart from the first orientation. In some instances, the operating console **1216** has a rotatable mount, like a swivel mount, connected to the kiosk **1212**, as represented by a circle **1235** in FIG. **12**E. The operating console **1216** may be configured to be rotated by one of the players without the use of any tool. A retention feature, like a latch or switch, may retain the operating console **1216** in the first or second orientation, and the latch or switch may be manually actuated by the player in order to rotate the operating console. In some implementations, a mechanical rotation mechanism may automatically rotate the operating console **1216**.

[0124] Features of the operating console are further illustrated in FIG. **13** which depicts the operating console of the electronic gaming system in FIG. **12**A. Here in FIG. **13**, two example configurations of the operating console **1216** of system **1200** are provided. In the lefthand portion of FIG. **13** is operating console **1216**A in a first configuration. In this first configuration, operating console **1216**A is in a first orientation and has two end pieces **1225**A and **1225**B. Each end piece **1225**A and **1225**B has a button **1227**A and **1227**B, respectively. The first orientation of the operating console **1216**A may be in reference to a first axis A**1** of the seating system and/or a position P**1** of a player on the seat (the first axis A**1** and position P**1** are also shown in FIG. **12**B). In the first orientation, a longitudinal axis LA**1** of the operating console **1216**A is oriented parallel to the first axis A**1** and a lateral axis LA**2** of the operating console **1216**A is oriented perpendicular to the first axis A**1**. In some instances, this first orientation may be considered a landscape or horizontal orientation.

[0125] As illustrated in FIG. **13**, the end pieces **1225**A-**1225**D of the operating console are configured to be uncoupled (i.e., detached) and coupled (i.e., attached) to the button deck **1223**. This detachability may provide numerous advantages, such as the use of buttons, displays, or other features at different orientations with respect to the button deck **1223** in order to provide such features in a desired orientation with respect to a player on the seat. For example, operating console **1216**B in the righthand portion of FIG. **13** is in a different orientation than operating console **1216**A, and operating console **1216**B has two different end pieces **1225**C and **1225**D. In this different orientation, or second orientation, the operating console **1216**B has been rotated 90 degrees with respect to the first orientation of operating console **1216**A. As can be seen for operating console **1216**B, the longitudinal axis LA**1** is oriented perpendicular to the first axis A**1** and the lateral axis LA**2** is oriented parallel to the first axis A**1**. In some instances, this second orientation may be considered a portrait or vertical orientation.

[0126] For operating console **1216**B, its end pieces **1225**C and **1225**D have been attached to the button deck **1223** and they have replaced end pieces **1225**A and **1225**B. The end piece **1225**C has a display **1229** that is configured to display information to a player, such as information about the game being played on the EGM or information related to the player's account. The other end piece **1225**D has a button **1227**C that has the same shape as buttons **1227**A and **1227**B, but this button has a different orientation with respect to the end piece and the button deck than buttons **1227**A and **1227**B. For a player in the position P**1** relative to operating console **1216**A and operating console **1216**B, the buttons **1227**A, **1227**B, and **1227**C are positioned in the same orientation with respect to this position P**1** despite rotation of the operating consoles **1216**A and **1216**B.

[0127] To provide the rotatable operating console with buttons in the same orientation with respect to a player, different end pieces may be used that position the buttons in different orientations with respect to the operating console. As illustrated in FIG. **13**, the operating console **1216** has different end pieces with buttons in different orientations, and these different orientations are configured to position the buttons in the same orientation with respect to the seat or position P**1**. More specifically, in the lefthand side of FIG. **13**, the first and second buttons **1227**A and **1227**B of end pieces **1225**A and **1225**B, respectively, are positioned in a first button orientation with respect to the button deck **1223** and in a second button orientation with respect to the first axis A**1** and position P**1**. To further illustrate this orientation, a representative dashed triangular outline of button **1227**A is depicted on end piece **1225**A. In this configuration, one triangular point TP**1** of the button is farther from the first axis A**1** than the other two points of the triangle. This one triangular point TP**1** is also facing towards the curved outer surface of the end piece **1225**A. In some instances, like shown, the other two triangular points may be arranged along an axis parallel to the longitudinal axis LA**1**.

[0128] When the operating console is rotated from the first orientation (e.g., the lefthand side of FIG. **13**) to the second orientation (e.g., the righthand side of FIG. **13**), the operating console **1216**B has another end piece **1225**D with button **1227**C. In some implementations, button **1227**C is the same as button **1227**A or **1227**B, although button **1227**C is positioned in a different orientation than buttons **1227**A and **1227**B with respect to the end piece and operating console **1216**B. As can be seen, the button **1227**C has been rotated with respect to the end piece **1225**D and the button deck **1223**. In comparison to buttons **1227**A and **1227**B, the button **1227**C has been rotated 90 degrees with respect to the end piece **1225**D and button deck **1223**. The triangular point TP**1** of button **1227**C is facing a linear section of the end piece **1225**D and facing the button deck **1223**. Further, the other two triangular points are arranged along an axis parallel to the lateral axis LA**2** of the operating console

1216B. With respect to the first axis A**1** and the position P**1**, the button **1227**C is positioned in the same orientation as buttons **1227**A and **1227**B.

[0129] In some implementations, an end piece may be oriented at a non-parallel angle with respect to the button deck. Referring back to the end piece **1225**C with the display **1229**, this end piece is oriented at a non-parallel angle θ1 with respect to the button deck **1223**. This angle may orient the display **1229** so it is more visible to a player in the seat than if the display **1229** was parallel to the button deck **1223**. In some implementations, this angle θ1 may be considered obtuse such that it is larger than 90 degrees and smaller than 180 degrees, such as about 120 degrees, about 135 degrees, or about 145 degrees, for example.

[0130] The ability to uncouple and couple the end pieces provides numerous advantages. For example, when an end piece needs repair or replacement (e.g., the button stops working or breaks), a new end piece can be positioned on the operating console **1216** without repairing the entire operating console **1216** or seating system. In another example, end pieces with different buttons, displays, or other functionality may be provided on the operating console **1216** so that it may be used with multiple EGMs and games without redesigning or building a new seating system for each EGM or game, and thereby reducing costs and time. In yet another example, as described herein, the buttons, displays, or other features of the end pieces can be oriented in the desired positions despite rotation or other movement of the operating console.

[0131] The rotation of the operating console is also illustrated in FIGS. **14** and **17**A-**17**C. FIG. **14** depicts seating system **1200** in another configuration. In some instances, this other configuration may be the same as that of operating console **1216**B of FIG. **13** described above. Here in FIG. **14**, the operating console **1216** has been rotated by 90 degrees as compared to that of FIGS. **12**A-**12**E, such as into the vertical or portrait orientation. The two end pieces **1225**C and **1225**D of the operating console **1216** of FIG. **14** are also different than the two end pieces **1225**A and **1225**B of the operating console **1216** of FIGS. **12**A-**12**F. Like in FIG. **13**, the two end pieces **1225**A and **1225**B have been decoupled and replaced with the end piece **1225**C having the display and end piece **1225**D having the button **1227**A in a different orientation, with respect to the button deck **1223**, than buttons **1227**A and **1227**B. Although FIGS. **12**A-**14** show the operating console being rotated between two orientations 90 degrees apart, the angular difference may be other angles in other implementations. For instance, the angular difference between the two orientations may be 90 degrees or within about +/−10 degrees from 90 degrees.

[0132] Configuring the operating console **1216** in multiple orientations provides for additional gameplay flexibility. For example, as illustrated in FIG. **12**B, the operating console **1216** is in a first orientation configured to provide for two-player gameplay. This first orientation may be that of the lefthand side of FIG. **13**. Here in FIG. **12**B, the first button **1227**A is closer to one side S**1** of the seat **1204** and the second button **1227**B is closer to the other side S**2** of the seat. As further illustrated, one button **1227**A is provided adjacent or closer to one player **1233**A and the other button **1227**B is provided adjacent or closer to the other player **1233**B. This configuration of the seating system **1200** and operating console **1216** provides each player with their own button for playing a game. Positioning the operating console in a different configuration, such as that illustrated in the righthand side of FIG. **13**, and in FIG. **14**, may provide for other gameplay functionality. In FIG. **14**, the operating console **1216** is positioned in the portrait orientation and this configuration may provide different or similar game play functionality than in FIGS. **12**A-**12**F. For example, in FIG. **14** a single button **1227**A is provided adjacent to the seat and may be the sole button for gameplay which may be used by one or both players on the seat **1204**. Additional excitement or gameplay may be had by providing both players access to only one gameplay button.

[0133] In some implementations, the operating console may also be configured to translate, or move, along an axis. This translation movement may be relative to the kiosk **1212**, the seat **1204**, or both. The operating console translation is illustrated in FIGS. **17**A-**17**C which depict off-angle views of the kiosk and the operating console in various configurations. Here, the kiosk **1212** and operating console **1216** are representationally depicted, along with a translation axis TAI. The kiosk **1212** and operating console **1216** in FIGS. **17**A-**17**C may be considered any of the kiosks and operating consoles provided herein, such as those illustrated in FIGS. **12**A-**14**.

[0134] The operating console **1216** is configured to be moved to one or more positions along the translation axis TAI. This movement may be by a player, a worker, a mechanical system (e.g., having a motor and a lead screw or linear translation screw), or a combination thereof. For example, the kiosk **1212** may have a slot or groove **1237** in which a connection feature of the operating console **1216** moves. The connection feature may be a connection to a linear translation screw, linear bearing, rail, or the like. In some instances, a spring friction or zero gravity display mechanism may be used. This can include a slotted groove along the translation axis TAI and a T-shaped bearing fitted into the slot of the slotted groove and attached to the operating console **1216**. One or more springs are provided between the T-shaped bearing and the slotted groove to apply a spring force against both components. This spring force causes the T-shaped bearing to contact the slotted groove and to create a friction force between surfaces of the T-shaped bearing and the slotted groove to prevent relative motion between these components. The T-shaped bearing is configured to be movable, such as by a player, by pushing the operating console **1216** and T-shaped bearing relatively closer to the slotted groove which exerts a force opposite the spring force and releases the frictional force and allows the T-shaped bearing to be moved within and along the slotted groove. When this pushing is released, the spring force again causes the T-shaped bearing to contact the slotted groove and to create the friction force to prevent relative motion between the T-shaped bearing and the slotted groove.

[0135] In another example, a linear bearing may be connected to the operating console, provided in or on a guide rail positioned along the translation axis TAI, and configured to be clamped to the guide rail by a clamping mechanism. Actuation of a trigger or latch is configured to release the clamping mechanism and provide for movement of the linear bearing along the guide rail and release or further actuation of the trigger or latch may reengage the clamping mechanism and prevent further movement of the linear bearing. In yet another example, the motor and lead screw may be used and actuated to move the operating console

1216 based on input received from a player or other person, such as through a button on the operating console **1216**.

[0136] Further, in some implementations, the translation axis TAI may be linear like illustrated in FIGS. **17**A-**17**C, and the resulting translation by the operating console may also be linear. In some other implementations, the translation axis TAI may be nonlinear, e.g., curved, such that is has one or more nonlinear sections, or one or more nonlinear sections and one or more linear sections. The resulting translation by the operating console with respect to the kiosk may be nonlinear, or partially linear and partially nonlinear. Even with a nonlinear translation axis, this axis may still be configured to provide linear translation of the operating console with respect to the seat. For example, the operating console is configured to translate linearly towards and away from the seat.

[0137] Referring first to FIG. **17**A, the operating console **1216** is in a first position, relative to the kiosk **1212**, along the translation axis TAI. In FIG. **17**B, the operating console **1216** is in a second position along the translation axis TAI that is different than the first position. The second position of FIG. **17**B may be considered closer to the seat than the first position.

[0138] The operating console may be configured to both rotate, e.g., like in FIGS. **12**A and **14**, and translate, e.g., like in FIGS. **17**A and **17**B. To illustrate this movement, FIG. **17**C further illustrates that the operating console **1216** may be rotated about a rotation axis RAI and translated along the translation axis TAI. The rotatable mount **1235**, which is configured to provide the rotational movement of the operating console, is also representationally shown with a dashed circle **1235** here in FIG. **17**C. The operating console **1216** may be configured to rotate about the rotation axis RAI at any point along the translation axis TAI.

[0139] Similar to above, translating the operating console to multiple positions provides for additional gameplay flexibility, as well as adaptive positioning for players. For example, a player can move the operating console to a comfortable distance away from the player, and the player can move the operating console farther away from the seat to provide clearance for the player's movement into and out of the seat. Additionally, this translation movement may assist with configuring the seating system into one-player or two-player configurations. For example, it may be desirable to position the operating console at the second position of FIG. **17**B and the first orientation in FIGS. **17**B and **12**A-**12**F, such as for some two-player games.

[0140] It may also be desirable to rotate the operating console to the second orientation of FIG. **14** for other gameplay or games. Given the size of the seating system and its components, it may be advantageous to move the operating console closer or farther from the seat to accommodate the second orientation and rotation to the second orientation. For instance, the operating console may be too close to the seat in the second orientation for a player to sit comfortably. Translating the operating console farther from the seat in this second orientation may provide for additional space and clearance for a player on the seat. In some instances, this movability of the operating console may provide players the option to configure the seating system for one-player or two-player gaming without the use of tools and at any time desired by the players. A player may push, pull, and/or rotate the operating console, without using a tool, between both orientations and positions with respect to the seat.

[0141] Some other aspects of the seating systems provided herein will now be discussed, such as flexible displays positioned on one or more areas of the seat. Each seating system has a seat with a seating surface and a backrest. The seating surface is configured to support, and be in contact with, a player's buttocks, and the backrest is configured to support, and be in contact with, the player's back while sitting on the seating surface. In some instances, the seat may have a side panel adjacent to the seating surface and backrest, an armrest also adjacent to the seating surface and backrest, or both. In some instances, the armrest may be a part of the side panel. As provided herein, some implementations of the seating systems may have one or more flexible displays configured to display various images and videos, and positioned on a seat side panel, an armrest, or both. The side panel and/or armrest may have one or more curved surfaces, and the one or more flexible displays may be curved and positioned on these one or more curved surfaces. In some implementations, the flexible displays are configured to display game information, attractions, a portion of a game instance, images, videos, or a combination thereof. These seating systems may also have any of the features provided herein, such as the kiosk **1212** and operating console **1216** described above. The flexible, curved displays provided herein may be flexible liquid crystal displays (LCD), flexible electrophoretic displays (EPD), flexible organic light emitting diode (OLED) displays, rollable TFT-driven OLED display boards, electronic paper displays, and the like.

[0142] Although additional details and examples of seating systems having flexible displays are provided farther below, a few brief examples will now be provided. In one example, the seating system of FIGS. **4**B and **4**D has a seat **405** with a side panel **439** and a flexible display **436** on the side panel **439**. The seat **405** also has an armrest **440** and a flexible display **420** on the armrest **440**. In another example, the seating system of FIGS. **11**A and **11**B has a seat **1104** with a side panel **1184** having a curved flexible display thereon. The seat **1104** also has an armrest **1140** with multiple parts including a flexible curved display **1148** positioned thereon. In yet another example, referring back to FIGS. **12**A-**12**F and **14**, the seating system **1200** may have at least one flexible curved display panel on the armrest, side panel, or both. In FIG. **12**E, a representative curved flexible display panel **1236** is illustrated, which is similar to flexible display panel **436** in FIG. **4**B. This curved flexible display panel **1236** is represented by the dashed shape. The flexible display panel **1236** may have various shapes that follow the contours and shapes of the side panels. This may include curved flexible display panels with outer boundaries having linear and non-linear edges.

[0143] In some implementations, the seating systems provided herein are configured to display various images and/or videos on the flexible displays positioned on the side panels and/or armrests. These videos and/or images may be based on, or related to, events or aspects of a game played on the EGM communicatively connected to the seating system. For instance, a game outcome, a bonus game trigger, a bonus game outcome, a secondary game trigger, or a secondary game outcome may cause one or more images or videos to be displayed on the flexible displays of the seat. These images or videos may be related to the trigger and/or outcome of such games. In some implementations, these outcomes or triggers may collectively represent a "trigger condition" for a controller that is configured to control the operation of the flexible display to cause the display to display images and/or videos related to the outcome, trigger, or other event in the game. In some instances, this may include images or videos that are not displayed on the one or more displays of the EGM and only displayed on the flexible displays of the seat.

[0144] The trigger condition, e.g., outcomes or triggers in a game, may be transmitted from the game controller of the EGM to a controller of the seating system in various ways. In some instances, this trigger condition may be transmitted as an instruction to the controller of the seating system. In other instances, this trigger condition may be transmitted as data that the controller of the seating system receives, analyzes, and interprets in order to cause the flexible displays on the seat to display images or videos.

[0145] In one example, when a secondary or bonus game is triggered during gameplay of the EGM, one or more images or videos may be displayed on the flexible displays of the seat indicating that the secondary or bonus game has been triggered. In one example, when a game on the EGM having a theme with buffalos triggers a secondary game, images or videos of buffalos may be seen moving on the flexible displays of the seat. In another example, when a game on the EGM having a theme with dragons triggers a secondary game, images or videos of flames or dragons flying may be seen moving on the flexible displays of the seat. In yet another example, when a game on the EGM having a theme with fireworks triggers a secondary game, images or videos of fireworks exploding may be seen moving on the flexible displays of the seat. In another example, when a game on the EGM having a theme with a sports game, like football, baseball, basketball, soccer, or the like, triggers a secondary game, images or videos of the sports game may be depicted on the flexible displays of the seat.

[0146] In some implementations, the images or videos displayed on the flexible displays of the seat may indicate the achievement of triggering a secondary or bonus game, as well as winning one or more awards, games, or jackpots. The images or videos displayed on the flexible displays of the seat may therefore be caused to be displayed based on the triggering event. For example, when the secondary or bonus game is triggered, various images or videos displayed in the flexible displays of the seat may celebrate or indicate the celebration of this triggering which may enhance the celebration of such event. These images or videos may also catch the attention of persons and prospective players around the seat that the triggering has occurred on the EGM and may cause such persons to become players.

[0147] In some implementations, the images or videos displayed on the flexible displays of the seat may display information related to the game being played on the EGM. This information may be any of the game information provided herein. For example, the wager amounts available for the game, or the available jackpots, may be displayed on the flexible display on the side panel and/or armrest. This can advantageously present the game information in a larger format than on the EGM itself which can attract potential players to play the EGM.

[0148] In some implementations, the images or videos displayed on the flexible displays of the seat may display information unrelated to the game being played on the EGM. For example, advertisements or events in the building where the seat is located may be displayed. This may include advertisements for food or beverage discounts, or concerts and tournaments in the building.

[0149] Other aspects of the seating systems will now be discussed, such as the physical coupling or uncoupling between a seating system and an EGM. In some implementations, the seating systems provided herein may be physically separate, or physically uncoupled, from the displays of the EGM to which the seating system is communicatively connected. For example, referring to FIG. **15**C, an electronic gaming system **1500** is provided that has a display **1511** of an EGM and three seating systems **1200** communicatively coupled to the EGM. Although the display of the EGM may have various shapes and sizes, the display **1511** here in FIG. **15**C is a large wheel **1508** having an indicator **1512**. In other implementations, the display may be differently shaped, sized, and spaced from the seating systems. In this example of FIG. **15**C, the three seating systems **1200** are not physically coupled to the display **1511** of the EGM.

[0150] Other seating systems provided herein may also be physically uncoupled from, or not physically coupled to, an EGM. Referring back to the seating systems **1200** described above, such as those in FIGS. **12**A-**14**, these seating systems may be considered physically uncoupled from, or not physically coupled to, the display of the EGM to which the seating system **1200** is communicatively coupled. In these instances, the EGM, including its displays, are physically separate, and physically uncoupled, from the seating system.

[0151] In some other implementations, the seating systems provided herein may be physically coupled with the displays of the EGM. This may include, for example, one or more EGM displays positioned on or coupled to one or more aspects of the seating system. Some examples are illustrated in FIGS. **16**A-**16**C which depict seating systems with one or more EGM displays. In FIG. **16**A, the seating system **1600**A has an operating console **1616** and one display **1611** of the EGM physically coupled to the seating system **1600**A, such as to the kiosk **1612**. In FIG. **16**B, the seating system **1600**B has two displays **1611**A and **1611**B of the EGM physically coupled to the seating system **1600**B, such as to the kiosk **1612**. In FIG. **16**C, the seating system **1600**C has a different display **1611**C of the EGM physically coupled to the seating system **1600**C, such as to the kiosk **1612**. Additional features of the seating system in FIGS. **16**A-**16**C are provided below.

[0152] As noted herein, the seating system may have one or more controllers that are communicatively coupled with one or more game controllers of the EGM or gaming system. In some implementations, this communication coupling may be hardwired via one or more cables or wires, it may be a wireless connection, or both. This communication coupling is configured to transmit the player input received at the seating system, such as on the operating console **1216**, to the game controller of the EGM. The game controller can use this player input from the seating system to execute one or more game instances. In some implementations, the one or more controllers of the seating system may also be configured to receive information and instructions from the game controller, such as information about the one or more game instances, like a trigger condition provided above. The one or more controllers of the seating system may cause, based on this information and/or instructions from the game controller, images and/or videos to be displayed on the flexible displays of the seat.

[0153] The subject disclosure also includes additional or alternative implementations and features of various seating systems, which are detailed below. Any of the following seating systems or seats may have any of the features of the seating systems described above. For example, benches, or seats, of seating systems **404**, **405**, **504**, **604**, **704**, **804**, **904**, **1004**, and **1104** may have the operating console provided above, such as the operating console **1216** illustrated in FIGS. **12A-14** and **17A-C**.

[0154] The additional or alternative seating system implementations will now be discussed. Some of these implementations may refer to a "gaming bench" or "bench" and these phrases may be used synonymously with the term "seat". Referring to FIGS. **4A-4G**, these Figures illustrate different views of a seating system **400** with a first seat **404**, which may also be considered a first gaming bench **404**. The first seat **404** is in the form of a lounger style bench, and includes a number of amenities including one or more displays, projectors, and audio-visual devices to engage players or non-players for a better overall customer experience, while providing a variety of upgrade, reconfiguration, and modularity capabilities. As shown in FIG. **4A**, the seat **404** provides an ability to connect to a new operating console and kiosk, such as those illustrated in FIGS. **12A-14** and **17A-17C**, for example. The seat **404** is also configured to be a part of a seating system communicatively connected to a large LED wheel display of an electronic gaming system, as well as to a standard gaming cabinet of an EGM, such as those shown in FIG. **1**.

[0155] Specifically, FIG. **4A** shows that the seat **404** also includes one or more surround sound speakers **408** embedded in a backrest **410**, and a subwoofer port **412** to reproduce or generate surround sounds and subwoofer bass. The backrest **410** may also include one or more hydraulic and/or acoustic mechanisms to produce vibration and/or massaging effects on the first gaming bench **404** that also may be synchronized with a game being played. In some examples, the backrest **410** may include RGB LED lit fabrics, thus embedding one or more RGB LED's. The RGB LEDs in the RGB LED lit fabrics may include configurable or controllable LED lighting patterns intended to match specific game characteristics. In the example shown, the first gaming bench **404** may have a width of about 49", a height of about 41", and a depth of 31". Other dimensions are also possible. Various ergonomic controls and adjustments may be possible. Thus, the first gaming bench **404** may be able to accommodate one or more players. Further, although not shown, the first gaming bench **404** may include foldable cup holders that may be tucked away within the backrest **410**.

[0156] The first gaming bench **404** also includes panels **416**, **418**, and **420**. Panel **416** may be a side panel, panel **418** may be considered a front seat panel, and panel **420** may be an armrest. In some examples, some or all the panels **416**, **418**, and **420** may include colored plastic panels. In other examples, some or all the panels **416**, **418**, and **420** may include laminated wraps such as cherry wood panels. In some examples, some or all the panels **416**, **418**, and **420** may be configured to have, project, or display static colors and/or artworks. That is, some or all the panels **416**, **418**, and **420** may include electronic display panels to display video.

[0157] In some other implementations, as shown in FIG. **4B**, another gaming bench **405** or seat **405** has side panels **439** (like panel **416**) that may include one or more flexible display panels **436** contoured about an armrest **440**, panel **420**, and/or side panel **439**. In FIG. **4B**, reference numerals are used that refer to the same parts in other figures of FIGS. **4A** AND **4C-4G**. In some instances, the flexible displays **436** on the side panels **439** may be configured to display images, videos, game information or attractions, or a portion of the game being played. The seat **405** may also include a footrest **424**. As shown, the seat **404** and seat **405** may be pivotably positioned on a swivel base **428**.

[0158] In some implementations, the swivel base **428** may have a swivel angle **429** of 30° in two different directions. FIG. **4G** shows the first gaming bench **404** may be configured to rotate to right, left, or both. In other examples, swivel base **428** may have other swivel angles, such as 45 degrees or 60 degrees. In some examples, the backrest **410** may recline with respect to the swivel base **428**.

[0159] In the implementations of FIGS. **4A-4G**, the seats **404** and **405** may also have a plug-and-play extension board **430** configured to couple to a cabinet, such as the main cabinet **116** of FIG. **1**, a pedestal, a display, or a kiosk similar to kiosk **1212** described above. In some examples, the plug-and-play extension board **430**, when coupled to the cabinet, pedestal, kiosk, or display, may interface data communications between the seat **404** or **405** and the cabinet, pedestal, kiosk, or display. For example, when the seat **404** or **405**, cabinet, pedestal, kiosk, or display is configured to play a specific game with specific sound, the seat **404** or **405** may transmit data to flexible displays **436** on the side panels **439**, the panels **418**, **420**, the surround sound speakers **408**, the backrest **410**, and the subwoofer port **412** for further manifestation of the specific game and sound.

[0160] In some instances, at least a portion of the data transmitted to control flexible displays **436** on the side panels **439**, the panels **418**, **420**, the surround sound speakers **408**, the backrest **410**, and the subwoofer port **412** may be received from the cabinet, pedestal, kiosk, or display. In some examples, the seat **404** or **405** may include some or all the components in the cabinet such as the main cabinet **116** of FIG. **1**. In such cases, when coupled to the seat **404** or **405**, the kiosk may simply provide display functionalities, while the seat **404** or **405** may provide processing and communication functionalities similar to those discussed above with respect to FIGS. **1** and **2**. That is, the seat **404** or **405** may include a bill validator (similar to the bill validator **234** of FIG. **2**), a player tracking device (similar to the player tracking interface **232**), one or more secondary display device, and the like, detailed below.

[0161] Referring to FIG. **4C**, the seat **404** may also include a display panel **444** configured to display game information, game offerings, and/or a game being played. In some examples, the display panel **444** may include flexible liquid crystal display (LCD) displays, flexible electrophoretic display (EPD) displays, and flexible organic light emitting diode (OLED) displays, rollable TFT-driven OLED displays, electronic paper displays, and the like. In some examples, the display panel **444** may also include RGB LED surround lighting. In other examples, display panel **444** may include one or more immersive LCD screens. Other displays are also envisioned. As shown, the first gaming bench **404** may also include a high-definition projector **448** operable to project customizable game information. In some examples, the first gaming bench **404** may also include RGB LED lighting about the armrest **440** and the swivel base **428**.

[0162] Similarly, in FIG. **4D**, the seat **405** includes RGB lighting displays **436** on the side panels **439** that may include one or more flexible display panels contoured about the armrest **440** and configured to display game information or attractions, or a portion of the game being played. The seat **404** or **405** may also include a secondary display device **456**. In some examples, the secondary display device **456** may include a custom tube display with RGB LED lighting. In some other examples, the secondary display device **456** may include a 3-D holographic LED tube. In still other examples, the secondary display device **456** may include a flexible display. In some examples, the secondary display device **456** may display at least a portion of the game being played in the seating system **400**, or other gaming information.

[0163] FIGS. **4**E and **4**F show a front and side view, respectively, of another variation of the seat **404** and **405** which may have the same or similar features as that of FIGS. **4**A-**4**D.

[0164] FIG. **5** illustrates different views of a second version of a seating system **500**. Similar to FIGS. **4A-4G**, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers in other figures (e.g., backrest **510** of FIG. **5** corresponds with backrest **410** of FIGS. **4**A-**4**G). Like with seat **404** or **405**, the seating system **500** of FIG. **5** includes an integrated modular gaming bench or seat, or a second gaming bench or seat **504**. The second seat **504** also includes one or more surround sound speakers **508** embedded in a backrest **510**, and a subwoofer port **512** to reproduce or generate surround sound and subwoofer bass. The backrest **510** may also include one or more acoustic and/or hydraulic mechanisms to produce vibration effects that may be synchronized with a game being played.

[0165] Like the backrest **410** in FIG. **4**A, the backrest **510** may also include RGB LED lit fabrics. In the example shown, the second gaming bench **504** may have a width of about 45", a height of about 40", and a depth of 29". Thus, the second gaming bench **504** may be able to accommodate one or more players. Further, although not shown, the second gaming bench **504** may include foldable cup holders that may be tucked away within the backrest **510**.

[0166] The second gaming bench **504** also includes front panel **516**, side panel **518** (which may have one or more curved flexible displays positioned thereon), armrest panel **520** (which may also have one or more curved flexible displays positioned thereon), and back panel **522**. The armrest panel **520** is similarly configured to contour around the backrest **510**. However, unlike panel **420** of FIG. **4**A, the armrest panel **520** does not wrap around toward the front panel **516**. In some other examples, the armrest panel **520** is contoured about an armrest **540** and configured to display game information or attractions, or a portion of the game being played. The second gaming bench **504** also includes a footrest **524**. As shown, the second seat **504** is pivotally positioned on a swivel base **528**. Although not shown, the second gaming bench **504** also includes a plug-and-play extension board (like the plug-and-play extension board **430** of FIG. **4**A) to be coupled to a cabinet such as the main cabinet **116** of FIG. **1**, or kiosk as detailed herein. The second seat **504** may also include an LED display **544** (like the display panel **444** of FIG. **4**C) configured to display game information, game offerings, and/or a game being played. The second gaming bench **504** may also include a high-definition projector **548** (like the high-definition projector **448** of FIG. **4**C). In some examples, the backrest **510** may recline with respect to the swivel base **528**.

[0167] FIG. **6** illustrates a third version of a seating system **600**. Again, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers in other figures (e.g., backrest **610** of FIG. **6** corresponds with backrest **410** of FIGS. **4**A-**4**G). Like the first gaming bench **404**, the seating system **600** includes an integrated modular gaming bench, or a third gaming bench **604** or seat **604**. The third gaming bench **604** also includes one or more surround sound speakers **608** embedded in a backrest **610**, and a subwoofer port **612** to reproduce or generate surround sounds and subwoofer bass. The backrest **610** may also include one or more acoustic and/or hydraulic mechanisms to produce vibration or massaging effects that may be synchronized with a game being played.

[0168] Like the backrest **410** in FIG. **4**A, the backrest **610** may also include RGB LED lit fabrics. The third gaming bench **604** may be able to accommodate one or more players. Further, although not shown, the third gaming bench **604** may include foldable cup holders that may be tucked away within the backrest **610**. The third seat **604** also includes front panel **616**, side panel **618** (which may have one or more curved flexible displays positioned thereon), and armrest panel **620** (which may have one or more curved flexible displays positioned thereon). Armrest panel **620** is similarly configured to contour around the backrest **610**. In some other examples, a flexible display panel is contoured about an armrest **640** and configured to display game information or attractions, or a portion of the game being played. The third gaming bench **604** also includes a footrest **624**. As shown, the third gaming bench **604** is pivotally positioned on a swivel base **628**. Although not shown, the third gaming bench **604** also includes a plug-and-play extension board (like the plug-and-play extension board **430** of FIG. **4**A) to be coupled to a cabinet such as the main cabinet **116** of FIG. **1**, or kiosk as detailed herein. Although not shown, the third gaming bench **604** may also include a flexible display on a side panel, armrest, or both (similar to the display panel **436** of FIG. **4**D) configured to display game information, game offerings, and/or a game being played, and a high-definition projector (similar to the high-definition projector **448** of FIG. **4**C). In some examples, the backrest **610** may recline with respect to the swivel base **628**.

[0169] FIG. **7** illustrates different views of a seating system **700**. Here, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers of other figures (e.g., backrest **710** of FIG. **7** corresponds with backrest **410** of FIGS. **4A-4G**). Like the first seat **404**, the seating system **700** includes an integrated modular gaming bench, or a fourth gaming bench **704** or seat **704**. The fourth gaming bench **704** includes one or more surround sound speakers **708** embedded in a backrest **710**, and a subwoofer port **712** to reproduce or generate surround sound and subwoofer bass. The backrest may also include RGB LED lit fabrics. In the example shown, the fourth gaming bench **704** may accommodate one or more players. Further, although not shown, the fourth gaming bench **704** may include foldable cup holders that may be tucked away within the backrest **710**. The fourth gaming bench **704** also includes front panel **716**, side panel **718**, armrest panel **720**, and back panel **722**. The armrest panel **720** is similarly configured to contour around the backrest **710**. In some other examples, the armrest panel **720** has one or more curved flexible displays positioned thereon, is contoured about an armrest **740**, and is configured to display game information or attractions, or a portion of the game being played. The fourth gaming bench **704** is pivotably positioned on a swivel base **728**.

[0170] Although not shown, the fourth seat **704** also includes a plug-and-play extension board (like the plug-and-play extension board **430** of FIG. **4**A) to be coupled to a cabinet such as the main cabinet **116** of FIG. **1**, or kiosk as detailed herein. The fourth gaming bench **704** may also include an LED display **744** (like the display panel **444** of FIG. **4**C) configured to display game information, game offerings, and/or a game being played. In some examples, the LED display **744** includes a 27" 4K high resolution LCD or LED display panel. The fourth gaming bench **704** may also include a high-definition projector **748** (like the high-definition projector **448** of FIG. **4**C) to display different gaming device information and/or games being offered. In some examples, the backrest **710** may recline with respect to the swivel base **728**.

[0171] FIGS. **8**A and **8**B depict yet another example seating system **800**. Here again, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers of other figures (e.g., backrest **810** of FIG. **8**A corresponds with backrest **410** of FIGS. **4**A-**4**G). Like the first gaming bench **404**, the seating system **800** includes an integrated modular gaming bench, or a fifth gaming bench **804** or seat **804**. The fifth gaming bench **804** is a three-quarter style bench that includes a pair of surround sound speakers **808** embedded in a backrest **810**, and a pair of subwoofer ports **812** to reproduce or generate surround sound and subwoofer bass. The backrest **810** may also include one or more acoustic and/or hydraulic mechanisms (not shown) to move the gaming bench **804** in various directions and/or to produce vibration or massaging effects that may be synchronized with a game being played. Like the backrest **410** in FIG. **4**A, the backrest **810** may also include RGB LED lit fabrics. In the example shown, the fifth gaming bench **804**, although not shown, may also include foldable cup holders that may be tucked away within the backrest **810**.

[0172] The fifth gaming bench **804** also includes front panel **816** that supports the pair of subwoofer ports **812**. As shown, the fifth gaming bench **804** may be pivotably positioned on a swivel base **828**. The fifth gaming bench **804** also includes a plug-and-play extension board **830** (like the plug-and-play extension board **430** of FIG. **4**A) to be coupled to a cabinet such as the main cabinet **116** of FIG. **1**, or kiosk as detailed herein. The fifth gaming bench **804** may also include an LED display **844** (like the display panel **444** of FIG. **4**C) configured to display game information, game offerings, and/or a game being played from the back of the fifth gaming bench **804**.

[0173] The fifth seat **804** may also include a high-definition projector **848** (like the high-definition projector **448** of FIG. **4**C). The high-definition projector **848** is depicted displaying images or videos downward, but it may display images or videos in other directions. As shown, the fifth gaming bench **804** may also include a secondary display device **856** (like the secondary display device **456** of FIG. **4**A), that may include a combination of a custom tube display with RGB LED lighting, a 3-D holographic LED tube, a flexible display, and the like. In some examples, the secondary display device **856** may display at least a portion of the game being played in the seating system **800**, or other gaming information. In some examples, the backrest **810** may recline with respect to the swivel base **828**.

[0174] FIGS. **9**A and **9**B depict another example seating system **900**. Here again, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers of other figures (e.g., backrest **910** of FIG. **9**A corresponds with backrest **410** of FIGS. **4**A-**4**G). Like the seating system **800** illustrated in FIGS. **8**A and **8**B, the seating system **900** includes an integrated modular gaming bench, or a sixth gaming bench **904** or seat **904**. The sixth gaming bench **904** is a full-size bench (unlike the three-quarter bench of FIG. **8**A), sometimes referred to as a high boy bench, that includes a pair of surround sound speakers **908** embedded in a backrest **910**, and a pair of subwoofer ports **912** on front panel **916** to reproduce or generate surround sound and subwoofer bass. The backrest **910** may also include one or more acoustic and/or hydraulic mechanisms (not shown) to produce vibration or massaging effects that may be synchronized with a game being played.

[0175] Like the backrest **410** in FIG. **4**A, the backrest **910** may also include RGB LED lit fabrics. In the example shown, the sixth gaming bench **904**, although not shown, may also include foldable cup holders that may be tucked away within the backrest **910**. The sixth gaming bench **904** also includes a plug-and-play extension board **930** (like the plug-and-play extension board **430** of FIG. **4**A) to be coupled to a cabinet such as the main cabinet **116** of FIG. **1**, or kiosk as detailed herein. The sixth seat **904** may also include an LED display **944** (like the display panel **444** of FIG. **4**C) configured to display game information, game offerings, and/or a game being played from the back of the sixth gaming bench **904**. The sixth gaming bench **904** may also include a high-definition projector **948** (like the high-definition projector **448** of FIG. **4**C), and a secondary display device **956** (like the secondary display device **456** of FIG. **4**A). In some examples, the secondary display device **956** may display at least a portion of the game being played in the seating system **900**, or other gaming information.

[0176] FIG. **10** depicts yet another example seating system **1000**. Here, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers of other figures (e.g., backrest **1010** of FIG. **10** corresponds with backrest **410** of FIGS. **4**A-**4**G). The seat **1004** of seating system **1000** has a backrest **1010**, a seating surface **1017**, and one or more flexible displays, such as display **1044** on the backrest **1010** or a flexible display **1036** on the side panel of the seat **1004**.

[0177] FIGS. **11**A and **11**B depict another example seating system **1100**. Here, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers of other figures (e.g., backrest **1110** of FIG. **11**A corresponds with backrest **410** of FIGS. **4**A-**4**G). These FIGS. **11**A and **11**B illustrate exploded views of seating system **1100** including a seat **1102** and a kiosk **1112** that may be coupled to a plug-and-play extension board **1108** (like the plug-and-play extension board **430** of FIG. **4**A). In the example shown, the kiosk **1112** may also be communicatively coupled to a display device that displays a game being played. The kiosk **1112** may include a button deck (such as, for example, the button deck **120** of FIG. **1** and operating console **1216** with button deck **1223** provided above). In some instances, the kiosk **1112** may be physically coupled to the display of the EGM or physically uncoupled from the display of the EGM. In still further examples, the kiosk **1112** may integrated on a track system or the like (not shown) to permit the lateral movement of the kiosk **1112**, while allowing the seating system **1100** to be a singular or modular unit. The movement of the kiosk **1112** toward, away, around or some combination of movements from the seat **1104** may facilitate player egress and/or improve ergonomics (e.g., reduce fatigue, increase comfort, etc.) for the player during operation.

[0178] FIG. **11**A illustrates a front exploded perspective view of a bench **1104**, which may also be considered a seat **1104** upon which one or more players may be seated, such as on a seating surface **1117**, like a seat pan, where a player may sit down. The seating surface **1117** is configured to support a player's buttocks. The seat **1104** may have fabrics **1116** embedded with RGB lighting (not shown). The fabrics **1116** may include full grain leather, boucle, chenille, Cryton, denim, canvas, linen, Matelasse, microfiber, micro-suede, and the like. In the example shown, seat **1104** may be supported by a swivel base **1120** (like the swivel base **428** of FIG. **4**A) with a bottom base **1124** and a bottom plate **1128** having a number of adjustable lever feet **1132**. In other examples, seat **1104** may be supported by a fixed base (not shown) with the bottom base **1124** and the bottom plate **1128** having the adjustable lever feet **1132**. As shown, seat **1104** also includes a footrest **1136** extending from the swivel base **1120**. Further, the swivel base **1120** may include one or more subwoofer ports (like the subwoofer port **412** of FIG. **4**A) that may include a vacuum formed acrylic cover **1140** lit with RGB LED. The seat also has a backrest **1110** connected to and adjacent to the seating surface **1117**.

[0179] In the example shown, seat **1104** may be surrounded by a number of panels including a reflective backplate **1144** covered by a flexible panel **1148** that may include a flexible LED panel **1148**, a flexible RGB panel, and the like. A vacuum formed acrylic panel **1152** is installed on top of the flexible panel **1148**. The vacuum formed acrylic panel **1152** may be translucent, clear, solid, and/or reflective. Further, seat **1104** may be lit with a controlled light strip **1156**. In some examples, seat **1104** may include a branding option **1160**.

[0180] FIG. **11**B illustrates a back exploded perspective view of seat **1104**. As shown, seat **1104** includes a siren style lighting tube **1164** (like the secondary display device **456** of FIG. **4**C). The siren style lighting tube **1164** may include a hologram tube, a transparent 12:3 OLED screen, a flexible LED panel, a backlit graphic or artwork, and the like. The seat **1104** also includes a an LED display **1168** (like the display panel **444** of FIG. **4**C) operable to display game information, game offerings, and/or a game being played. The LED display **1168** may include one of a flexible LED panel, an immersive LCD panel, an LCD display, a backlit graphic or artwork, and the like. As shown, the LED display **1168** is shielded by a cover **1172** that may include a vacuum formed acrylic cover with RGB LED's, which may also be translucent, clear, solid color, and/or reflective. Seat **1104** also includes an optional high-definition projector **1176** (like the high-definition projector **448** of FIG. **4**C) to display different gaming device information and/or games being offered, as discussed. The seat **1104** also includes a back panel **1180** (like the back panel **522** of FIG. **5**), and a side panel **1184** (like the side panel **518** of FIG. **5**) covered by a side panel rail **1188** and that may have a curved flexible display.

[0181] FIGS. **15**A-**15**C depict another example seating system **1500**. The electronic gaming system **1500** includes a plurality of seating systems **1200** (like the seating system **1200** in FIGS. **12**A-**12**F or seating system **1400** in FIG. **14**) arranged for playing a group game. In this example, the seating systems **1200** are facing a giant wheel **1508** with an indicator **1512**. Although FIGS. **15**A-**15**C show that the giant wheel **1508** as a physical wheel, the giant wheel **1508** may also be a projected wheel projected from the front projector **1252** (of FIG. **12**A), or a combination of the front projectors **1252** (of FIG. **12**C). The physical wheel could take the form of a different shaped display device.

[0182] FIGS. **16**A-**16**C depict another example seating system **1600**. Here, the last two numbers of each reference numeral refer to the same features with reference numerals having the same last two numbers of other figures (e.g., operating console **1616** of FIG. **16**A corresponds with operating console **121** of FIGS. **12**A-**12**F). FIG. **16**A includes an electronic gaming system **1600**A comprising a kiosk **1612** comprising a operating console **1616** and a power meter display **1611**A. FIG. **16**B is another configuration of the electronic gaming system **1600**B comprising a kiosk **1612** comprising an operating console **1616**, a power meter display **1611**A, and a game display unit **1611**B, which is illustrated as a mini-wheel unit, where the kiosk **1612**, a power meter display **1611**A, and a game display unit **1611**B are integrated a single modular assembly that can easily reconfigured to add, remove or replace parts of the assembly. FIG. **16**C is like the example of FIG. **16**B and illustrates an electronic gaming system **1600**C further comprising a game display unit **1611**C, which is illustrated as a wonder wheel unit. The electronic gaming systems **1600**A-**1600**C may also be integrated into a single unit that may be moved as a single unit, e.g., on wheels or tracks. Such an arrangement may permit an improved ability to replace one or more electronic gaming systems **1600**A-**1600**C on gaming establishment floors. As provided herein, the seating systems **1600**A-**1600**C may have other displays of the EGM, such as rectangular or curved displays similar to those illustrated in FIG. **1** above.

[0183] When an element, such as a layer, is referred to as being "on," "connected to," or "coupled to" another element, it may be directly on, directly connected to, or directly coupled to the other element or at least one intervening element may be present. When, however, an element is referred to as being "directly on," "directly connected to," or "directly coupled to" another element, there are no intervening elements present. Other terms and/or phrases if used herein to describe a relationship between elements should be interpreted in a like fashion, such as "between" versus "directly between," "adjacent" versus "directly adjacent," "on" versus "directly on," etc. Further, the term "connected" may refer to physical, electrical, and/or fluid connection.

[0184] For the purposes of this disclosure, "at least one of X, Y, . . . , and Z" and "at least one selected from the group consisting of X, Y, . . . , and Z" may be construed as X only, Y only, . . . , Z only, or any combination of two or more of X, Y, . . . , and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0185] Although the terms "first," "second," "third," etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure. To this end, use of such identifiers, e.g., "a first element," should not be read as suggesting, implicitly or inherently, that there is necessarily another instance, e.g., "a second element." Further, the use, if any, of ordinal indicators, such as (a), (b), (c), . . . , or (1), (2), (3), . . . , or the like, in this disclosure and accompanying claims, is to be understood as not conveying any particular order or sequence, except to the extent that such an order or sequence is explicitly indicated. For example, if there are three steps labeled (i), (ii), and (iii), it is to be understood that these steps may be performed in any order (or even concurrently, if not otherwise contraindicated), unless indicated otherwise. For example, if step (ii) involves the handling of an element that is created in step (i), then step (ii) may be viewed as happening at some point after step (i). In a similar manner, if step (i) involves the handling of an element that is created in step (ii), the reverse is to be understood.

[0186] Spatially relative terms, such as "beneath," "below," "under," "lower," "above," "upper," "over," "higher," "side" (e.g., as in "sidewall"), and the like, may be used herein for descriptive purposes, and, thereby, to describe one element's spatial relationship to at least one other element as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as "below" or "beneath" other elements or features would then be oriented "above" or "over" the other elements or features. Thus, the term "below" can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

[0187] The term "between," as used herein and when used with a range of values, is to be understood, unless otherwise indicated, as being inclusive of the start and end values of that range. For example, between 1 and 5 is to be understood as inclusive of the numbers **1**, **2**, **3**, **4**, and **5**, not just the numbers **2**, **3**, and **4**.

[0188] As used herein, the phrase "operatively connected" is to be understood as referring to a state in which two components and/or systems are connected, either directly or indirectly, such that, for example, at least one component or system can control the other. For instance, a controller may be described as being operatively connected with (or to) a resistive heating unit, which is inclusive of the controller being connected with a sub-controller of the resistive heating unit that is electrically connected with a relay that is configured to controllably connect or disconnect the resistive heating unit with a power source that is capable of providing an amount of power that is able to power the resistive heating unit so as to generate a desired degree of heating. The controller itself likely will not supply such power directly to the resistive heating unit due to the current(s) involved, but it is to be understood that the controller is nonetheless operatively connected with the resistive heating unit.

[0189] As used herein, the singular forms, "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is also to be understood that the phrases "for each <item> of the one or more <items>," "each <item> of the one or more <items>," and/or the like, if used herein, are inclusive of both a single-item group and multiple-item groups, i.e., the phrase "for . . . each" is used in the sense that it is used in programming languages to refer to each item of whatever population of items is referenced. For example, if the population of items referenced is a single item, then "each" would refer to only that single item (despite dictionary definitions of "each" frequently defining the term to refer to "every one of two or more things") and would not imply that there must be at least two of those items. Similarly, the term "set" or "subset" should not be viewed, in itself, as necessarily encompassing a plurality of items—it is to be understood that a set or a subset can encompass only one member or multiple members (unless the context indicates otherwise). In addition, the terms "comprises," "comprising," "includes," and/or "including," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0190] While the invention has been described with respect to the figures, it will be appreciated that many modifications and changes may be made by those skilled in the art without departing from the spirit of the invention. Any variation and derivation from the above description and figures are included in the scope of the present invention as defined by the claims.

What is claimed is:

1. A seating system for electronic gaming on an electronic gaming machine, the seating system comprising:

a seat with a seating surface and a backrest;

a kiosk directly or indirectly coupled to the seat; and

an operating console rotatably coupled to the kiosk and having:

a button deck,

a first end piece detachably coupled to a first end of the operating console and having a first button, and

a second end piece detachably coupled to a second end of the operating console, opposite the first end, and having a second button or a display, wherein:

the first end piece and the second end piece are configured to be detached from the operating console,

the operating console is configured to receive a third end piece at the first end or the second end when the first end piece or the second end piece, respectively, has been detached,

the operating console is configured to be rotated between a first orientation and a second orientation 90 degrees apart from the first orientation, and

the button deck, the first end piece, and the second end piece are configured to be communicatively coupled to a game controller of the electronic gaming machine.

2. The system of claim **1**, wherein the button deck, the first end piece, the second end piece, or a combination thereof are configured to:
receive input from a user, and
provide information related to the input from the user to the game controller of the electronic gaming machine.

3. The system of claim **1**, wherein the operating console is further configured to translate along a translation axis.

4. The system of claim **3**, wherein the translation axis is perpendicular to the axis of rotation.

5. The system of claim **3**, wherein the operating console is configured to translate along the translation axis without the use of tools.

6. The system of claim **3**, wherein:
the translation axis has a first end and a second end,
the second end is closer to the seat than the first end, and
the operating console is closer to the seat when positioned at the second end than when positioned at the first end.

7. The system of claim **1**, wherein the second end piece is oriented at a non-parallel angle with respect to the button deck.

8. The system of claim **1**, wherein the first button is a physical button.

9. The system of claim **1**, wherein the second end piece has a display that is configured to display player related information.

10. The system of claim **1**, further comprising:
a display of the electronic gaming machine coupled to the kiosk and/or the seat, and
the game controller that is configured to cause a game instance to be displayed on the display.

11. The system of claim **1**, wherein the electronic gaming machine is without any physical coupling to the seat or the kiosk.

12. The system of claim **1**, wherein the seat further comprises at least one display panel configured to display game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

13. The system of claim **12**, further comprising a controller communicatively coupled with the game controller, wherein the controller is configured to display, based on an outcome of a game instance on the electronic gaming machine, game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

14. The system of claim **13**, wherein the game information, attractions, a portion of a game instance, images, videos, or a combination thereof displayed based on the outcome of the game instance is related to a game theme of the game instance.

15. The system of claim **12**, wherein the game information includes one or more wager amounts of the game instance.

16. The system of claim **12**, wherein:
the at least one display panel comprise one or more curved flexible display panel,
the seat further includes at least one side panel adjacent to the seating surface and the backrest, and
the curved flexible display panel is positioned on the side panel.

17. The system of claim **16**, wherein the curved flexible display panel comprises a flexible liquid crystal display (LCD), a flexible electrophoretic display (EPD), a flexible organic light emitting diode (OLED) display, a rollable TFT-driven OLED display, or an electronic paper display.

18. The system of claim **16**, wherein:
the side panel includes an armrest, and
the curved flexible display panel is positioned on the armrest.

19. A seating system for electronic gaming on an electronic gaming machine, the seating system comprising:
a seat with a seating surface, a backrest, and at least one armrest adjacent to the seating surface and the backrest;
at least one flexible curved display panel on the at least one armrest;
a kiosk directly or indirectly coupled to the seat; and
an operating console coupled to the kiosk and having a button deck configured to be communicatively coupled to a game controller of the electronic gaming machine, wherein the at least one flexible curved display panel is configured to display game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

20. The system of claim **19**, further comprising a controller communicatively coupled with the game controller, wherein the controller is configured to display, based on an outcome of a game instance on the electronic gaming machine, the game information, attractions, a portion of a game instance, images, videos, or a combination thereof.

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